

Digital Signal Analysis and Processing

Complete Numerical Problems & Solutions

CT 704 (BEI/BCT, IV/I) · EX 753 (BEX, IV/II)

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Every computational question in all **45 past papers (2066–2082 BS)**, worked, in one volume. **48 worked families** over the seven CT 704 chapters.

The grouping is the point. This subject asks the same six procedures thirty to forty times each, so a family is worked **once in full** — opening with the question exactly as its paper prints it, the paper's own spelling and slips included — and the sibling papers that ask the same thing then follow as a **data-and-answer table**, split out only where the calculation genuinely differs. Nothing is dropped: every computational paper of the 45 is below.

Theory lives in the companion volume, **Digital Signal Analysis and Processing — Complete Exam Notes**. **Chapter 8 is not in either volume**: it is the EX 753 material outside the CT 704 syllabus.

Ch	Topic	Families	Biggest family
1	Discrete-Time Signals and Systems	12	Convolution sum, 45 questions, 277 marks
2	Z-Transform	7	Inverse Z-transform, 32 questions
3	Analysis of LTI Systems in the Frequency Domain	5	Pole-zero plot and magnitude response, 24 papers
4	Discrete Filter Structures	6	Lattice step-down, 38 papers, 314 of 449 marks
5	FIR Filter Design	4	Kaiser-window and optimum/Remez designs
6	IIR Filter Design	7	Digital Butterworth by bilinear transformation, 32 papers
7	Discrete Fourier Transform and FFT	7	8-point DFT by DIT/DIF (41 papers) and circular convolution (35)
Total		48	

Every answer in this volume is machine checked. `src/verify.py` recomputes each published result independently from the problem statement and asserts it against the printed value — convolutions and correlations, ROC-aware inverse Z-transforms, pole-zero and magnitude responses, the lattice step-down in **exact rational arithmetic** (`lattice.py`) and the quantization figures (`quant.py`), every window value, filter length, Kaiser β and Bessel figure (`fir.py`), every order, cut-off, pole, $H(s)$ and $H(z)$ (`iir.py`, itself self-tested against Oppenheim examples 7.2, 7.3 and 7.6), and every DFT, butterfly stage and circular convolution (`dft.py`, which checks DIT and DIF against the direct DFT on nine sequences before it prints). **1743 checks, all passing.**

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Ch 1 · Discrete-Time Signals and Systems

Numerical Problems & Solutions

Convolution alone is 34 of the 45 papers. The four convolution sections below come first for that reason. Within a section the method is worked once, in full, and every other paper that ran the same method is listed under it with its data and its answer: **same process, different numbers, so only the numbers change.**

What you need before starting

- $y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$, and it is commutative, so fold whichever sequence is shorter.
- **Length and position check, use it every time:** for two finite sequences the output starts at $n_x + n_h$ and has $L_x + L_h - 1$ samples.
- $\sum_{k=0}^{N-1} a^k = \frac{1-a^N}{1-a}$, $\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}$, $\sum_{k=m}^p a^k = \frac{a^m - a^{p+1}}{1-a}$ ($a \neq 1$).
- $u[n+A] - u[n-B]$ is a **window**: it is 1 for $-A \leq n \leq B-1$ and 0 elsewhere, so it has $A+B$ taps. **Getting these two limits right is most of the marks.**

1 Convolution of two finite sequences

TOP 16/45 70 Ch **6**, 73 Shr **6**, 70 Bh **2+4**, 70 Asa **2+4**, 74 Bh **6**, 74 Ma **5**, 71 Ch **6**, 79 Bh **5**, 81 Bh **5**, 79 Ba **5**, 74 Ash **3+2**, 82 Bh **5**, 72 Ka **5**, 75 Ash **5**, 79 Ch **4+6**, 75 Bh **4+6** · **one method, sixteen papers**

QUESTION AS PRINTED 2070 Chaitra · Q2 · [6]

Find output, $y(n)$ when: $h(n) = \{5, 4, 3, 2\}$ and $x(n) = \{1, 0, 3, 2\}$

Method (the tabular method, used for every problem in this section)

1. Write both sequences with their starting index. Here neither is marked, so both start at $n = 0$.
2. **Predict the answer's shape first:** start = $n_x + n_h = 0 + 0 = 0$, length = $4 + 4 - 1 = 7$, so $y[n]$ runs $n = 0$ to 6.
3. Build the product table: x across the top, h down the side, each cell = $x[i] h[j]$.
4. $y[n]$ is the sum of the cells on the anti-diagonal $i + j = n$ (shaded steps below).
5. **Check:** $\sum y[n]$ must equal $(\sum x[n])(\sum h[n])$.

Step 1: product table

	$x[0] = 1$	$x[1] = 0$	$x[2] = 3$	$x[3] = 2$
$h[0] = 5$	5	0	15	10
$h[1] = 4$	4	0	12	8
$h[2] = 3$	3	0	9	6
$h[3] = 2$	2	0	6	4

Step 2: sum each anti-diagonal

n	cells with $i + j = n$	$y[n]$
0	5	5
1	4 + 0	4
2	3 + 0 + 15	18
3	2 + 0 + 12 + 10	24
4	0 + 9 + 8	17
5	6 + 6	12
6	4	4

Final answer

$$y[n] = \{5, 4, 18, 24, 17, 12, 4\} \quad n = 0 \dots 6$$

The underline marks $n = 0$. Check: $\sum x = 6$, $\sum h = 14$, $6 \times 14 = 84$ and $5 + 4 + 18 + 24 + 17 + 12 + 4 = 84 \checkmark$

Every other paper that runs this same method

A window such as $(1/2)^n \{u[n+2] - u[n-2]\}$ is written out as its samples first: that is the only extra step, and the underlined sample is $n = 0$.

Paper	$h[n]$	$x[n]$	$y[n]$
73 Shr 6	$\{1, 1, 1\}$	$\{1, 1, 1, 1\}$	$\{\underline{1}, 2, 3, 3, 2, 1\}$
70 Bh 2+4	$\{1, 1, 1\}$	$\{1, -2, 2, 3, 4\}$	$\{\underline{1}, -1, 1, 3, 9, 7, 4\}$
70 Asa 2+4	$\{1, 0, 1\}$	$\{1, -2, -2, 3, 4\}$	$\{\underline{1}, -2, -1, 1, 2, 3, 4\}$
74 Bh 6	$\{2, 2, -1, 1\}$	$\{1, 2, 1, 2\}$	$\{2, 6, 5, 5, 5, -1, 2\}$
74 Ma 5	$\{5, 3, 4, 2, 0\}$	$\{0, 2, 4, 6\}$	$\{\underline{0}, 10, 26, 50, 38, 32, 12, 0\}$
71 Ch 6	$\{\underline{1}, 3, 2, -1, 1\}$ at $n = -1$	$2\delta[n] - \delta[n-1] = \{2, -1\}$	$\{2, \underline{5}, 1, -4, 3, -1\}$, from $n = -1$
79 Bh, 81 Bh 5	$2\delta[n+1] + 2\delta[n-1] = \{2, 0, 2\}$ at $n = -1$	$\delta[n] + 2\delta[n-1] - \delta[n-3]$	$\{2, \underline{4}, 2, 2, 0, -2\}$, from $n = -1$
79 Ba 5	$(\frac{1}{2})^n \{u[n+2] - u[n-2]\} = \{4, 2, 1, \frac{1}{2}\}$ at $n = -2$	$\{2, 1, 0, -1, 4\}$	$\{8, 8, \underline{4}, -2, 14.5, 7, 3.5, 2\}$, from $n = -2$
74 Ash 3+2	same h as 79 Ba	$\{2, 1, 0.5, -1\}$	$\{8, 8, \underline{6}, -1, -1, -0.75, -0.5\}$, from $n = -2$
82 Bh 5	$(0.5)^n \{u[n] - u[n-3]\} = \{1, \frac{1}{2}, \frac{1}{4}\}$	$\{2, 1, 0.5, -1\}$	$\{2, 2, 1.5, -0.5, -0.375, -0.25\}$
72 Ka 5	$(\frac{1}{3})^n \{u[n+1] - u[n-2]\} = \{3, 1, \frac{1}{3}\}$ at $n = -1$	$\{2, 1, 0.5, 3\}$	$\{6, \underline{5}, \frac{19}{6}, \frac{59}{6}, \frac{19}{6}, 1\}$, from $n = -1$
75 Ash 5	$2^n \{u[n] - u[n-3]\} = \{1, 2, 4\}$	$\delta[n] + \delta[n-1] + \delta[n-2]$	$\{\underline{1}, 3, 7, 6, 4\}$
79 Ch, 75 Bh 4+6	same h as 75 Ash	$\delta[n] + \delta[n-1] - \delta[n-3]$	$\{\underline{1}, 3, 6, 3, -2, -4\}$

The two traps in this table. (1) A $\delta[n+1]$ or a $u[n+2]$ puts samples at *negative* n , so the answer starts left of the origin: mark $n = 0$ or lose the marks. (2) $u[n+2] - u[n-2]$ has 4 taps ($n = -2, -1, 0, 1$), not 5: the upper limit $u[n-B]$ is exclusive.

2 Convolution where one sequence is infinite

TOP 12/45 78 Bh 5, 73 Ma 2+6, 78 Ch 6, 76 Ch 6, 69 Ch 1+4, 81 Ch 2+2+6, 80 Ch 2+2+6, 72 Ma 8, 76 Bh 2+6, 77 Ch 3+7, 81 Ba 6, 68 Bh 5, 76 Ash 6, 73 Bh 8 · the answer is a formula per range, not a list of samples

QUESTION AS PRINTED 2078 Bhadra · Q2 · [5]

Find the output of LTI system having impulse response $h[n] = u[n] - u[n-4]$ and input signal $x[n] = (1/2)^n u[n]$.

Method (used for every problem in this section)

1. Turn the window into limits. $h[n] = u[n] - u[n-4]$ is 1 for $0 \leq n \leq 3$, so $h[n-k] = 1$ for $0 \leq n-k \leq 3$, i.e. $n-3 \leq k \leq n$.
2. Combine with the other sequence's own limits. $x[k] = (1/2)^k$ needs $k \geq 0$. So k runs from $\max(0, n-3)$ to n .
3. The max is why there are two ranges. Solve $\max(0, n-3)$:
 - $0 \leq n \leq 3$: lower limit is 0 (the window hangs off the left of x),
 - $n \geq 4$: lower limit is $n-3$ (the window is fully inside x),
 - $n < 0$: no overlap at all, $y[n] = 0$.
4. Sum the geometric series on each range separately.
5. Check: the two formulas must agree in value at the last sample of the first range against a direct count.

Range A, $0 \leq n \leq 3$

$$y[n] = \sum_{k=0}^n \left(\frac{1}{2}\right)^k = \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} = 2 \left[1 - \left(\frac{1}{2}\right)^{n+1}\right] = \boxed{2 - \left(\frac{1}{2}\right)^n}$$

Sample values: $y[0] = 1$, $y[1] = 1.5$, $y[2] = 1.75$, $y[3] = 1.875$.

Range B, $n \geq 4$

$$y[n] = \sum_{k=n-3}^n \left(\frac{1}{2}\right)^k = \frac{\left(\frac{1}{2}\right)^{n-3} - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} = 2 \left(\frac{1}{2}\right)^{n-3} \left[1 - \left(\frac{1}{2}\right)^4\right] = \boxed{15 \left(\frac{1}{2}\right)^n}$$

Sample values: $y[4] = 0.9375$, $y[5] = 0.46875$, and it decays by half each step. **Why the peak is at $n = 3$:** that is the last n at which the whole window still sits over the large early samples of x .

Final answer

$$y[n] = \begin{cases} 0, & n < 0 \\ 2 - \left(\frac{1}{2}\right)^n, & 0 \leq n \leq 3 \\ 15 \left(\frac{1}{2}\right)^n, & n \geq 4 \end{cases}$$

Every other paper that runs this same method

Read the table as: window limits $\rightarrow$ two ranges $\rightarrow$ two geometric sums. Only the numbers move.

Paper	Given	Answer
73 Ma 2+6	$h = u[n+1] - u[n-4]$ (5 taps, $-1 \dots 3$), $x = \left(\frac{1}{2}\right)^n u[n]$	$y = 2 - \left(\frac{1}{2}\right)^{n+1}$ for $-1 \leq n \leq 3$; $y = 31\left(\frac{1}{2}\right)^{n+1}$ for $n \geq 4$
78 Ch 6	$h = u[n+1] - u[n-3]$ (4 taps, $-1 \dots 2$), $x = (0.5)^n u[n]$	$y = 2 - \left(\frac{1}{2}\right)^{n+1}$ for $-1 \leq n \leq 2$; $y = 15\left(\frac{1}{2}\right)^{n+1}$ for $n \geq 3$
76 Ch 6	$h = \left(\frac{1}{2}\right)^n u[n-1]$, $x = u[n+1] - u[n-4]$ (5 taps, $-1 \dots 3$)	$y = 1 - \left(\frac{1}{2}\right)^{n+1}$ for $0 \leq n \leq 4$; $y = 31\left(\frac{1}{2}\right)^{n+1}$ for $n \geq 5$
69 Ch 1+4	$h = 2u[n] - 2u[n-4]$, $x = \left(\frac{1}{3}\right)^n u[n]$	$y = 3 \left[1 - \left(\frac{1}{3}\right)^{n+1}\right]$ for $0 \leq n \leq 3$; $y = 80\left(\frac{1}{3}\right)^n$ for $n \geq 4$
81 Ch, 80 Ch, 72 Ma	$h = 2u[n] - 2u[n-5]$, $x = \left(\frac{1}{3}\right)^n u[n]$	$y = 3 \left[1 - \left(\frac{1}{3}\right)^{n+1}\right]$ for $0 \leq n \leq 4$; $y = 242\left(\frac{1}{3}\right)^n$ for $n \geq 5$
76 Bh 2+6	$h = 2u[n+2] - 2u[n-3]$ (5 taps, $-2 \dots 2$), $x = \left(\frac{1}{3}\right)^n u[n-1]$	$y = 1 - \left(\frac{1}{3}\right)^{n+2}$ for $-1 \leq n \leq 3$; $y = 242\left(\frac{1}{3}\right)^{n+2}$ for $n \geq 4$
77 Ch 3+7	$h = 2u[n+3] - 2u[n-4]$ (7 taps, $-3 \dots 3$), $x = \left(\frac{1}{2}\right)^n u[n+1]$	$y = 8 - \left(\frac{1}{2}\right)^{n+2}$ for $-4 \leq n \leq 2$; $y = 127\left(\frac{1}{2}\right)^{n+2}$ for $n \geq 3$

Where the 2 in front goes: an amplitude of 2 on the window (69 Ch, 81 Ch, 76 Bh, 77 Ch) multiplies every term, which is why those constants are 3, 80, 242, 127 rather than 2, 40, 121, 63.5. Multiply at the end, once.

2b The four that do not fit the pattern

Each of these changes the calculation, so each is worked separately rather than tabulated.

81 Ba **6** : two different exponentials

- Given $h(n) = \left(\frac{1}{3}\right)^n u(n-3)$, $x(n) = \left(\frac{1}{6}\right)^{n-6} u(n)$.
- First tidy x : $\left(\frac{1}{6}\right)^{n-6} = 6^6 \left(\frac{1}{6}\right)^n = 46656 \left(\frac{1}{6}\right)^n$.
- Overlap needs $k \geq 0$ and $n-k \geq 3$, so $0 \leq k \leq n-3$: **one range only**, $n \geq 3$.

$$y[n] = 46656 \sum_{k=0}^{n-3} \left(\frac{1}{6}\right)^k \left(\frac{1}{3}\right)^{n-k} = 46656 \left(\frac{1}{3}\right)^n \sum_{k=0}^{n-3} \left(\frac{1}{2}\right)^k$$

- The ratio is $\frac{1/6}{1/3} = \frac{1}{2}$: **two different bases collapse to one geometric sum**, which is the whole trick here.

$$y[n] = 3456 \left(\frac{1}{3}\right)^{n-3} - 1728 \left(\frac{1}{6}\right)^{n-3}, \quad n \geq 3$$

- Check: $y[3] = 3456 - 1728 = 1728$,
 $y[4] = 1152 - 288 = 864$, $y[6] = 128 - 8 = 120$.

68 Bh **5** : both sides infinite, one left-sided

- Given $x[n] = 3^n u[-n-5]$, $h[n] = u[n-5]$.
- $u[-n-5]$ is 1 for $n \leq -5$: **a left-sided signal**, and 3^n decays as $n \rightarrow -\infty$, so the sum converges.
- Overlap needs $k \leq -5$ and $n-k \geq 5$ i.e. $k \leq n-5$. Upper limit is $\min(-5, n-5)$, lower limit $-\infty$.
- $n \leq 0$: upper limit $n-5$, so

$$y[n] = \sum_{k=-\infty}^{n-5} 3^k = \frac{3^{n-5}}{1 - \frac{1}{3}} = \boxed{\frac{1}{2} 3^{n-4}}$$

- $n \geq 0$: upper limit is fixed at -5 , so $y[n]$ stops changing:

$$y[n] = \sum_{k=-\infty}^{-5} 3^k = \frac{3}{2} \cdot 3^{-5} = \boxed{\frac{1}{162}}$$

- Running the series backwards** is the only new idea: for a left-sided sum put $k = -m$ and it is an ordinary $\sum (1/3)^m$.

73 Bh 8 : finite × finite, but symbolic

- Given $x[n] = 1$ for $0 \leq n \leq 4$, $h[n] = a^n$ for $0 \leq n \leq 6$, $a > 1$.
- Both finite, so the answer runs $n = 0$ to 10 : but a is symbolic, so it is written as sums, not samples.
- Overlap needs $0 \leq k \leq 4$ and $0 \leq n - k \leq 6$, i.e. $\max(0, n - 6) \leq k \leq \min(4, n)$.
- **Two min / max switches means three ranges**, not two.

$$\text{With } S(k) = \frac{a^{k+1} - 1}{a - 1}:$$

$$y[n] = \begin{cases} S(n), & 0 \leq n \leq 4 \\ S(n) - S(n - 5), & 5 \leq n \leq 6 \\ S(6) - S(n - 5), & 7 \leq n \leq 10 \end{cases}$$

- Check with $a = 2$: $y[0] = 1$, $y[4] = 31$,
 $y[6] = 127 - 31 = 96$, $y[10] = 127 - 63 = 64$.

76 Ash 6 : the question as printed cannot be solved

- The paper prints $x[n] = 2^n 4[-n]$, $0 < a < 1$ and $h[n] = 4[n]$: the sequence symbols are mangled, and it is meant to read $x[n] = a^n u[-n]$, $h[n] = u[n]$.
- **With $0 < a < 1$ that diverges.** $a^n u[-n]$ grows without bound as $n \rightarrow -\infty$, so $\sum_{k \leq \min(0, n)} a^k$ is infinite for every n .
- **Reading A**, the standard textbook problem and the one that fits the printed $0 < a < 1$: $x[n] = a^n u[n]$, $h[n] = u[n]$.

$$y[n] = \sum_{k=0}^n a^k = \boxed{\frac{1 - a^{n+1}}{1 - a}} \quad n \geq 0$$

As $n \rightarrow \infty$ this tends to $1/(1 - a)$: a running sum that settles.

- **Reading B**, keeping $u[-n]$ literally, which needs $a > 1$ instead:

$$y[n] = \frac{a^{n+1}}{a - 1} \quad (n \leq 0), \quad y[n] = \frac{a}{a - 1} \quad (n \geq 0)$$

- **In the exam:** state the inconsistency in one line, then solve reading A. verified/added

3 Response to a complex exponential

TOP 10/45 70 Ch 4, 72 Ka 4, 80 Ba 5, 82 Ba 5, 71 Shr 2+3+5, 69 Bh 2+4, 73 Shr 3+7 · **never convolve these** · 5 marks in about 90 seconds

QUESTION AS PRINTED 2080 Baishakh · Q2 · [5]

Find the output of LTI system having impulse response $h[n] = (1/2)^n * u[n]$ and input $x[n] = 5e^{j\pi n/3}$ for $-\infty < n < \infty$.

The paper prints a stray “*” between $(1/2)^n$ and $u[n]$; it is a product, not a convolution.

Method (used for every problem in this section)

1. Recognise the input: a **single complex exponential running over all n** is an eigenfunction, so $y[n] = H(e^{j\omega_0}) x[n]$ and no convolution is needed.
2. Find the frequency response once:

$$H(e^{j\omega}) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n e^{-j\omega n} = \sum_{n=0}^{\infty} \left(\frac{1}{2} e^{-j\omega}\right)^n = \frac{1}{1 - \frac{1}{2} e^{-j\omega}}$$

converging because $|\frac{1}{2} e^{-j\omega}| = \frac{1}{2} < 1$.

3. Substitute ω_0 from the input, here $\omega_0 = \pi/3$.
4. Put the denominator into rectangular form, then into polar form.
5. Answer = input × that complex number.

Step-by-step

- $e^{-j\pi/3} = \cos 60^\circ - j \sin 60^\circ = 0.5 - j0.866$.
- Denominator = $1 - \frac{1}{2}(0.5 - j0.866) = 1 - 0.25 + j0.433 = 0.75 + j0.433$.
- $|0.75 + j0.433| = \sqrt{0.5625 + 0.1875} = \sqrt{0.75} = 0.8660$, $\angle = \tan^{-1} \frac{0.433}{0.75} = 30^\circ$.
- So $H(e^{j\pi/3}) = \frac{1}{0.866 \angle 30^\circ} = 1.1547 \angle -30^\circ$.

$$y[n] = 5 \times 1.1547 e^{-j\pi/6} e^{j\pi n/3} = 5.7735 e^{j(\pi n/3 - \pi/6)}$$

Read the answer: the sinusoid keeps its frequency, its amplitude is multiplied by 1.1547 and it is delayed in phase by 30° . That sentence is worth a mark on its own.

Every other paper that runs this same method

Paper	$h[n], x[n]$	$H(e^{j\omega_0})$	$y[n]$
70 Ch 4	$(\frac{1}{3})^n u[n], 5e^{j\pi n/2}$	$\frac{1}{1+j/3} = 0.9 - j0.3 = 0.9487\angle -18.43^\circ$	$4.7434 e^{j(\pi n/2 - 0.3217)}$
72 Ka, 80 Ba, 82 Ba 4 5	$(\frac{1}{2})^n u[n], 5e^{j\pi n/3}$	$1.1547\angle -30^\circ$	$5.7735 e^{j(\pi n/3 - \pi/6)}$
69 Bh 2+4	$(\frac{1}{2})^n u[n], Ae^{j\pi n/2}$	$\frac{1}{1+j0.5} = 0.8944\angle -26.57^\circ$	$0.8944A e^{j(\pi n/2 - 0.4636)}$
71 Shr 2+3+5	$(\frac{1}{2})^n u[n], Ae^{j\pi n}$	$\frac{1}{1+\frac{1}{2}} = \frac{2}{3}$ (real, $e^{-j\pi} = -1$)	$\frac{2}{3}A e^{j\pi n}$

73 Shr 3+7 : a sum of three terms, so use superposition

- Given $h[n] = (\frac{1}{2})^n u[n]$, $x[n] = 10 - 5 \sin(\pi n/2) + 20 \cos(\pi n)$.
- **Do not convolve.** Split into frequencies and scale each by H at that frequency.
- $\omega = 0$ (the constant 10): $H(e^{j0}) = \frac{1}{1-\frac{1}{2}} = 2 \rightarrow$ contributes 20.
- $\omega = \pi$ (the $20 \cos \pi n$): $H(e^{j\pi}) = \frac{1}{1+\frac{1}{2}} = \frac{2}{3} \rightarrow$ contributes $\frac{40}{3} \cos(\pi n) = 13.333 \cos(\pi n)$.
- $\omega = \pi/2$ (the $-5 \sin$): $H(e^{j\pi/2}) = 0.8944\angle -26.57^\circ \rightarrow$ contributes $-4.472 \sin(\pi n/2 - 26.57^\circ)$.

$$y[n] = 20 + 13.333 \cos(\pi n) - 4.472 \sin(\frac{\pi n}{2} - 26.57^\circ)$$

- **Why a real sinusoid keeps its shape:** $\cos \omega_0 n$ is two exponentials at $\pm \omega_0$, and for real $h[n]$ the two responses are conjugates, so they recombine into one real sinusoid with the magnitude and phase of H .

4 Convolution from tabulated or stem-plot data

HOT 4/45 73 Ch 6, 75 Ch 5+2, 72 Ch 3+2, 71 Bh 7 · same arithmetic as §1, **the marks are in the index bookkeeping**

QUESTION AS PRINTED 2073 Chaitra · Q2 · [6]

Calculate $y[n]$, if $x[n]$ is $x[-2] = 0.5, x[0] = 1, x[1] = 0.75, x[3] = 0.5$ and $n[n]$ is $n[0] = 1, n[1] = 0.75$ and $n[2] = 0.5$ and verify your result.

The paper writes the impulse response as " $n[n]$ "; it means $h[n]$.

Method (the only difference from §1)

1. Fill in the gaps with zeros. The question lists only the non-zero samples, and a missing index is a zero, not an absent sample.
 - $x[n] = \{0.5, 0, \underline{1}, 0.75, 0, 0.5\}$ for $n = -2 \dots 3$
 - $h[n] = \{\underline{1}, 0.75, 0.5\}$ for $n = 0 \dots 2$
2. Predict the shape: start $= -2 + 0 = -2$, length $= 6 + 3 - 1 = 8$, so $n = -2 \dots 5$.
3. Product table and anti-diagonals, exactly as in §1.
4. "Verify your result" is worth marks: use the sum check.

Product table

	$x[-2] \ 0.5$	$x[-1] \ 0$	$x[0] \ 1$	$x[1] \ 0.75$	$x[2] \ 0$	$x[3] \ 0.5$
$h[0] = 1$	0.5	0	1	0.75	0	0.5
$h[1] = 0.75$	0.375	0	0.75	0.5625	0	0.375
$h[2] = 0.5$	0.25	0	0.5	0.375	0	0.25

Anti-diagonal sums

n	terms	$y[n]$
-2	0.5	0.5
-1	0.375 + 0	0.375
0	0.25 + 0 + 1	1.25
1	0 + 0.75 + 0.75	1.5
2	0.5 + 0.5625 + 0	1.0625
3	0.375 + 0 + 0.5	0.875
4	0 + 0.375	0.375
5	0.25	0.25

Final answer and the verification the question asks for

$$y[n] = \{0.5, 0.375, \underline{1.25}, 1.5, 1.0625, 0.875, 0.375, 0.25\}$$

$$n = -2 \dots 5$$

$\sum x = 2.75$, $\sum h = 2.25$, product = 6.1875; and $0.5 + 0.375 + 1.25 + 1.5 + 1.0625 + 0.875 + 0.375 + 0.25 = 6.1875$ ✓ **This sum check is the standard way to “verify your result”** and takes one line.

Every other paper that runs this same method

Paper	$h[n]$	$x[n]$	$y[n]$
75 Ch 5+2	$h[-2] = 3, h[0] = 2, h[1] = 1$	$x[n] = 2^n$ for $-1 \leq n \leq 3$	$\{1.5, 3, 7, \underline{14.5}, 29, 10, 20, 8\}$, from $n = -3$
72 Ch 3+2	$h[-2] = 1, h[0] = 2, h[1] = 3$	$x[0] = 0.5, x[2] = 2, x[3] = 3$	$\{0.5, 0, \underline{3}, 4.5, 4, 12, 9\}$, from $n = -2$
71 Bh 7	$h[0] = 0.5, h[1] = 0.75, h[2] = 1$ (given as a stem plot)	$x[-2] = 1, x[0] = 0.5, x[2] = 1,$ $x[3] = 0.75$	$\{0.5, 0.75, \underline{1.25}, 0.375, 1, 1.125, 1.5625, 0.75\}$ from $n = -2$

75 Ch is the one people get wrong: $x[n] = 2^n$ for $-1 \leq n \leq 3$ starts at $n = -1$ with the value 0.5, not at $n = 0$ with the value 1. With h starting at -2 , the output starts at -3 .

5 Energy and power of a given signal

TOP 10/45 79 Bh, 69 Ch, 81 Bh, 74 Ash, 79 Ba, 82 Bh, 72 Ka, 68 Bh, 75 Ch, 74 Ma · **almost always the same question as §6**, asked as one 4-mark part

QUESTION AS PRINTED 2079 Bhadra · Q1 · [2+2]

Define energy and power signal. Determine whether the signal $x[n] = \cos(2\pi n/5) + \sin(\pi n/3)$ is periodic or not. If periodic, find its fundamental period.

Method (used for every problem in this section)

- Look at $|x[n]|$ before computing anything.
 - Bounded and never dying away $\rightarrow$ power signal, $E = \infty$.
 - Dying away or finite length $\rightarrow$ energy signal, $P = 0$.
- Only then evaluate the one quantity that is finite. Computing both wastes time.
- For a **periodic** signal use one period: $P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$.
- For a **decaying** signal use $E = \sum_n |x[n]|^2$ and the geometric sum.
- State the verdict as a sentence: “ $E = \infty$ and P is finite and non-zero, so it is a **power signal**.”

Worked: the 79 Bh signal

- $x[n] = \cos(2\pi n/5) + \sin(\pi n/3)$ is a sum of two sinusoids: bounded, never decays.
- Both parts are periodic (§6 gives $N = 30$), so the sum is periodic $\rightarrow$ **power signal**, $E = \infty$.
- Power without summing: two sinusoids at *different* frequencies are orthogonal over the common period, so their powers add.

$$P = \frac{A_1^2}{2} + \frac{A_2^2}{2} = \frac{1}{2} + \frac{1}{2} = \boxed{1}$$

- Quotable:** a real sinusoid of amplitude A has $P = A^2/2$; a complex exponential of amplitude A has $P = A^2$ (no factor of 2, because $|e^{j\theta}| = 1$ rather than an oscillating cosine).

Every other paper that runs this same method

68 Bh 5: “find the energy and power of $x[n] = u[n]$ ”

- Energy:** $E = \sum_{n=-\infty}^{\infty} |u[n]|^2 = \sum_{n=0}^{\infty} 1 = \boxed{\infty}$
- Power:** the limit runs over $2N + 1$ samples but only $N + 1$ of them are non-zero:

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N + 1} \sum_{n=0}^N 1 = \lim_{N \rightarrow \infty} \frac{N + 1}{2N + 1} = \boxed{\frac{1}{2}}$$

- $P = \frac{1}{2}$, **not 1**. This is the single most common slip in the chapter, and 72 Ka asks the same thing beside $\delta[n]$.
- $x[n] = \delta[n]$ (72 Ka): $E = 1$, and $P = \lim_{N \rightarrow \infty} \frac{1}{2N + 1} = 0 \rightarrow$ **energy signal**.

Paper	$x[n]$	Answer
79 Bh, 69 Ch, 81 Bh 2+2 4	$\cos(2\pi n/5) + \sin(\pi n/3)$	power signal, $P = 1$, $E = \infty$; periodic, $N = 30$
74 Ash 2+2	$e^{j(\pi n/3 + \pi/4)}$	$ x[n] = 1 \rightarrow$ power signal, $P = 1$, $E = \infty$; periodic, $N = 6$
79 Ba, 82 Bh 2+2	$e^{j(\pi n/2 + 4\pi/7)}$	power signal, $P = 1$, $E = \infty$; periodic, $N = 4$
72 Ka 2+3	$u[n]$ and $\delta[n]$	$u[n]$: power, $P = \frac{1}{2}$. $\delta[n]$: energy, $E = 1$
68 Bh 5	$u[n]$	$E = \infty$, $P = \frac{1}{2}$

75 Ch 3+4 and 74 Ma 3+3 ask the *definitions* only (power Vs energy, Fourier series Vs Fourier transform; energy Vs power, periodic Vs aperiodic, symmetric Vs antisymmetric), plus "is $y[n] = x^2[n]$ linear": that last part is §9. **A constant phase offset like $+\pi/4$ or $+4\pi/7$ changes nothing:** it shifts the signal, and neither power nor period depends on a shift.

6 Periodicity and fundamental period

TOP 17/45 80 Bh 5, 80 Ba 2+2, 78 Bh 4, 70 Ch 3, 69 Bh 4, 67 Mng 4, 66 Ma 4, and the 10 papers of §5 · **the most reliable 4 marks in the paper**

QUESTION AS PRINTED 2080 Bhadra · Q1 · [5]

Determine whether the signal $x[n] = e^{j\pi n/16} \cos(n\pi/17)$ is periodic or not. If periodic, find its fundamental period.

Method (used for every problem in this section)

1. Read off ω_0 for each factor or term.
2. Form $\frac{\omega_0}{2\pi}$. **Irrational $\rightarrow$ aperiodic, stop there.**
3. Rational: write it as $\frac{m}{N}$ in **lowest terms**; N is that term's period.
4. Combine: for a sum or a product of periodic terms, the overall period is $\text{LCM}(N_1, N_2)$.
5. **Check by substitution** if unsure: confirm $x[n + N] = x[n]$ and that no smaller divisor of N works.

Worked: the 80 Bh signal

- First factor $e^{j\pi n/16}$: $\omega_1 = \frac{\pi}{16}$, so $\frac{\omega_1}{2\pi} = \frac{1}{32} \rightarrow N_1 = 32$.
- Second factor $\cos(n\pi/17)$: $\omega_2 = \frac{\pi}{17}$, so $\frac{\omega_2}{2\pi} = \frac{1}{34} \rightarrow N_2 = 34$.
- Both rational, so the product is periodic.
- $32 = 2^5$, $34 = 2 \times 17$, so $\text{LCM} = 2^5 \times 17$.

$$N = 544$$

Do not multiply $32 \times 34 = 1088$. The LCM is the answer; multiplying is only right when the two are coprime.

The three traps, each a real paper

- **70 Ch (i)** $\cos(\pi n^2/8)$: n^2 , so it is **not a sinusoid in n** and the $\omega_0/2\pi$ rule does not apply. Test directly:

$$\cos\left(\frac{\pi(n+8)^2}{8}\right) = \cos\left(\frac{\pi n^2}{8} + 2\pi n + 8\pi\right) = \cos\left(\frac{\pi n^2}{8}\right)$$

since $2\pi n$ and 8π are whole multiples of 2π . $N = 4$ fails, so $N = 8$.

- **70 Ch (ii)** $\cos(n/2) \cos(\pi n/4)$: the first factor has $\omega = \frac{1}{2}$ with **no π** , so $\omega/2\pi = 1/4\pi$ is irrational $\rightarrow$ the product is **aperiodic**. One irrational factor is enough.
- **80 Ba (a)** $\sin(n\pi) + \cos(n\pi)$: **$\sin(n\pi) = 0$ for every integer n** , so the signal is just $\cos(n\pi) = (-1)^n$ and $N = 2$. Treating it as $\text{LCM}(2, 2)$ gives the same answer, but say why the sine vanished.

Every other paper that runs this same method

Paper	$x[n]$	$\omega_0/2\pi$ and answer
78 Bh 4	$\cos(\pi n/2) \cos(\pi n/4)$	$\frac{1}{4}$ and $\frac{1}{8} \rightarrow N_1 = 4, N_2 = 8, N = \text{LCM} = 8$
80 Ba (b) 2	$\sin(3\pi n/5) + \cos(4\pi n/7)$	$\frac{3}{10}$ and $\frac{2}{7} \rightarrow N_1 = 10, N_2 = 7, N = 70$
69 Bh (a) 4	$\cos(11\pi n/3)$	$\frac{11}{6} \rightarrow N = 6$
69 Bh (b)	$e^{j(7/5)n}$	$\frac{7}{10\pi}$, irrational $\rightarrow$ aperiodic
67 Mng (a) 4	$\cos(5\pi n/3)$	$\frac{5}{6} \rightarrow N = 6$
67 Mng (b)	$\sin(\pi n/\sqrt{2} + \pi/8)$	$\frac{1}{2\sqrt{2}}$, irrational $\rightarrow$ aperiodic
66 Ma (a) 4	$\cos(3\pi n/8 + \pi/4)$	$\frac{3}{16} \rightarrow N = 16$
66 Ma (b)	$\sin(0.8n)$: printed "sing(0.8n)"	$\frac{0.4}{\pi}$, irrational $\rightarrow$ aperiodic
\$5 papers	$\cos(2\pi n/5) + \sin(\pi n/3)$	$\frac{1}{5}, \frac{1}{6} \rightarrow N = \text{LCM}(5, 6) = 30$
\$5 papers	$e^{j(\pi n/3 + \pi/4)}, e^{j(\pi n/2 + 4\pi/7)}$	$\frac{1}{6} \rightarrow N = 6; \frac{1}{4} \rightarrow N = 4$

The rule in one line: a π in ω_0 that cancels against the 2π leaves a ratio of whole numbers and the signal is periodic. No π , or a $\sqrt{\quad}$ left over, and it is not.

7 Even and odd parts

HOT 4/45 71 Ch 3, 70 Asa 3, 71 Shr 4+5, 67 Mng 2

QUESTION AS PRINTED 2071 Chaitra · Q1 · [3]

Find the even and odd part of the signal $x[n] = 1$ for $-4 \leq n \leq 0$; 2 for $1 \leq n \leq 4$.

Worked, in the tabular form (use this layout every time)

n	-4	-3	-2	-1	0	1	2	3	4
$x[n]$	1	1	1	1	1	2	2	2	2
$x[-n]$	2	2	2	2	1	1	1	1	1
$x_e[n] = \frac{1}{2}(x[n] + x[-n])$	1.5	1.5	1.5	1.5	1	1.5	1.5	1.5	1.5
$x_o[n] = \frac{1}{2}(x[n] - x[-n])$	-0.5	-0.5	-0.5	-0.5	0	0.5	0.5	0.5	0.5

Reading the table

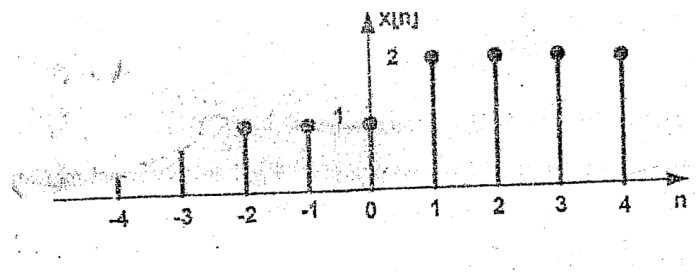
- Row 2 is row 1 **reversed about** $n = 0$: $x[-4]$ takes the old $x[4]$, and so on.
- Checks that cost nothing:** x_e is symmetric about $n = 0$ ✓, and $x_o[0] = 0$ ✓
- Add the last two rows back together and row 1 must reappear ✓

$$x_e[n] = \{1.5, 1.5, 1.5, 1.5, \underline{1}, 1.5, 1.5, 1.5, 1.5\}$$

$$x_o[n] = \{-0.5, -0.5, -0.5, -0.5, \underline{0}, 0.5, 0.5, 0.5, 0.5\}$$

67 Mng 2: $x(n) = 2u[n] - 2u[n-4]$

- That window is $\{2, 2, 2, 2\}$ for $n = 0 \dots 3$, zero elsewhere.
- $x[-n] = \{2, 2, 2, 2\}$ for $n = -3 \dots 0$.
- $x_e = \{1, 1, 1, \underline{2}, 1, 1, 1\}$ and $x_o = \{-1, -1, -1, \underline{0}, 1, 1, 1\}$ for $n = -3 \dots 3$.
- The $n = 0$ sample doubles in x_e** , because it overlaps itself.



71 Shr 4+5 gives the signal as a stem plot instead of a formula. Read the samples off the plot, then run exactly the same table. CT 704 2071 Shrawan Q1

Second half of 71 Shr

- The same paper then gives $x[n] = h[n] = 1$ for $-1 \leq n \leq 1$, zero elsewhere, and asks for the output by the graphical method.
- Both are $\{1, 1, 1\}$ starting at $n = -1$, so the output starts at -2 and has $3 + 3 - 1 = 5$ samples.

$$y[n] = \{1, 2, \underline{3}, 2, 1\} \quad n = -2 \dots 2$$

- The triangle** is what convolving two equal rectangles always gives: worth recognising on sight.

8 Plotting and transformation of the independent variable

HOT 4/45 76 Ch **2+3**, 74 Ch **3+2**, 66 Ma **2**, 69 Bh **3** · **pure marks**: no formula to recall, only care with indices

QUESTION AS PRINTED 2076 Chaitra · Q1 · [2+3]

Define even and odd type discrete time signals with suitable example. Plot the signal $x[-2n + 3]$ where $x[n] = \{1, 2, 0, -1, -3, -4\}$.

Method for $x[an + b]$ (index map, never “shift then scale”)

1. Take $x[n] = \{1, 2, 0, -1, -3, -4\}$ starting at $n = 0$, so $x[0] = 1, \dots, x[5] = -4$.
2. For each output index n , compute the argument $m = -2n + 3$ and copy $x[m]$.
3. Keep only those n for which m lands on an existing sample, $0 \leq m \leq 5$:

$$0 \leq -2n + 3 \leq 5 \Rightarrow -1 \leq n \leq 1.5 \Rightarrow n = -1, 0, 1$$

4. **Only three samples survive**, because scaling by 2 throws away every second one.

n	$m = -2n + 3$	$x[m]$	$y[n]$
-1	5	-4	-4
0	3	-1	-1
1	1	2	2

$$x[-2n + 3] = \{-4, \underline{-1}, 2\} \quad n = -1, 0, 1$$

The even-indexed originals ($x[0], x[2], x[4]$) are lost: that is decimation, and it is not reversible.

The three plotting questions

- **74 Ch 3+2**: $x[n] = u[n] - u[n - 3] + 5\delta[n - 4] + nu[n - 6]$
 - $u[n] - u[n - 3] = 1$ for $n = 0, 1, 2$
 - $5\delta[n - 4] = 5$ at $n = 4$
 - $nu[n - 6] = n$ for $n \geq 6$: 6, 7, 8, ... **a ramp that never stops**
 - $x[n] = \{1, 1, 1, 0, 5, 0, 6, 7, 8, \dots\}$
- **66 Ma 2**:
 - $x(n) = u(n) - u(n - 5) + 5\delta(n - 6) + nu(n - 7) - nu(n - 9)$
 - 1 for $n = 0 \dots 4$; 5 at $n = 6$; the two ramps leave only $n = 7$ and $n = 8 \rightarrow$ values 7 and 8.
 - $x[n] = \{1, 1, 1, 1, 1, 0, 5, 7, 8\}$, zero after.
 - **$nu(n - 7) - nu(n - 9)$ is a windowed ramp**, not an infinite one: the second term cancels everything from $n = 9$ on.
- **69 Bh 3**: $x[n] = u[n + 8] - u[n - 4]$
 - A rectangle of height 1 from $n = -8$ to $n = 3$: **12 samples**.
 - **The upper limit is exclusive**: $u[n - 4]$ switches on at $n = 4$, so $n = 3$ is the last non-zero sample.

9 Testing a system for its properties

TOP 12/45 76 Ch **5**, 74 Ch **3+3**, 82 Ba **2+2**, 78 Ch **3**, 75 Ash **3+3**, 71 Bh **4**, 70 Bh **5**, 68 Bh **5**, 67 Mng, 66 Ma **10**, 69 Bh **5**, 80 Bh **2+2+2**, 81 Ba **5**

QUESTION AS PRINTED 2068 Bhadra · Q1 · [5]

State whether or not the system $y[n] = e^{x[2n]}$ is (a) linear (b) time invariant (c) memoryless (d) causal, where $x[n]$ is input to system and $y[n]$ is output of system.

Worked in full: $y[n] = e^{x[2n]}$

- **(a) Linear?**
 - $x_1 \rightarrow e^{x_1[2n]}, x_2 \rightarrow e^{x_2[2n]}$.
 - Input $ax_1 + bx_2$ gives $e^{ax_1[2n] + bx_2[2n]} = e^{ax_1[2n]} \cdot e^{bx_2[2n]}$.
 - Required: $a e^{x_1[2n]} + b e^{x_2[2n]}$. A product is not a weighted sum.
 - **Not linear** ×
- **(b) Time invariant?**
 - Delayed input: $x_1[n] = x[n - n_0]$ gives $y_1[n] = e^{x[2n - 2n_0]}$.
 - Delayed output: $y[n - n_0] = e^{x[2(n - n_0)]} = e^{x[2n - 2n_0]}$.
 - $2n - n_0 \neq 2n - 2n_0$, so **not time invariant** ×
 - **This is the general rule**: any time-scaled index $x[2n]$, $x[n^2]$, $x[-n]$ makes a system time varying.
- **(c) Memoryless?**
 - At $n = 3$ the output needs $x[6]$, a different instant.
 - **Not memoryless** × (it is dynamic).
- **(d) Causal?**
 - For $n > 0$, $2n > n$: the output at $n = 3$ needs $x[6]$, a future input.
 - **Not causal** ×
- **Stable?** (not asked here, but 66 Ma asks it)
 - $|x[n]| \leq M$ gives $|y[n]| \leq e^M$, finite $\rightarrow$ **BIBO stable** ✓

Answer shape that scores: one line of algebra per property, then the verdict with its reason. “Not linear” alone gets nothing.

Every other system the papers set, with the working reduced to its deciding line

Paper	System	Verdict, and the line that decides it
76 Ch 5	$y[n] = x[-n]$ (TI?)	Delayed input $x[-n-n_0]$; delayed output $x[-(n-n_0)] = x[-n+n_0]$. Different → time varying
76 Ch 5	$y[n] = x[n^2]$ (linear?)	$ax_1[n^2] + bx_2[n^2]$ is exactly what superposition demands → linear ✓ (but time varying)
74 Ch, 78 Ch 3	$y[n] = y[n-4] + x[n-4]$ (TI?)	Every index is a plain delay, no explicit $n \rightarrow$ time invariant ✓
74 Ch, 75 Ash 3	$y[n] = x^2[n]$ (linear?)	$(ax_1 + bx_2)^2$ leaves the cross term $2abx_1x_2 \rightarrow$ not linear
82 Ba 2+2	$y[n] = x[n] + x[-n]$ (causal? TI?)	At $n = -2$ it needs $x[2] \rightarrow$ non-causal ; the fold makes it time varying
75 Ash 3	$y[n] = \cos(\frac{5\pi}{8}n + \frac{\pi}{4})$	No $x[n]$ anywhere : zero input gives non-zero output → not linear
71 Bh 4	$y[n] = x[n] - 3x[n-1]$; $y[n] = x[n+1] + 4x[n]$	Both linear ✓; first causal ✓, second needs $x[n+1] \rightarrow$ non-causal
70 Bh 5	$y(n) = \log_{10}[x(n)]$	$\log(ax_1 + bx_2) \neq a \log x_1 + b \log x_2 \rightarrow$ non-linear ; TI ✓; causal ✓
70 Bh 5	$y(n) = x(-n-2)$	Linear ✓; fold → time varying ; at $n = -3$ needs $x[1] \rightarrow$ non-causal
67 Mng	$y(n) = n x[2(n-2)]$, $n > 0$	Linear ✓; explicit n and scaled index → time varying ; needs $x[2n-4] \rightarrow$ not memoryless
66 Ma 10	$y(n) = 2^{\log_2 x(n)} + 2^{\log_2 x(n)}$	This simplifies to $y(n) = 2x(n)$: linear , TI , memoryless , causal , stable : simplify before testing
66 Ma 10	$y(n) = \sin\{x(n) - x(n-1)\}$	Non-linear (sine of a sum); TI ✓; dynamic; causal ✓; stable ✓ since $ \sin \leq 1$
80 Bh 2+2+2	$y(n) = \sum_{k=0}^n x(k)$ (accumulator)	Needs all past inputs → not memoryless ; TI ✓; $x[n] = u[n]$ gives $y[n] = n+1 \rightarrow \infty \rightarrow$ unstable
81 Ba 5	$y(n) = x(n) + e^a y(n-1)$	$h[n] = (e^a)^n u[n]$, so $\sum h $ converges only for $ e^a < 1 \rightarrow$ BIBO stable iff $a < 0$

10 Sampling an analog signal

HOT 7/45 81 Ch 1+3+3, 74 Ma 7, 78 Ch 2+2+3, 80 Ch 3+4, 73 Bh 3+4, 73 Ma 4, 76 Bh 2+1+1+2+2 · EX 753 sets this nearly every year

QUESTION AS PRINTED 2081 Chaitra · Q1 · [1+3+3]

Consider the analog signal $x_a(t) = 3 \cos 100\pi t$.

- Determine the minimum sampling rate required to avoid aliasing.
- Suppose that signal is sampled at $F_s = 75$ Hz. What is the discrete time signal obtained after sampling?
- Assume now that we sample this signal using a sampling rate $F_s = 5,000$ samples/sec. What is discrete time signal after sampling?

Method (used for every problem in this section)

- Read the analog frequency off the argument: $\cos(2\pi F t)$, so $2\pi F = 100\pi$ gives $F = 50$ Hz. **Divide by 2π , not by π .**
- Nyquist rate $= 2F_{max}$.
- Substitute $t = nT = n/F_s$: the digital frequency is $\omega = \frac{2\pi F}{F_s}$ rad/sample.
- Fold ω into $-\pi \leq \omega < \pi$ by subtracting multiples of 2π . If you had to subtract anything, **that term has aliased**: say so.
- Write the answer as $x[n] = A \cos(\omega n)$ with the folded ω .

(a) Minimum sampling rate

$$- F = \frac{100\pi}{2\pi} = 50 \text{ Hz.}$$

$$F_s \geq 2F = \boxed{100 \text{ Hz}}$$

(b) $F_s = 75 \text{ Hz}$: below Nyquist

$$- \omega = \frac{2\pi \times 50}{75} = \frac{4\pi}{3} \text{ rad/sample.}$$

$$- \frac{4\pi}{3} > \pi, \text{ so fold: } \frac{4\pi}{3} - 2\pi = -\frac{2\pi}{3}.$$

- Cosine is even, so

$$\boxed{x[n] = 3 \cos\left(\frac{2\pi n}{3}\right)}$$

- **Aliased**: this is the same sequence a 25 Hz analog cosine would have given at 75 Hz. 75 Hz is below the 100 Hz Nyquist rate, exactly as part (a) warned.

(c) $F_s = 5000 \text{ Hz}$: well above Nyquist

$$- \omega = \frac{2\pi \times 50}{5000} = \frac{\pi}{50}, \text{ already inside } [-\pi, \pi).$$

$$\boxed{x[n] = 3 \cos\left(\frac{\pi n}{50}\right)}$$

no aliasing.

Every other paper that runs this same method

- 78 Ch 2+2+3 : same signal, $F_s = 200 \text{ Hz}$. $\omega = \frac{2\pi \cdot 50}{200} = \frac{\pi}{2}$
 $\rightarrow x[n] = 3 \cos(\pi n/2)$, no aliasing.

- 80 Ch 3+4 :

$$x(t) = \sin 2000\pi t - 5 \cos 12000\pi t + 10 \sin 6000\pi t,$$

$$F_s = 5000.$$

$$- F = 1000, 6000, 3000 \text{ Hz} \rightarrow \omega = 0.4\pi, 2.4\pi, 1.2\pi.$$

$$- \text{Fold: } 2.4\pi - 2\pi = 0.4\pi \text{ and } 1.2\pi - 2\pi = -0.8\pi.$$

$$x[n] = \sin(0.4\pi n) - 5 \cos(0.4\pi n) - 10 \sin(0.8\pi n)$$

- **Two of the three aliased**, and the 6000 Hz term landed on top of the 1000 Hz term: they are now indistinguishable. The sign of the last term flips because sin is odd.

- For $0 \leq n \leq 3$: $x[n] = \{-5, -6.472, 14.143, -6.053\}$ to 3 dp.

- 73 Bh 3+4 : identical but with $+5 \cos 12000\pi t$, so the middle term keeps its + sign.

- 73 Ma 4 :

$$x(t) = \cos 1000\pi t + 3 \sin 3000\pi t + 5 \cos 5000\pi t,$$

$$F_s = 6000.$$

- $F = 500, 1500, 2500 \rightarrow \omega = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$: **all already inside $[-\pi, \pi)$, no aliasing**, because
 $F_s = 6000 > 2 \times 2500.$

- 76 Bh 2+1+1+2+2 :

$$x(t) = 3 \cos 50\pi t + 10 \sin 300\pi t - 0.7 \cos 100\pi t,$$

$$F_s = 5000.$$

- $F = 25, 150, 50 \rightarrow F_{max} = 150$, **Nyquist = 300 Hz**.

- $\omega = \frac{\pi}{100}, \frac{3\pi}{50}, \frac{\pi}{50}$, no aliasing.

- **Reconstruction**: since F_s is above Nyquist the original $x(t)$ comes back exactly, $x_r(t) = x(t)$. **That is the whole answer to part (c)**: the sampling theorem guarantees it.

11 Difference equation: zero-input and zero-state response

PIN 1/45 78 Bh 4 · the LCCDE theory is in Ch 1 §1.9; the transform route to the same answer is Ch 2

QUESTION AS PRINTED 2078 Bhadra · Q4 · [4]

Determine the zero-input response for a second order system given by: $y[n] - 3y[n-1] - 4y[n-2] = x[n]$

Method

1. **Zero input** means set $x[n] = 0$, leaving the homogeneous equation $y[n] - 3y[n-1] - 4y[n-2] = 0$.
2. Try $y[n] = \lambda^n$ and divide through by λ^{n-2} to get the **characteristic equation**.
3. Solve for the roots.
4. The response is a weighted sum of λ_i^n ; fix the weights from the initial conditions.

Step-by-step

- $\lambda^n - 3\lambda^{n-1} - 4\lambda^{n-2} = 0 \Rightarrow \lambda^2 - 3\lambda - 4 = 0$
- $(\lambda - 4)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 4, \lambda_2 = -1$

$$y_{zi}[n] = C_1(4)^n + C_2(-1)^n$$

Fixing the constants

- **The paper gives no initial conditions**, so the boxed form above is the full answer. If $y[-1]$ and $y[-2]$ are supplied, solve

$$\frac{C_1}{4} - C_2 = y[-1], \quad \frac{C_1}{16} + C_2 = y[-2]$$

$$C_1 = \frac{16(y[-1] + y[-2])}{5}, \quad C_2 = \frac{4y[-2] - y[-1]}{5}$$

verified/added

- **Read the roots:** $|\lambda_1| = 4 > 1$, so the natural response **grows without bound** and the system is unstable. Saying that earns the last mark.

The other half, if asked

- **Zero-state** response: set the initial conditions to zero and convolve, $y_{zs}[n] = x[n] * h[n]$, or use $H(z)$ (Ch 2).
- **Total** = $y_{zi} + y_{zs}$. An LTI system must be relaxed, so a question that gives non-zero initial conditions is not asking about an LTI system.

Ch 2 · Z-Transform

Numerical Problems & Solutions

32 of the 45 papers ask for an inverse Z-transform. Every one of them is below. They split into five methods; the method is worked once in full and the other papers follow with their own data and answer.

The one procedure, used in §1, §2, §3 and §5

- 1. Proper?** If the numerator's degree is $\geq$ the denominator's, divide first. The quotient is a polynomial and becomes impulses; the remainder is proper.
- 2. Divide by z ,** expand $X(z)/z$ in partial fractions, multiply back by z . Every term then reads $\frac{Az}{z-a}$, which is a standard pair.
- 3. Assign each pole a side** from the ROC: pole radius *below* the ROC $\rightarrow$ right-sided; *above* $\rightarrow$ left-sided.
- 4. Write the answer term by term.**

The pairs (the whole table)

Term	$ z > a $	$ z < a $
$\frac{z}{z-a}$	$a^n u[n]$	$-a^n u[-n-1]$
$\frac{z}{(z-a)^2}$	$n a^{n-1} u[n]$	$-n a^{n-1} u[-n-1]$
C	$C \delta[n]$	
z^k	$\delta[n+k]$	
z^{-k}	$\delta[n-k]$	

The three checks that cost nothing

- 1. Initial value:** for a causal $x[n]$, $x[0] = \lim_{z \rightarrow \infty} X(z)$.
- 2. Sum of residues:** the partial-fraction constants must add back to the original numerator's leading coefficient.
- 3. Substitute one number:** compute $x[0]$ or $x[1]$ from your answer and from a direct long division of $X(z)$. If they disagree, the ROC assignment is the usual culprit.

A right-sided answer that contains $u[-n-1]$, or a causal answer with samples at negative n , is wrong on sight.

1 Partial fractions, and how the ROC picks the answer

TOP 10/45 82 Ba **1+5**, 81 Ba **2+2+3**, 73 Bh **6**, 72 Ma **6**, **71 Ch 1+5**, 76 Bh **2+6**, 77 Ch **6**, 70 Ma **6**, 80 Ch **6**, 70 Bh **6** · **one $X(z)$, three answers:** the point of the whole chapter

QUESTION AS PRINTED 2082 Baishakh · Q3 · [1+5]

Define the ROC. Find the inverse z-transform of $X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$ using partial fraction method for (a) $|z| > 1$ (b) $|z| < 0.5$ (c) $0.5 < |z| < 1$.

Step 1: factor the denominator and mark the poles

- $1 - 1.5z^{-1} + 0.5z^{-2} = (1 - z^{-1})(1 - 0.5z^{-1})$, so the poles are at $z = 1$ and $z = 0.5$.
- Numerator degree 0 < denominator degree 2, so it is already proper: no division.
- Three ROCs are possible, and the question asks for all three: $|z| < 0.5$, $0.5 < |z| < 1$, $|z| > 1$.

Step 2: partial fractions, once, for all three parts

$$\frac{1}{(1 - z^{-1})(1 - 0.5z^{-1})} = \frac{A}{1 - z^{-1}} + \frac{B}{1 - 0.5z^{-1}}$$

- Put $w = z^{-1}$: $1 = A(1 - 0.5w) + B(1 - w)$.
- $w = 1$ kills B : $1 = A(1 - 0.5) \Rightarrow \boxed{A = 2}$.
- $w = 2$ kills A : $1 = B(1 - 2) \Rightarrow \boxed{B = -1}$.
- Check: $A + B = 1$, the numerator's constant term ✓

$$X(z) = \frac{2}{1 - z^{-1}} - \frac{1}{1 - 0.5z^{-1}}$$

Step 3: the ROC decides each term's side, and nothing else changes

Part	pole $z = 1$	pole $z = 0.5$	$x[n]$
(a) $ z > 1$	outside both $\rightarrow$ right	right	$[2 - (0.5)^n]u[n]$
(b) $ z < 0.5$	inside both $\rightarrow$ left	left	$[-2 + (0.5)^n]u[-n - 1]$
(c) $0.5 < z < 1$	ROC is inside it $\rightarrow$ left	ROC is outside it $\rightarrow$ right	$-2u[-n - 1] - (0.5)^n u[n]$

Reading the three answers

- (a) is **causal**: $x[0] = 1, x[1] = 1.5, x[2] = 1.75 \dots$ climbing to 2. Contains the unit circle? **No**: so this one is not stable.
- (b) is **anticausal**: $x[-1] = 0, x[-2] = 2, x[-3] = 6,$ growing as $n \rightarrow -\infty$.
- (c) is **two-sided** and is the only one whose ROC contains $|z| = 1$, so **(c) is the stable one**: $x[0] = -1, x[1] = -0.5,$ and $x[n] = -2$ for every $n \leq -1$.
- **All three come from the same two fractions.** Only the choice of pair changes. That is the whole examinable idea.
- If the question gives *one* ROC, do steps 1 and 2 identically and read one row of the table.
- If it says “causal” instead of giving an ROC, that means the outermost row: every pole right-sided.

Every other paper that runs this same method

Paper	$X(z)$	ROC	$x[n]$
81 Ba 2+2+3	$\frac{1}{1 - 0.8z^{-1} + 0.12z^{-2}},$ poles 0.6, 0.2; $A = 1.5, B = -0.5$	(i) $ z > 0.6$ (ii) $ z < 0.2$ (iii) $0.2 < z < 0.6$	(i) $[1.5(0.6)^n - 0.5(0.2)^n]u[n]$ (ii) $[-1.5(0.6)^n + 0.5(0.2)^n]u[-n - 1]$ (iii) $-1.5(0.6)^n u[-n - 1] - 0.5(0.2)^n u[n]$
73 Bh 6 , 72 Ma 6	same as 82 Ba	$ z > 1$ $ z < 0.5$	$[2 - (0.5)^n]u[n]$ $[-2 + (0.5)^n]u[-n - 1]$
71 Ch 1+5	$\frac{1}{(z - 0.5)(z + 2)};$ divide by z first, giving $-1 + \frac{0.8z}{z - 0.5} + \frac{0.2z}{z + 2}$	(i) $0.5 < z < 2$ (ii) $ z < 0.5$ (iii) $ z > 2$	(i) $-\delta[n] + 0.8(0.5)^n u[n] - 0.2(-2)^n u[-n - 1]$ (ii) $-\delta[n] - [0.8(0.5)^n + 0.2(-2)^n]u[-n - 1]$ (iii) $-\delta[n] + [0.8(0.5)^n + 0.2(-2)^n]u[n]$

71 Ch is the one that needs the $X(z)/z$ step: its numerator has no z , so expanding directly would leave $\frac{A}{z-a}$ terms. Dividing by z first produces the constant -1 (an impulse) plus two proper $\frac{Az}{z-a}$ terms. 72 Ma prints its first ROC as $|z| > 1.5$; for this $X(z)$ that is the same region as $|z| > 1$, since both lie outside every pole, so the answer is unchanged.

2 Improper in z^{-1} : divide first, the quotient becomes impulses

TOP 7/45 80 Bh **1+6**, 80 Ba **1+5**, 71 Shr **6**, 69 Bh **6**, 80 Ch **6**, 70 Bh **6**, 70 Ma **6** · **numerator and denominator both degree 2 in z^{-1}**

QUESTION AS PRINTED 2080 Bhadra · Q3 · [1+6]

Define ROC. Find the inverse z -transform of $H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}},$ ROC: $0.25 < |z| < 0.5$.

Step 1: spot that it is improper, and divide

- Both numerator and denominator have degree 2 in z^{-1} , so a constant must come out first. Divide the *highest* powers: $z^{-2} \div 0.125z^{-2} = 8$.

$$N - 8D = (1 - 8) + (2 + 6)z^{-1} + (1 - 1)z^{-2} = -7 + 8z^{-1}$$

$$H(z) = 8 + \frac{-7 + 8z^{-1}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

- **The 8 is already an answer:** a constant is $8\delta[n]$.

Step 2: factor, then partial fractions on the remainder

- $1 - 0.75z^{-1} + 0.125z^{-2} = (1 - 0.5z^{-1})(1 - 0.25z^{-1})$: poles 0.5 and 0.25.

- With $w = z^{-1}$: $-7 + 8w = A(1 - 0.25w) + B(1 - 0.5w)$.
- $w = 2$: $-7 + 16 = 9 = A(0.5) \Rightarrow A = 18$. $w = 4$: $-7 + 32 = 25 = B(-1) \Rightarrow B = -25$.
- Check $A + B = -7 \checkmark$

$$H(z) = 8 + \frac{18}{1 - 0.5z^{-1}} - \frac{25}{1 - 0.25z^{-1}}$$

Step 3: the ROC $0.25 < |z| < 0.5$ splits the two poles

- Pole 0.5: the ROC lies *inside* it $\rightarrow$ **left-sided** $\rightarrow -18(0.5)^n u[-n-1]$.
- Pole 0.25: the ROC lies *outside* it $\rightarrow$ **right-sided** $\rightarrow -25(0.25)^n u[n]$.
- **Check:** $h[0] = 8 + 0 - 25 = -17$;
 $h[1] = -25(0.25) = -6.25$; $h[-1] = -18(2) = -36$;
 $h[-2] = -18(4) = -72$.
- The ROC is a ring not containing $|z| = 1$, so this system is **neither causal nor stable**: worth one line if the question asks.

$$h[n] = 8\delta[n] - 18(0.5)^n u[-n-1] - 25(0.25)^n u[n]$$

Every other paper that runs this same method

Paper	$X(z)$ and ROC	After dividing	$x[n]$
80 Ba 1+5	$\frac{1 + 2z^{-1} + z^{-2}}{1 - 1.5z^{-1} + 0.5z^{-2}}, z > 1$	$2 + \frac{8}{1 - z^{-1}} - \frac{9}{1 - 0.5z^{-1}}$	$2\delta[n] + [8 - 9(0.5)^n]u[n]$
71 Shr 6	$\frac{1 + 2z^{-1} + z^{-2}}{1 + 1.5z^{-1} + 0.5z^{-2}}, z > 1$	$2 - \frac{1}{1 + 0.5z^{-1}}$ one factor cancels	$2\delta[n] - (-0.5)^n u[n]$
69 Bh 6	$\frac{1 - \frac{1}{2}z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})}, z > \frac{1}{2}$	$-4 + \frac{-2}{1 - 0.5z^{-1}} + \frac{7}{1 - 0.25z^{-1}}$	$-4\delta[n] + [-2(0.5)^n + 7(0.25)^n]u[n]$
80 Ch, 70 Bh 6	$\frac{1 + 2z^{-1} + z^{-2}}{1 + 4z^{-1} + 4z^{-2}},$ causal	$0.25 + \frac{0.5}{1 + 2z^{-1}} + \frac{0.25}{(1 + 2z^{-1})^2}$	$0.25\delta[n] + (0.25n + 0.75)(-2)^n u[n]$

71 Shr is the free one. Its numerator is $(1 + z^{-1})^2$ and its denominator $(1 + z^{-1})(1 + 0.5z^{-1})$, so after the division the remainder $-1 - z^{-1}$ is exactly $-(1 + z^{-1})$ and **one factor cancels**. Look for a common factor before grinding through partial fractions.

80 Ch / 70 Bh has a repeated pole. Its denominator is $(1 + 2z^{-1})^2$, a double pole at $z = -2$. Match $0.75 + w = A(1 + 2w) + B$ to get $A = 0.5$, $B = 0.25$, then use $\frac{1}{(1 - az^{-1})^2} \leftrightarrow (n+1)a^n u[n]$. "Causal" in the question replaces a stated ROC and means $|z| > 2$.

3 Improper in z : the z^4/z^2 family

TOP 11/45 79 Bh **6**, 76 Ch **2+6**, 81 Ch **6**, 71 Bh **6**, 79 Ch **6**, 72 Ash **5**, 81 Bh **1+5**, 74 Ch **1+3+5**, 73 Ch **1+3+5**, 74 Bh **1+3+5** · the same question with the numerator changed, eleven times

QUESTION AS PRINTED 2079 Bhadra · Q3 · [6]

Find the inverse z -transform of $X(z) = \frac{2z^4 + 2z^3 - 3z + 2}{z^2 - 1.5z - 1}$, ROC: $|z| < 0.5$, using partial fraction expansion.

Step 1: poles, and what the ROC means

- $z^2 - 1.5z - 1 = (z - 2)(z + 0.5)$: poles at $z = 2$ and $z = -0.5$, radii 2 and 0.5.
- ROC $|z| < 0.5$ is inside *both*, so **both poles are left-sided** and the whole answer carries $u[-n-1]$.

Step 2: long division in z , because the numerator is degree 4

Written as ordinary polynomial division, highest power first. The missing z^2 term is carried as $0z^2$.

Quotient term	subtract	new remainder
$2z^2$	$2z^2(z^2 - 1.5z - 1) = 2z^4 - 3z^3 - 2z^2$	$5z^3 + 2z^2 - 3z + 2$
$5z$	$5z(z^2 - 1.5z - 1) = 5z^3 - 7.5z^2 - 5z$	$9.5z^2 + 2z + 2$
9.5	$9.5(z^2 - 1.5z - 1) = 9.5z^2 - 14.25z - 9.5$	$16.25z + 11.5$

$$X(z) = \underbrace{2z^2 + 5z + 9.5}_{\text{quotient}} + \underbrace{\frac{16.25z + 11.5}{(z - 2)(z + 0.5)}}_{G(z), \text{ proper}}$$

Step 3: divide $G(z)$ by z and expand

$$\frac{G(z)}{z} = \frac{16.25z + 11.5}{z(z-2)(z+0.5)} = \frac{C}{z} + \frac{D}{z-2} + \frac{E}{z+0.5}$$

$$\begin{aligned} - C &= \frac{11.5}{(-2)(0.5)} = -11.5 \\ - D &= \frac{16.25(2) + 11.5}{2(2.5)} = \frac{44}{5} = 8.8 \\ - E &= \frac{16.25(-0.5) + 11.5}{(-0.5)(-2.5)} = \frac{3.375}{1.25} = 2.7 \end{aligned}$$

$$G(z) = -11.5 + \frac{8.8z}{z-2} + \frac{2.7z}{z+0.5}$$

Step 4: apply the pairs

- $2z^2 \rightarrow 2\delta[n+2]$, $5z \rightarrow 5\delta[n+1]$, $9.5 \rightarrow 9.5\delta[n]$ (**powers of z are advances**).
- $-11.5 \rightarrow -11.5\delta[n]$, which merges with the 9.5 to give $-2\delta[n]$.
- Both poles left-sided, so $\frac{Az}{z-a} \rightarrow -A a^n u[-n-1]$.

Check, sample by sample

n	from the answer	$x[n]$
1	nothing is non-zero	0
0	-2	-2
-1	$5 - [8.8(0.5) + 2.7(-2)]$	6
-2	$2 - [8.8(0.25) + 2.7(4)]$	-11
-3	$0 - [8.8(0.125) + 2.7(-8)]$	20.5

$x[n] = 0$ for all $n \geq 1$, as a purely left-sided answer must be. If your answer has a tail to the right, the ROC was read backwards.

Every other paper in this family

Identical denominator $(z-2)(z+0.5)$ and identical ROC $|z| < 0.5$; only the numerator changes. Q is the quotient, and the impulse at $n=0$ is $Q_0 + C$ after C is folded in.

Paper	Numerator	Quotient $Q(z)$, and C	$x[n]$
76 Ch 2+6	$z^4 + 5z^3 - 3z + 4$	$z^2 + 6.5z + 10.75$; $C = -14.75$	$\delta[n+2] + 6.5\delta[n+1] - 4\delta[n] - [10.8(2)^n + 3.95(-0.5)^n]u[-n-1]$
81 Ch, 71 Bh 6	$z^4 + z^3 - 3z + 5$	$z^2 + 2.5z + 4.75$; $C = -9.75$	$\delta[n+2] + 2.5\delta[n+1] - 5\delta[n] - [4.6(2)^n + 5.15(-0.5)^n]u[-n-1]$
79 Ch 6	$z^4 - 2z^3 - z + 4$	$z^2 - 0.5z + 0.25$; $C = -4.25$	$\delta[n+2] - 0.5\delta[n+1] - 4\delta[n] - [0.4(2)^n + 3.85(-0.5)^n]u[-n-1]$
72 Ash 5	$z^4 + 2z^3 - z + 4$	$z^2 + 3.5z + 6.25$; $C = -10.25$	$\delta[n+2] + 3.5\delta[n+1] - 4\delta[n] - [6.8(2)^n + 3.45(-0.5)^n]u[-n-1]$

74 Ch, 73 Ch, 74 Bh 1+3+5 : different denominator

- $X(z) = \frac{2z^3 + 2z^2 + 3z + 5}{z^2 - 0.1z - 0.2}$, ROC $|z| < 0.4$.
- **All three papers misprint the leading term as $2z^2$** , which would make the fraction proper and the question trivial. Solved here as $2z^3$. **verified/added**
- $z^2 - 0.1z - 0.2 = (z-0.5)(z+0.4)$, both left-sided under $|z| < 0.4$.
- Division gives $Q = 2z + 2.2$, remainder $\frac{3.62z + 5.44}{(z-0.5)(z+0.4)}$.
- $C = -27.2$, $D = 16.111$, $E = 11.089$.

$$x[n] = 2\delta[n+1] - 25\delta[n] - [16.111(0.5)^n + 11.089(-0.4)^n]u[-n-1]$$

- Check: $x[0] = -25$, $x[-1] = -2.5$, $x[-2] = -133.75$.

81 Bh 1+5 : the clean one

- $X(z) = \frac{2z^3 - 5z^2 + z + 3}{(z-1)(z-2)}$, ROC $|z| < 1$: both poles left-sided.
- Denominator = $z^2 - 3z + 2$. Division gives quotient $2z + 1$ and **remainder exactly 1**:

$$X(z) = 2z + 1 + \frac{1}{(z-1)(z-2)}$$

$$- \frac{1}{z(z-1)(z-2)} = \frac{0.5}{z} - \frac{1}{z-1} + \frac{0.5}{z-2}, \text{ so}$$

$$\frac{1}{(z-1)(z-2)} = 0.5 - \frac{z}{z-1} + \frac{0.5z}{z-2}$$

$$x[n] = 2\delta[n+1] + 1.5\delta[n] + [1 - 0.5(2)^n]u[-n-1]$$

- Check: $x[0] = 1.5$, $x[-1] = 2.75$, $x[-2] = 0.875$, $x[-3] = 0.9375$.

4 Inverse by long division (power series)

HOT 5/45 73 Bh 6, 72 Ma 6, 76 Bh 2+6, 77 Ch 6, 70 Ma 6 · only when the question names the method

QUESTION AS PRINTED 2073 Bhadra · Q3 · [6]

Determine the inverse z-transform of $X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$ using long division method, when ROC: $|z| > 1$ and when ROC: $|z| < \frac{1}{2}$.

Method (the direction of the division is the whole question)

1. **ROC outside** → right-sided → arrange both polynomials in **ascending powers of z^{-1}** and divide.
2. **ROC inside** → left-sided → rewrite in **ascending powers of z** first, then divide.
3. Read the quotient's coefficients off as $x[0], x[1], x[2] \dots$ (or $x[-1], x[-2] \dots$).
4. **Give the recurrence too:** a division gives numbers, and the recurrence is what shows the pattern continues.

(a) ROC $|z| > 1$, right-sided

- Divide 1 by $1 - 1.5z^{-1} + 0.5z^{-2}$ in ascending z^{-1} :

step	quotient term	remainder
1	1	$1.5z^{-1} - 0.5z^{-2}$
2	$1.5z^{-1}$	$1.75z^{-2} - 0.75z^{-3}$
3	$1.75z^{-2}$	$1.875z^{-3} - 0.875z^{-4}$
4	$1.875z^{-3}$	...

$$x[n] = \{1, 1.5, 1.75, 1.875, 1.9375, \dots\}$$

- Recurrence from the denominator:
 $x[n] = 1.5x[n-1] - 0.5x[n-2], x[0] = 1.$
- **Closed form, if you want it:** partial fractions give
 $x[n] = [2 - (0.5)^n]u[n]$, which matches every term. This is 82 Ba part (a).

(b) ROC $|z| < \frac{1}{2}$, left-sided

- Rewrite with z ascending: multiply top and bottom by z^2 ,

$$X(z) = \frac{z^2}{z^2 - 1.5z + 0.5} = \frac{z^2}{0.5 - 1.5z + z^2}$$

and divide z^2 by $0.5 - 1.5z + z^2$.

- First term $\frac{z^2}{0.5} = 2z^2$, and the division continues in rising powers of z .

$$x[n] = \{\dots, 6, 2, 0, 0\} \text{ for } n = -3, -2, -1, 0$$

- **Closed form:** $x[n] = [-2 + (0.5)^n]u[-n-1]$, so
 $x[-1] = 0, x[-2] = 2, x[-3] = 6, x[-4] = 14.$
- **$x[-1] = 0$ is not a mistake:** the two terms cancel exactly at $n = -1$.

The other three, and the two ROCs that cannot be right

Paper	$X(z)$	Printed ROC	Answer
72 Ma 6	$\frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$	(a) $ z > 1.5$ (b) $ z < 0.5$	Same as 73 Bh: $ z > 1.5$ is outside every pole, so it is the same region as $ z > 1$
76 Bh 2+6	$\frac{1}{1 - 1.58z^{-1} + 0.5z^{-2}}$	$ z > 1$ and $ z < \frac{1}{2}$	Right-sided division gives $\{1, 1.58, 1.9964, 2.3643, 2.7374, \dots\}$ with $x[n] = 1.58x[n-1] - 0.5x[n-2]$
77 Ch 6	$\frac{1}{1 - 0.5z^{-1} + 1.5z^{-2}}$	$ z > 1$	$\{1, 0.5, -1.25, -1.375, 1.1875, 2.65625\}$, $x[n] = 0.5x[n-1] - 1.5x[n-2]$
70 Ma 6	$\frac{1 + 2z^{-1} + z^{-2}}{1 + 4z^{-1} + 4z^{-2}}$	all possible ROCs	$ z > 2$: the causal answer of §2. $ z < 2$: the same partial fractions with $u[-n-1]$ and a sign change

76 Bh: the printed ROCs are impossible

- 1.58 puts the poles at $z = 1.1423$ and $z = 0.4377$, not at 1 and 0.5.
- So " $|z| > 1$ " **contains** the pole at 1.1423, and " $|z| < \frac{1}{2}$ " contains the pole at 0.4377. **An ROC can never contain a pole** (§2.2, property 3), so neither region is a legal ROC.
- With 1.5 instead of 1.58 the poles are exactly 1 and 0.5 and both printed ROCs are correct: and 73 Bh asks the identical question with $\frac{3}{2}$. **1.58 is a misprint for 1.5.** verified/added
- In the exam: divide as printed (the division does not care), and add one line saying the ROC must be $|z| > 1.1423$ or $|z| < 0.4377$.

77 Ch: same defect, different cause

- $z^2 - 0.5z + 1.5$ has complex roots, $z = 0.25 \pm j1.198$, both of modulus $\sqrt{1.5} = 1.2247$.
- "ROC $|z| > 1$ " again contains them, so it is not a legal ROC; the causal ROC is $|z| > 1.2247$. verified/added
- The poles are outside the unit circle, so this sequence **grows**: the printed terms alternate and increase in size, which is correct, not an arithmetic slip.
- **Long division is unaffected** by all of this: it only needs to know the sequence is right-sided.

5 Repeated poles: the $X(z)/z$ method earning its keepHOT 3/45 76 Ash 3+6, 73 Shr 1+5, 70 Asa 1+5

QUESTION AS PRINTED 2076 Ashwin · Q3 · [3+6]

State Convolution property of Z-transform. Find inverse Z-transform of $X(z) = \frac{z}{(z - 0.6)(z + 0.5)^2}$, ROC: $|z| > 0.6$.

Step 1: divide by z , which removes the numerator entirely

$$\frac{X(z)}{z} = \frac{1}{(z - 0.6)(z + 0.5)^2} = \frac{A}{z - 0.6} + \frac{B}{z + 0.5} + \frac{C}{(z + 0.5)^2}$$

- A : cover $(z - 0.6)$ and put $z = 0.6$: $A = \frac{1}{(0.6 + 0.5)^2} = \frac{1}{1.21} = 0.8264$.
- C : cover $(z + 0.5)^2$ and put $z = -0.5$: $C = \frac{1}{-0.5 - 0.6} = \frac{1}{-1.1} = -0.9091$.
- B : clear denominators, $1 = A(z + 0.5)^2 + B(z - 0.6)(z + 0.5) + C(z - 0.6)$, and match the z^2 coefficient: $0 = A + B$, so $B = -A = -0.8264$.
- Matching one coefficient is faster than differentiating, and it is the step to use for every repeated pole.

Step 2: multiply back by z and apply the pairs

$$X(z) = \frac{0.8264z}{z - 0.6} - \frac{0.8264z}{z + 0.5} - \frac{0.9091z}{(z + 0.5)^2}$$

- ROC $|z| > 0.6$ is outside both poles, so **every term is right-sided**.
- $\frac{z}{z - a} \rightarrow a^n u[n]$ and $\frac{z}{(z - a)^2} \rightarrow n a^{n-1} u[n]$, here with $a = -0.5$.

$$x[n] = \left[0.8264(0.6)^n - 0.8264(-0.5)^n - 0.9091 n(-0.5)^{n-1} \right] u[n]$$

- Tidier, using $(-0.5)^{n-1} = -2(-0.5)^n$:

$$x[n] = \left[0.8264(0.6)^n + (1.8182n - 0.8264)(-0.5)^n \right] u[n]$$

Check

n	substitution	$x[n]$
0	$0.8264 - 0.8264$	0
1	$0.4959 + (1.8182 - 0.8264)(-0.5)$	0
2	$0.2975 + (3.6364 - 0.8264)(0.25)$	1
3	$0.1785 + (5.4545 - 0.8264)(-0.125)$	-0.4

$x[0] = x[1] = 0$ is correct, not a slip. $X(z)$ has one z on top and a cubic underneath, so the first non-zero sample cannot appear before $n = 2$: a quick sanity check before any arithmetic.

The two left-sided siblings

Paper	$X(z)$, ROC	Constants	$x[n]$
73 Shr 1+5	$\frac{z}{(z - 0.4)(z + 1.5)^2}$, $ z < 0.4$	$A = \frac{1}{3.61} = 0.2770$, $B = -0.2770$, $C = \frac{1}{-1.9} = -0.5263$	$\left[-0.2770(0.4)^n + 0.2770(-1.5)^n + 0.5263 n(-1.5)^{n-1} \right] u[-n - 1]$
70 Asa 1+5	$\frac{z}{(z - 1)(z - 2)^2}$, $ z < 1$	$A = 1$, $B = -1$, $C = 1$	$\left[-1 + (2)^n - n(2)^{n-1} \right] u[-n - 1]$

Both ROCs lie inside every pole, so every term flips to $u[-n - 1]$ with a sign change: including the repeated one. Check 70 Asa: $x[-1] = -1 + 0.5 + 0.25 = -0.25$, $x[-2] = -1 + 0.25 + 0.25 = -0.5$, $x[-3] = -0.6875$.

6 Finding $X(z)$ and locating the ROC

HOT 6/45 70 Ch **2+4**, 70 Asa **2+4**, 75 Ash **3+6**, 75 Ch **4+6**, 75 Bh **4+6**, 73 Ma **1+4**

QUESTION AS PRINTED 2075 Chaitra · Q3 · [4+6]

State the properties of ROC. Find the Z-transform and locate the ROC of the signal $x[n] = (0.1)^n u[n] + (0.3)^n u[-n - 1]$.

Method

1. Transform each term with its own ROC.
2. **Mind the sign on the left-sided term:** the standard pair is $-b^n u[-n - 1] \leftrightarrow \frac{1}{1 - bz^{-1}}$, so a $+b^n u[-n - 1]$ in the question picks up a minus.
3. Intersect the two ROCs.

Worked

- $(0.1)^n u[n] \rightarrow \frac{1}{1 - 0.1z^{-1}}$, ROC $|z| > 0.1$.
- $(0.3)^n u[-n - 1] \rightarrow -\frac{1}{1 - 0.3z^{-1}}$, ROC $|z| < 0.3$.

$$X(z) = \frac{1}{1 - 0.1z^{-1}} - \frac{1}{1 - 0.3z^{-1}}, \quad 0.1 < |z| < 0.3$$

- The ring is non-empty because the right-sided pole (0.1) is the *inner* one. It does not contain $|z| = 1$, so this signal has **no DTFT**.

70 Ch and 70 Asa **2+4** : the ROC comes out empty

QUESTION AS PRINTED 2070 Chaitra · Q3 · [2+4]

List out the properties of Region of Convergence. Find the Z-transform and locate the ROC of the signal. $x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{3}\right)^n u[-n - 1]$

- First term: $\frac{1}{1 + \frac{1}{3}z^{-1}}$, ROC $|z| > \frac{1}{3}$.
- Second term is already in the $-b^n u[-n - 1]$ form, so $\frac{1}{1 - \frac{1}{3}z^{-1}}$, ROC $|z| < \frac{1}{3}$.
- **Both radii are $\frac{1}{3}$.** The intersection of $|z| > \frac{1}{3}$ and $|z| < \frac{1}{3}$ is **empty**.

The Z-transform of this $x[n]$ does not exist.

- That is a complete, correct answer, and it is the point property 8 of §2.2 is making: an ROC must be a connected, non-empty region.

The siblings

- **75 Ash **3+6**** : $x[n] = (0.6)^n u[n] + (0.25)^n u[n]$: **both right-sided**, so both ROCs are “outside” and the intersection is the larger one:

$$X(z) = \frac{1}{1 - 0.6z^{-1}} + \frac{1}{1 - 0.25z^{-1}}, \quad |z| > 0.6$$

- **75 Bh **4+6**** : $x[n] = (0.25)^n u[n] + (0.6)^n u[-n - 1]$: the mirror of 75 Ch:

$$X(z) = \frac{1}{1 - 0.25z^{-1}} - \frac{1}{1 - 0.6z^{-1}}, \quad 0.25 < |z| < 0.6$$

- **73 Ma **1+4**** : $x[n] = (0.5 + j0.2)^n u[n] + (-j)^n u[-n - 1]$.
 - Radii are $|0.5 + j0.2| = \sqrt{0.29} = 0.5385$ and $|-j| = 1$, so **ROC $0.5385 < |z| < 1$** .
 - $z = 0.4$ is **outside the ROC, so $X(0.4)$ does not converge**: that is the answer, not a number.
 - $z = 0.7$ is inside:

$$X(0.7) = \frac{1}{1 - \frac{0.5 + j0.2}{0.7}} - \frac{1}{1 + \frac{j}{0.7}} = 1.4211 + j2.2198.$$

The intended question verified/added

- This is BB Sir's worked example with one digit changed: his version is $x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n - 1]$.
- Then the radii are $\frac{1}{3}$ and $\frac{1}{2}$, the right-sided pole is the inner one, and

$$X(z) = \frac{1}{1 + \frac{1}{3}z^{-1}} + \frac{1}{1 - \frac{1}{2}z^{-1}} = \frac{2z(z - \frac{1}{12})}{(z + \frac{1}{3})(z - \frac{1}{2})}$$

with **ROC $\frac{1}{3} < |z| < \frac{1}{2}$** .

- **In the exam:** give the empty-ROC answer for the signal as printed, in two lines, then solve the $\frac{1}{2}$ version. Both halves earn marks; neither alone is the full answer.

7 Inverse by inspection: $X(z)$ is already a polynomial

PIN 2/45 74 Ash 3+3, 78 Ch 4 · no partial fractions, no ROC decision: 30 seconds if you spot it

74 Ash 3+3

- $X(z) = z^2 \left[1 - \frac{3}{2}z^{-1}\right] (1 + z^{-1})(1 - z^{-1})$.
- $(1 + z^{-1})(1 - z^{-1}) = 1 - z^{-2}$, so

$$X(z) = z^2 \left(1 - \frac{3}{2}z^{-1}\right) (1 - z^{-2})$$

- Expand:
 $z^2 \left(1 - \frac{3}{2}z^{-1} - z^{-2} + \frac{3}{2}z^{-3}\right) = z^2 - 1.5z - 1 + 1.5z^{-1}$.
- Read the coefficients as impulses, **a power of z is an advance**:

$$x[n] = \delta[n+2] - 1.5\delta[n+1] - \delta[n] + 1.5\delta[n-1]$$

- As a sequence: $x[n] = \{1, -1.5, \underline{-1}, 1.5\}$ for $n = -2 \dots 1$.

78 Ch 4

- $X(z) = z^2(1 - 1.5z^{-1})(1 - z^{-1})(1 + z^{-2})$.
- $(1 - 1.5z^{-1})(1 - z^{-1}) = 1 - 2.5z^{-1} + 1.5z^{-2}$.
- Times $(1 + z^{-2})$: $1 - 2.5z^{-1} + 2.5z^{-2} - 2.5z^{-3} + 1.5z^{-4}$.
- Times z^2 : $z^2 - 2.5z + 2.5 - 2.5z^{-1} + 1.5z^{-2}$.

$$x[n] = \{1, -2.5, \underline{2.5}, -2.5, 1.5\}, \quad n = -2 \dots 2$$

Why there is no ROC to worry about: a finite-length sequence converges everywhere except possibly $z = 0$ and $z = \infty$ (§2.2, property 4). Both of these have positive and negative powers, so their ROC is $0 < |z| < \infty$.

Ch 3 · Analysis of LTI Systems in the Frequency Domain

Numerical Problems & Solutions

35 of the 56 questions in this chapter are the same question: plot the poles and zeros, then sketch the magnitude response. All of them are below. They split into two shapes, according to whether the paper gives you a difference equation or the pole and zero locations themselves, and the second half of the work is identical in both.

The one procedure, used in §1 and §2

- 1. Get $H(z)$.** From an LCCDE, Z-transform both sides and divide. Multiply top and bottom by z^N so both are polynomials in z , not in z^{-1} .
- 2. Factor.** Numerator roots are zeros, denominator roots are poles. These are almost always quadratics, so the quadratic formula finishes it, and complex roots always arrive in conjugate pairs.
- 3. Plot:** unit circle, $\circ$ for each zero, $\times$ for each pole, and the zeros at the origin if the numerator is the shorter polynomial.
- 4. Three anchor values,** which is all a not-to-scale sketch needs: $|H(1)|$ at $\omega = 0$, $|H(j)|$ at $\omega = \pi/2$, $|H(-1)|$ at $\omega = \pi$.
- 5. Peak at the pole angle, dip at the zero angle.** Join smoothly, then say in words whether it is low-pass, high-pass, band-pass or a notch.

The geometry, in one line

$$|H(e^{j\omega})| = |K| \frac{\prod_k |e^{j\omega} - z_k|}{\prod_k |e^{j\omega} - p_k|}$$

- Distances from the zeros multiply on top, distances from the poles divide underneath.
- A pole at radius r and angle θ makes $|H|$ peak at $\omega = \theta$, and the peak is sharper the closer r is to 1.
- A zero on the circle forces $|H|$ to **exactly zero** at its angle.

The checks that cost nothing

- 1.** $|H(1)|$ is the sum of the coefficients over the sum of the denominator coefficients; $|H(-1)|$ is the same with alternating signs. Two lines.
- 2.** Complex roots must be **conjugate pairs**. A lone complex pole is an arithmetic slip, every time.
- 3.** Product of the roots of $z^2 + \alpha z + \beta$ is β , sum is $-\alpha$. Check your factorisation against the coefficients before plotting.
- 4. If any pole has $|p| \geq 1$ and the system is stated causal, stop:** the ROC excludes the unit circle and there is no frequency response to draw.

1 Difference equation given: find $H(z)$, plot, sketch $|H|$

TOP 24/45 80 Ch 8, 72 Ash 7, 78 Bh 2+4, 81 Bh 3+7, 80 Bh 3+7, 81 Ba 3+7, 76 Ash 2+8, 79 Bh 3+7, 79 Ba 3+7, 74 Ash 3+7, 72 Ka 6, 72 Ch 2+4, 70 Ch 6, 71 Bh 6, 73 Ma 7, 69 Ch 2+2+6, 76 Bh 2+5, 75 Ch 2+7, 69 Bh 3+3+2+4, 70 Asa 2+6+2, 70 Ma 8

QUESTION AS PRINTED 2080 Chaitra · 2072 Ashwin · [8/7]

Plot the pole-zero in the z -plane and draw the magnitude response (not to scale) of the system described by the difference equation $y[n] - 0.4y[n-1] + 0.25y[n-2] = x[n] + 0.5x[n-1]$ (magnitude response only).

Step 1: transform and divide

- Z-transform term by term, using $y[n-k] \leftrightarrow z^{-k}Y(z)$:

$$Y(z)(1 - 0.4z^{-1} + 0.25z^{-2}) = X(z)(1 + 0.5z^{-1})$$

$$H(z) = \frac{1 + 0.5z^{-1}}{1 - 0.4z^{-1} + 0.25z^{-2}} = \frac{z(z + 0.5)}{z^2 - 0.4z + 0.25}$$

- Multiplying by z^2/z^2 is what exposes the zero at the origin. **Do it before factoring, every time.**

Step 2: factor

- **Zeros**, from the numerator $z(z + 0.5)$:

$$z = 0 \quad \text{and} \quad z = -0.5$$

- **Poles**, from $z^2 - 0.4z + 0.25 = 0$:

$$z = \frac{0.4 \pm \sqrt{0.16 - 1}}{2} = 0.2 \pm j0.4583$$

- Check: product of the poles is $0.04 + 0.21 = 0.25$, the constant term. Sum is 0.4. Both agree.

- In polar form the pole pair is

$$|p| = \sqrt{0.2^2 + 0.4583^2} = \mathbf{0.5}, \quad \theta = \arctan \frac{0.4583}{0.2} = 66.4^\circ = 1.1$$

- $|p| = 0.5 < 1$, so the causal system is **stable** and the frequency response exists.
- $|p| = 0.5$ **exactly**, because the constant term of a monic quadratic is the product of the roots, and here that product is $|p|^2 = 0.25$.

Step 3: the three anchor values

- At $\omega = 0$, put $z = 1$: $|H| = \left| \frac{1 + 0.5}{1 - 0.4 + 0.25} \right| = \frac{1.5}{0.85} = \mathbf{1.765}$
- At $\omega = \pi/2$, put $z = j$: $|H| = \left| \frac{1 - 0.5j}{1 + 0.4j - 0.25} \right| = \frac{1.118}{0.850} = \mathbf{1.315}$
- At $\omega = \pi$, put $z = -1$: $|H| = \left| \frac{1 - 0.5}{1 + 0.4 + 0.25} \right| = \frac{0.5}{1.65} = \mathbf{0.303}$

Step 4: read the shape off the plot

- The pole pair sits at 66.4° , so $|H|$ **peaks** near there. The true peak is **1.96** at **50.9°**, pulled back towards $\omega = 0$ by the zero at $z = -0.5$ sitting on the far side.
- The zero at $z = -0.5$ lies at angle 180° , so $|H|$ is **smallest** at $\omega = \pi$, and 0.303 is the minimum.
- The zero at the origin is equidistant from every point of the circle and **does not shape the response at all**. It still belongs on the plot.
- Verdict: a **low-pass** response, rising a little to a broad hump around 50° before falling to 0.303 at $\omega = \pi$.

The same poles with the zero moved: 78 Bh and 81 Bh

Two more papers ask the identical question with $x[n] - 0.4x[n-1]$ on the right instead of $x[n] + 0.5x[n-1]$. **The denominator is unchanged, so the poles are the same $0.2 \pm j0.4583$** ; only the zero moves, from -0.5 to $+0.4$, that is from angle 180° to angle 0° .

- The zero now sits nearest $\omega = 0$, so it **pulls DC down** instead of $\omega = \pi$: $|H(1)| = 0.6/0.85 = \mathbf{0.706}$, $|H(j)| = \mathbf{1.267}$, $|H(-1)| = 1.4/1.65 = \mathbf{0.849}$.
- Peak **1.34** at **73.9°**, and the response is now **high-pass** in character.
- **One sign in the numerator flips the filter from low-pass to high-pass, with the poles untouched**. That is the lesson of the whole section.

Every other paper in this group

Same five steps. Poles from the y coefficients, zeros from the x coefficients, then the three anchor values.

Papers	Difference equation	Zeros	Poles ($ p $)	$ H $ at $0, \frac{\pi}{2}, \pi$	Character
80 Bh	$y(n) = x(n) + 0.8x(n-1) + 0.8x(n-2) + 0.49y(n-2)$	$-0.4 \pm j0.8$ (0.894)	± 0.7 (0.7)	5.10 0.55 1.96	low-pass, dip at 116°
79 Bh	$y[n] - 0.35y[n-1] + 0.25y[n-2] = x[n] - 0.75x[n-1]$	0.75, 0	$0.175 \pm j0.468$ (0.5)	0.28 1.51 1.09	high-pass, peak 1.53 at 83°
79 Ba	$y[n] - 0.3y[n-1] + 0.2y[n-2] = x[n] - 0.5x[n-1]$	0.5, 0	$0.15 \pm j0.421$ (0.447)	0.56 1.31 1.00	high-pass, peak 1.31 at 85°
74 Ash	$y[n] - 0.4y[n-1] + 0.2y[n-2] = x[n] + 0.5x[n-1] + 0.6x[n-2] + 0.8x[n-3]$	$0.185 \pm j0.942$ (0.96), -0.869	$0.2 \pm j0.4$ (0.447)	3.63 0.56 0.19	low-pass, dip at 79°
72 Ka	$y[n] - 0.5y[n-1] + 0.3y[n-2] = x[n] + 0.7x[n-1]$	$-0.7, 0$	$0.25 \pm j0.487$ (0.548)	2.13 1.42 0.17	low-pass, peak 2.44 at 50°
72 Ch	$y[n] - 0.4y[n-1] + 0.1y[n-2] = x[n] + 0.6x[n-1]$	$-0.6, 0$	$0.2 \pm j0.245$ (0.316)	2.29 1.18 0.27	low-pass
70 Ch	$y[n] - 0.3y[n-1] + 0.225y[n-2] = x[n] + 0.5x[n-1]$	$-0.5, 0$	$0.15 \pm j0.45$ (0.474)	1.62 1.35 0.33	low-pass, peak 1.80 at 55°

Papers	Difference equation	Zeros	Poles ($ p $)	$ H $ at $0, \frac{\pi}{2}, \pi$	Character
71 Bh	$y[n] - 0.3y[n-1] + 0.225y[n-2] = x[n] - 0.4x[n-1]$	0.4, 0	$0.15 \pm j0.45$ (0.474)	0.65 1.30 0.92	high-pass, peak 1.32 at 81°
73 Ma	$y[n] - 0.3y[n-1] + 0.2y[n-2] = x[n] + 0.7x[n-1]$	-0.7, 0	$0.15 \pm j0.421$ (0.447)	1.89 1.43 0.20	low-pass, peak 2.00 at 48°
69 Ch	$y[n] - 0.4y[n-1] + 0.2y[n-2] = x[n] + 0.1x[n-1] - 0.06x[n-2]$	-0.3, 0.2	$0.2 \pm j0.4$ (0.447)	1.30 1.19 0.53	low-pass, peak 1.51 at 55°
76 Bh	$y[n] - 0.6y[n-1] + 0.35y[n-2] = x[n] - 0.1x[n-1] - 0.2x[n-2]$	0.5, -0.4	$0.3 \pm j0.510$ (0.592)	0.93 1.36 0.46	band-pass, peak 1.91 at 61°
75 Ch	$y(n) - 0.3y(n-1) = 4x(n) + 0.7x(n-1) + 2x(n-2)$	$-0.0875 \pm j0.702$ (0.707)	0.3 (0.3)	9.57 2.03 4.08	low-pass, dip at 99°
69 Bh	$y[n] = x[n] + 1.21x[n-2] - 0.8y[n-1]$	$\pm j1.1$ (1.1)	-0.8 (0.8)	1.23 0.16 11.05	high-pass, deep dip at 90°
70 Asa, 70 Ma	$y(n) = x(n) + x(n-2)$ (FIR, no feedback)	$\pm j$ on the circle	none (all-zero)	2.00 0 2.00	$ H = 2 \cos \omega $, exact null

75 Ch prints $2x(n-2) + 0.7x(n-1) + 4x(n)$ in that order Read the delays, not the position on the page. The constant term is 4, so $H(z) = (4 + 0.7z^{-1} + 2z^{-2})/(1 - 0.3z^{-1})$ and $|H(1)| = 6.7/0.7 = 9.57$. Writing it as $2 + 0.7z^{-1} + 4z^{-2}$ instead gives 9.57 as well, because the coefficients still sum to 6.7, **but every other value is wrong** and the zeros move from $-0.0875 \pm j0.702$ to $-0.0875 \pm j0.351$.

70 Asa and 70 Ma also ask whether the system is causal and stable Yes to both, and the reason is one line: $y(n) = x(n) + x(n-2)$ uses only present and past inputs, so it is causal; its impulse response is $h = \{1, 0, 1\}$, a finite sum, so $\sum |h[n]| = 2 < \infty$ and it is stable. **Every FIR system is stable**, which is why the question is worth only 2 marks.

81 Ba and 76 Ash: the pole is at $z = 2.75$ $y(n) = 0.67x(n) - 0.3x(n-1) + 2.75y(n-1)$ rearranges to $y(n) - 2.75y(n-1) = 0.67x(n) - 0.3x(n-1)$, so $H(z) = (0.67z - 0.3)/(z - 2.75)$: one zero at $z = 0.448$ and **one pole at $z = 2.75$, far outside the unit circle**. If the system is causal its ROC is $|z| > 2.75$, which does not contain the unit circle, so **the frequency response does not exist** and neither the magnitude (81 Ba) nor the phase (76 Ash) can be drawn. Only the anticausal reading, ROC $|z| < 2.75$, is stable; on that reading $|H|$ climbs gently from 0.211 at $\omega = 0$ to 0.259 at $\omega = \pi$. **State which reading you are using**. A pole at 0.275 rather than 2.75 would make the causal system stable, and is the likeliest intended value.

2 Poles and zeros given directly

TOP 11/45 82 Ba **3+7**, 80 Ba **4+6**, 82 Bh **2+8**, 76 Ch **2+6**, 75 Ash **2+8**, 81 Ch **8**, 73 Bh **8**, 74 Ch **2+8**, 75 Bh **2+6**, 73 Ch **2+8**, 71 Ch **2+8**
 · **no algebra at all**: the factoring is done for you

QUESTION AS PRINTED 2080 Baishakh · 2082 Bhadra · [4+6]

A filter has poles at $0.45 \pm j1.6$ and zeros at $0.58 \pm j2.06$. Map them in the z -plane and plot the magnitude response (not to scale).

Step 1: radii and angles, which is all the geometry needs

– **Poles** $0.45 \pm j1.6$:

$$|p| = \sqrt{0.45^2 + 1.6^2} = \mathbf{1.662}, \quad \theta_p = \arctan \frac{1.6}{0.45} = 74.3^\circ$$

– **Zeros** $0.58 \pm j2.06$:

$$|z| = \sqrt{0.58^2 + 2.06^2} = \mathbf{2.140}, \quad \theta_z = \arctan \frac{2.06}{0.58} = 74.3^\circ$$

- **Poles and zeros share the same angle**, 74.3° , and the zeros sit further out. So they partly cancel: the peak the poles want to make is flattened by the dip the zeros want to make.
- Both lie **outside** the unit circle, so nothing is close to the circle and **the response is gentle, with no sharp feature anywhere**.
- That is the answer the 6 marks are for. The sketch is nearly flat.

Step 2: the three anchor values

– Build $H(z) = \frac{(z - 0.58 - j2.06)(z - 0.58 + j2.06)}{(z - 0.45 - j1.6)(z - 0.45 + j1.6)} = \frac{z^2 - 1.16z + 4.578}{z^2 - 0.9z + 2.763}$, using (sum, product) of each conjugate pair.

- $|H(1)| = \left| \frac{1 - 1.16 + 4.578}{1 - 0.9 + 2.763} \right| = \frac{4.418}{2.863} = \mathbf{1.544}$
- $|H(j)| = \mathbf{1.902}$, $|H(-1)| = \left| \frac{1 + 1.16 + 4.578}{1 + 0.9 + 2.763} \right| = \frac{6.738}{4.663} = \mathbf{1.446}$
- Peak **2.04** at **74°**, exactly the pole angle. Range 1.45 to 2.04: **a shallow bump, not a resonance.**

Step 3: the phase, when asked

82 Ba asks for the phase as well. Add the angle of each vector from a zero and subtract the angle of each vector from a pole: $\arg H = \sum_k \arg(e^{j\omega} - z_k) - \sum_k \arg(e^{j\omega} - p_k)$. With conjugate pairs the imaginary parts cancel at $\omega = 0$ and $\omega = \pi$, so **the phase is zero at both ends** and odd in between.

The rest of the family

Papers	Poles ($ p $)	Zeros ($ z $)	$ H $ at $0, \frac{\pi}{2}, \pi$	Shape
82 Ba, 80 Ba, 82 Bh	$0.45 \pm j1.6$ (1.662)	$0.58 \pm j2.06$ (2.140)	1.54 1.90 1.45	shallow bump, peak 2.04 at 74°
76 Ch , 75 Ash, 81 Ch, 73 Bh	$0.45 \pm j1.06$ (1.152)	$0.58 \pm j2.06$ (2.140)	3.10 3.93 2.09	real peak 11.50 at 67°
74 Ch	$0.45 - j0.77$ (0.892), $-2 \pm j0.3$ (2.022)	none	0.20 0.18 0.63	rises towards $\omega = \pi$
75 Bh, 71 Ch	$0.45 \mp j0.77$ (0.892), $-2 \pm j0.3$ (2.022)	$1.2 \pm j3$ (3.231)	1.81 1.76 8.76	high-pass, peak 8.76 at π
73 Ch	$0.45 + j0.77$ (0.892), $2 \pm j0.7$ (2.119)	$1.2 \pm j0.43$ (1.275)	0.27 0.43 0.37	broad bump at 72°

Every pole in this section except $0.45 \pm j0.77$ lies outside the unit circle $|p|$ runs 1.152, 1.662, 2.022, 2.119. So **none of these systems is both causal and stable**. The questions only ask you to map the points and sketch $|H|$, which is legitimate as long as the ROC is taken to be an annulus that contains the unit circle, making the system stable but **not causal**. **Say so in one line and the mark is safe**. The $0.45 \pm j0.77$ pair, at $|p| = 0.892$, is the only one inside.

74 Ch, 75 Bh, 73 Ch and 71 Ch each print a lone complex pole They give $0.45 - 0.77i$ or $0.45 + 0.77i$ with **no conjugate**, alongside a pair $-2 \pm 0.3i$ or $2 \pm 0.7i$ that is properly conjugate. **A difference equation with real coefficients cannot have an unpaired complex pole**, so the intended pole is the pair $0.45 \pm j0.77$ and the printed sign is a dropped $\pm$. Restoring the conjugate changes the numbers, so **state which reading you used**: with the pair restored, 74 Ch has $|H| = 0.12, 0.21, 0.34$ at $0, \pi/2, \pi$ and peaks at 0.80 near 60°. The values tabulated above are for the question exactly as printed.

Where $0.45 \pm j0.77$ comes from The local worked example these questions are built on specifies its poles and zeros as a **radius and an angle**: a zero of radius 0.9 at 1.0376 rad is $0.9 \cos(1.0376) + j0.9 \sin(1.0376) = 0.4575 + j0.7751$, **which the source rounds to $0.45 + j0.77$** , and a pole of radius 0.892 at 2.5158 rad is $-0.728 + j0.522$. **Recognising the pair saves a conversion**, and it explains why the same two decimals keep reappearing across six different papers.

3 All-pole systems, and the steady-state response to a sinusoid

PIN 3/45 69 Bh 6, 67 Mng 10, 68 Bh 7

QUESTION AS PRINTED 2068 Bhadra · [7]

For the system $y[n] - \frac{10}{24}y[n-1] + \frac{1}{24}y[n-2] = x[n]$, find the frequency response, then determine $y[n]$ for the input $x[n] = \sin(\frac{\pi}{3}n) + \sin(\frac{\pi}{5}n)$.

Step 1: the frequency response

- There is no x delay, so the numerator is 1: the system is **all-pole**.

$$H(z) = \frac{1}{1 - \frac{10}{24}z^{-1} + \frac{1}{24}z^{-2}} = \frac{z^2}{z^2 - \frac{5}{12}z + \frac{1}{24}}$$

- Factor: $z^2 - \frac{5}{12}z + \frac{1}{24} = (z - \frac{1}{4})(z - \frac{1}{6})$, so the **poles are $\frac{1}{4}$ and $\frac{1}{6}$** , both real, both well inside the circle. Two zeros sit at the origin.

$$H(e^{j\omega}) = \frac{1}{(1 - \frac{1}{4}e^{-j\omega})(1 - \frac{1}{6}e^{-j\omega})}$$

- $|H|$ falls from 1.60 at $\omega = 0$ to 0.686 at $\omega = \pi$: a gentle **low-pass**, with no peak because both poles are far from the circle.

Step 2: use the eigenfunction result, do not convolve

- A sinusoid in gives the same sinusoid out, scaled by $|H|$ and shifted by $\angle H$ at that frequency. Evaluate H twice and stop.
- At $\omega = \pi/3$: $|H| = \mathbf{1.1955}$, $\angle H = \mathbf{-0.3987}$ rad = -22.85° .
- At $\omega = \pi/5$: $|H| = \mathbf{1.4159}$, $\angle H = \mathbf{-0.2949}$ rad = -16.90° .

$$y[n] = 1.1955 \sin\left(\frac{\pi}{3}n - 0.3987\right) + 1.4159 \sin\left(\frac{\pi}{5}n - 0.2949\right)$$

- **The lower frequency is amplified more**, which is the low-pass character showing up in the answer.

69 Bh: the same method, one line shorter

- $y[n] - 0.8y[n-1] + 0.15y[n-2] = x[n]$ gives $z^2 - 0.8z + 0.15 = (z - 0.5)(z - 0.3)$, so the **poles are 0.5 and 0.3** with a double zero at the origin.
- $|H|$: **2.857** at $\omega = 0$, 0.857 at $\pi/2$, **0.513** at π . Monotonically falling, so **low-pass**, peak at DC.

67 Mng as printed has a pole exactly on the unit circle The paper writes $y[n] - \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] - x[n] = 0$. That denominator factors as $z^2 - \frac{5}{6}z - \frac{1}{6} = (z-1)(z+\frac{1}{6})$, putting **a pole at $z = 1$** . A pole on the circle means $|H(e^{j0})|$ is infinite, the system is only marginally stable and it is not BIBO stable, so strictly **there is no frequency response to plot**. Change the one sign, $-\frac{1}{6} \rightarrow +\frac{1}{6}$, and the poles become $\frac{1}{2}$ and $\frac{1}{3}$, giving an ordinary low-pass with $|H(1)| = 1/[(1-\frac{1}{2})(1-\frac{1}{3})] = 3$. **68 Bh asks the identical question with the + sign** and poles $\frac{1}{4}, \frac{1}{6}$, which is strong evidence that the minus is a misprint. Answer the + version and note the printed one in a line.

4 Zero-input, zero-state and total response

PIN 2/45 78 Bh **4**, 66 Ma **10**

QUESTION AS PRINTED 2066 Magh · [10]

A system is characterized by $y(n) - 0.5y(n-1) - 0.25y(n-2) - x(n) = 0$, the input is $x(n) = 0.2^n u(n)$ and the initial conditions are $y(-1) = 2$, $y(-2) = 4$. Find (a) the zero input response, (b) the zero state response, (c) the total response, (d) the system function $H(z)$, (e) the poles of $H(z)$.

(d) and (e) first, because everything else uses them

$$H(z) = \frac{1}{1 - 0.5z^{-1} - 0.25z^{-2}} = \frac{z^2}{z^2 - 0.5z - 0.25}$$

- Roots: $z = \frac{0.5 \pm \sqrt{0.25 + 1}}{2} = \frac{1 \pm \sqrt{5}}{4}$, that is

$$p_1 = \mathbf{0.8090}, \quad p_2 = \mathbf{-0.3090}$$

- Both have $|p| < 1$, so the causal system is **stable**.

(a) Zero-input response: the input is switched off, the initial conditions drive it

- The form is fixed by the poles: $y_{zi}[n] = C_1(0.8090)^n + C_2(-0.3090)^n$.
- Impose $y(-1) = 2$ and $y(-2) = 4$ on that form and solve the pair:

$$C_1 = 1 + \frac{2}{\sqrt{5}} = \mathbf{1.8944}, \quad C_2 = 1 - \frac{2}{\sqrt{5}} = \mathbf{0.1056}$$

$$y_{zi}[n] = 1.8944(0.8090)^n + 0.1056(-0.3090)^n, \quad n \geq 0$$

- Quick check by running the recurrence $y[n] = 0.5y[n-1] + 0.25y[n-2]$ with no input: $y[0] = 0.5(2) + 0.25(4) = 2$, $y[1] = 1.5$, $y[2] = 1.25$, $y[3] = 1$. The closed form gives the same four numbers.

(b) Zero-state response: the initial conditions are zero, the input drives it

- $X(z) = \frac{1}{1 - 0.2z^{-1}}$, so $Y_{zs}(z) = H(z)X(z)$ has three poles: 0.8090, -0.3090 and 0.2.

- Partial-fraction $Y_{zs}(z)/z$ and multiply back:

$$y_{zs}[n] = 0.9612 (0.8090)^n + 0.1678 (-0.3090)^n - 0.1290 (0.2)^n, \quad n \geq 0$$

- The three residues sum to 1, which must happen because $y_{zs}[0] = 1$. **That is the free check.** The last residue is exactly $-\frac{4}{31}$.

(c) Total response

$$y[n] = y_{zi}[n] + y_{zs}[n] = 2.8557 (0.8090)^n + 0.2734 (-0.3090)^n - 0.1290 (0.2)^n$$

- First few values: 3, 2.2, 1.89, 1.503, 1.2256. Running the original recurrence with $y(-1) = 2$, $y(-2) = 4$ and $x(n) = 0.2^n$ gives the same sequence.
- Natural against forced is a different split from zero-input against zero-state.** The natural part is everything in $(0.8090)^n$ and $(-0.3090)^n$, which draws from both; the forced part is only the $(0.2)^n$ term.

78 Bh gives no initial conditions $y[n] - 3y[n-1] - 4y[n-2] = x[n]$ has $z^2 - 3z - 4 = (z-4)(z+1)$, so the zero-input response is $y_{zi}[n] = C_1(4)^n + C_2(-1)^n$ and **that form is the whole answer**, since no $y(-1)$ or $y(-2)$ is printed to pin C_1 and C_2 down. Worth adding: $|4| > 1$ and $|-1| = 1$, so the zero-input response **never decays** and the system is unstable. Say it, because the 4 marks are for recognising the form and its behaviour.

5 Linear phase

PIN 3/45 70 Asa **2+4**, 67 Mng **6**, 69 Bh **6** · the derivation for general M is in §3.6

QUESTION AS PRINTED 2070 Ashad · [2+4]

What is the condition satisfied by a linear phase FIR filter? Show that the filter with $h(n) = \{-1, 0, 1\}$ is a linear phase filter.

The condition, for the 2 marks

$$h[n] = \pm h[M-1-n], \quad 0 \leq n \leq M-1$$

- The impulse response must be **symmetric** (+) or **antisymmetric** (−) about its midpoint $\alpha = \frac{M-1}{2}$.
- That α is then the constant group delay in samples.

Now show the phase really is linear

Check the given $h(n)$

- $M = 3$, so $M-1-n$ maps $0 \leftrightarrow 2$ and fixes 1.
- $h[0] = -1$ and $h[2] = +1$, so $h[0] = -h[2]$; and $h[1] = 0 = -h[1]$.
- So $h[n] = -h[2-n]$: the sequence is **antisymmetric**, and the condition is met with the minus sign.

$$H(e^{j\omega}) = \sum_{n=0}^2 h[n]e^{-j\omega n} = -1 + e^{-2j\omega}$$

- Factor out the midpoint delay $e^{-j\omega}$, which is the standard trick:

$$H(e^{j\omega}) = e^{-j\omega}(e^{-j\omega} - e^{+j\omega}) = e^{-j\omega}(-2j \sin \omega) = 2 \sin \omega e^{-j(\omega+\pi/2)}$$

$$|H(e^{j\omega})| = 2|\sin \omega|, \quad \arg H(e^{j\omega}) = -\omega - \frac{\pi}{2}$$

- The phase is a **straight line** in ω with slope -1 , so

$$\text{grd } H = -\frac{d}{d\omega} \left(-\omega - \frac{\pi}{2} \right) = 1 \text{ sample} = \alpha = \frac{M-1}{2} = 1 \checkmark$$

- The constant $-\pi/2$ is why this is generalized linear phase.** It comes from the j in $-2j \sin \omega$, that is from the antisymmetry, and it does not affect the group delay, so it causes no distortion.
- The magnitude $2|\sin \omega|$ is zero at $\omega = 0$: an antisymmetric filter always kills DC, which is why this one is a differencer.

67 Mng: show that $h[n] = h[N - 1 - n]$, $0 \leq n \leq N - 1$, is linear phase. This is the general case of the same argument and it is worked in full in §3.6: pair each term with its mirror, factor out $e^{-j\omega\alpha}$ with $\alpha = (N - 1)/2$, and what is left is a real cosine sum. **Real bracket, so the whole phase is the exponential**, giving $\arg H = -\alpha\omega$. 69 Bh asks for the same derivation restricted to odd M , which is the easier case because the midpoint sample $h[\alpha]$ stands alone instead of pairing.

Ch 4 · Discrete Filter Structures

Numerical Problems & Solutions

42 of the 68 questions are the lattice, and they are all one recursion. Every paper is below, grouped by what the recursion is run on. Sections 3 to 5 share a single calculation; learn it once.

The step-down, which is §3, §4 and §5

1. Take the polynomial $A_M(z) = 1 + a_M(1)z^{-1} + \dots + a_M(M)z^{-M}$. For an FIR $H(z) = B(z)$ it is the numerator; for an all-pole or pole-zero $H(z)$ it is **always the denominator**.
2. $k_M = a_M(M)$, read straight off the last coefficient.
- 3.

$$a_{m-1}(i) = \frac{a_m(i) - k_m a_m(m-i)}{1 - k_m^2}, \quad i = 1, \dots, m-1$$

which folds the polynomial against its reverse and drops the degree by one.

4. Repeat until A_1 , whose one coefficient is k_1 .
5. **Stable** if and only if every $|k_m| < 1$. Nothing else to compute.

Two shortcuts worth having

- The last step is always $k_1 = \frac{a_2(1)}{1 + k_2}$, because $a_2(2-1) = a_2(1)$ makes the numerator $a_2(1)(1 - k_2)$ and $1 - k_2^2 = (1 - k_2)(1 + k_2)$.
- **Keep fractions**. $1 - k^2$ breeds denominators, and a decimal answer is visibly wrong by the third stage.

The two checks

1. **Step back up:** $a_m(i) = a_{m-1}(i) + k_m a_{m-1}(m-i)$ must return the polynomial you started from. Two lines, and it catches every arithmetic slip.
2. If any $|k_m| = 1$ exactly, **stop**: the polynomial is symmetric and no lattice exists. See the box in §3.

1 Direct Form I and Direct Form II

TOP 9/45 79 Bh 2+2, 79 Ba 4, 82 Bh 2+2, 74 Ch 5, 72 Ch 5, 72 Ka 4, 69 Ch 4, 68 Bh 9, 70 Ch 4

QUESTION AS PRINTED 2079 Bhadra · [2+2]

Obtain the Direct Form I and Direct Form II realization of the system $y[n] - 0.25y[n-2] = x[n] + 0.4x[n-1] + 0.5x[n-2]$.

Step 1: put it in computing form and read off the taps

$$y[n] = x[n] + 0.4x[n-1] + 0.5x[n-2] + 0.25y[n-2]$$

- Feedforward $b = \{1, 0.4, 0.5\}$, feedback $a = \{1, 0, -0.25\}$, so the taps are $-a_1 = 0$ and $-a_2 = +0.25$.
- $a_1 = 0$, **so there is no $y[n-1]$ branch at all**. Leave the tap out; do not draw a multiplier by zero.
- $M = N = 2$, so Direct Form I needs $2 + 2 = 4$ delays and Direct Form II needs $\max(2, 2) = 2$.

Step 2: Direct Form I

- Input into a chain of two delays, tapped 1, 0.4, 0.5 into an adder.
- That adder's output goes into a second chain of two delays, tapped 0 and 0.25 back into the same adder.
- Reading left to right it is $B(z)$ then $1/A(z)$.

Step 3: Direct Form II

- Swap the two halves and share one chain of two delays carrying $w[n]$:

$$w[n] = x[n] + 0.25w[n-2], \quad y[n] = w[n] + 0.4w[n-1] + 0.5w[n-2]$$

- Left of the chain: one branch, 0.25 from $w[n-2]$ back to the top adder. Right of the chain: three branches, 1, 0.4, 0.5 into the output adder.
- **Two delays instead of four, same arithmetic**. That is the entire answer to "why Direct Form II".

Every other paper in this group

Papers	System	Taps	Watch for
79 Ba, 82 Bh	$y[n] - 0.75y[n-1] - 0.25y[n-2]$ $= x[n] + 0.5x[n-1]$	$b = \{1, 0.5\}$ $-a = \{0.75, 0.25\}$	both feedback taps positive
74 Ch	$3y[n] + y[n-1] + 2y[n-4]$ $= 2x[n] + x[n-3]$	$b = \{\frac{2}{3}, 0, 0, \frac{1}{3}\}$ $-a = \{-\frac{1}{3}, 0, 0, -\frac{2}{3}\}$	divide by $a_0 = 3$ first; 4 delays, most taps zero
72 Ch	$y(n) = -0.1y(n-1) + 0.2y(n-2)$ $+ 3x(n) + 3.6x(n-2) + 0.6x(n-2)$	$b = \{3, 0, 4.2\}$ $-a = \{-0.1, 0.2\}$	$x(n-2)$ is printed twice; add them, $3.6 + 0.6 = 4.2$
72 Ka, 69 Ch	$y(n) = -0.1y(n-1) + 0.72y(n-2)$ $+ 0.7x(n) - 0.252x(n-2)$	$b = \{0.7, 0, -0.252\}$ $-a = \{-0.1, 0.72\}$	Direct Form II only
70 Ch	$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2]$ $-x[n] - 2x[n-1] = 0$	$b = \{1, 2\}$ $-a = \{\frac{3}{4}, -\frac{1}{8}\}$	move the x terms across: they are positive on the right
68 Bh	$H(z)$ given factored, 3 zeros over 4 poles	multiply out first	see the box below

68 Bh gives $H(z)$ already factored, and wants Direct Form I and II. The zeros are $\frac{1}{5}e^{\pm j\pi/5}$ and $\frac{1}{3}$; the poles are $\frac{4}{5}$, $\frac{1}{5}$ and $\frac{1}{7}e^{\pm j\pi/7}$. A direct form needs the expanded polynomials, not the factors, so multiply each conjugate pair out first using $(1 - re^{j\theta}z^{-1})(1 - re^{-j\theta}z^{-1}) = 1 - 2r\cos\theta z^{-1} + r^2z^{-2}$: the zero pair gives $1 - \frac{2}{5}\cos 36^\circ z^{-1} + \frac{1}{25}z^{-2}$ and the pole pair $1 - \frac{2}{7}\cos(180/7)^\circ z^{-1} + \frac{1}{49}z^{-2}$. Multiply through and you have a 3rd-order numerator over a 4th-order denominator, so Direct Form II needs 4 delays. The same factored form is exactly what S2 wants left alone, which is the joke of this chapter: cascade wants factors, direct form wants them multiplied out.

2 Cascade and parallel realizations

TOP 8/45 81 Ba 5, 74 Ma 7, 80 Ba 4, 76 Ch 4, 77 Ch 7, 69 Bh 6, 67 Mng, 78 Ch 6

QUESTION AS PRINTED 2081 Baishakh · [5]

Realize the following system in cascade form of second-order sections (signal flow graph): $H(z) = \frac{10(1 - 0.25z^{-1})(1 - 0.667z^{-1})}{(1 - 0.75z^{-1})(1 - 0.125z^{-1})\{1 - (0.5 + j0.5)z^{-1}\}}$

Step 1: pair the poles, then pair the zeros to match

- The conjugate pair must stay together. With $\sigma = 0.5$ and $\omega = 0.5$:

$$\{1 - (0.5 + j0.5)z^{-1}\}\{1 - (0.5 - j0.5)z^{-1}\} = 1 - z^{-1} + 0.5z^{-2}$$

using $-2\sigma = -1$ and $\sigma^2 + \omega^2 = 0.5$.

- The two real poles pair with each other: $(1 - 0.75z^{-1})(1 - 0.125z^{-1}) = 1 - 0.875z^{-1} + 0.09375z^{-2}$.
- Three zeros over four poles, so one section's numerator is first order. Pair the two "small" zeros and leave $(1 + 2z^{-1})$ to ride with the real-pole section: $(1 - 0.25z^{-1})(1 - 0.667z^{-1}) = 1 - 0.917z^{-1} + 0.167z^{-2}$.

Step 2: write the two sections and place the gain

$$H(z) = 10 \cdot \underbrace{\frac{1 - 0.917z^{-1} + 0.167z^{-2}}{1 - 0.875z^{-1} + 0.09375z^{-2}}}_{\text{section 1}} \cdot \underbrace{\frac{1 + 2z^{-1}}{1 - z^{-1} + 0.5z^{-2}}}_{\text{section 2}}$$

- Draw each section in Direct Form II and chain them. The gain 10 goes on the input branch, or is split between sections to keep the internal signals from overflowing.
- Every coefficient is real. If a j survives, a pair was split.

The rest of the cascade family

All the same two steps. Only 74 Ma also wants the parallel form, which needs partial fractions instead of factors.

Papers	Poles	Sections
74 Ma	same as 81 Ba, gain 5	identical sections, gain 5; parallel form also asked, so partial-fraction the expanded $H(z)$
80 Ba	$0.5e^{\pm j\pi/3}, -0.7e^{\pm j\pi/4}$	$1 - 0.5z^{-1} + 0.25z^{-2}$ and $1 + 0.99z^{-1} + 0.49z^{-2}$
76 Ch	$0.6e^{\pm j\pi/3}, -0.5e^{\pm j2\pi/7}$	the paper prints a stray n in every exponent; read it as $n = 1$
77 Ch	$-0.2, 0.25, -0.5 \pm j0.15$	real pair $1 - 0.05z^{-1} - 0.05z^{-2}$, complex pair $1 + z^{-1} + 0.2725z^{-2}$
69 Bh	$0.5, 0.3, 0.7e^{\pm j\pi/4}$	$1 - 0.8z^{-1} + 0.15z^{-2}$ and $1 - 0.99z^{-1} + 0.49z^{-2}$; all-pole, numerator 1
67 Mng	$\frac{5}{6}, \frac{1}{6}, \frac{3}{4}e^{\pm j\pi/4}$	$1 - z^{-1} + \frac{5}{36}z^{-2}$ and $1 - 1.0607z^{-1} + 0.5625z^{-2}$

78 Ch asks for the parallel form from an impulse response $h[n] = (0.5)^n u[n] + n(0.2)^n u[n]$. Transform each term with the standard pairs: $\frac{1}{1 - 0.5z^{-1}}$ and $\frac{0.2z^{-1}}{(1 - 0.2z^{-1})^2}$. **Parallel form is what partial fractions already give you**, so no further work is needed: two sections added, the first a one-pole section, the second a repeated-pole section needing two delays. The repeated pole is the only wrinkle: it cannot be split further, so that section is second order with denominator $1 - 0.4z^{-1} + 0.04z^{-2}$.

67 Mng prints the same exponent twice Its last two denominator factors both read $e^{-j\pi/4}$. A real filter needs the conjugate, so the second is $e^{+j\pi/4}$, and the pair multiplies to $1 - \frac{3}{2} \cos 45^\circ z^{-1} + \frac{9}{16} z^{-2} = 1 - 1.0607z^{-1} + 0.5625z^{-2}$. Note it in one line and continue.

3 The step-down, run on an FIR numerator

TOP 9/45 81 Ba 5, 78 Bh 3+7, 72 Ch 5, 73 Bh 6, 74 Ch 5, 74 Ash 3+7, 73 Shr 3+7, 76 Bh 2+5, 72 Ka 6, 69 Ch 6, 70 Ch 6

QUESTION AS PRINTED 2081 Baishakh · 2078 Bhadra · [5]

Determine the lattice coefficients corresponding to the FIR filter with system function $H(z) = A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$, and check its stability.

Stage 3

- $k_3 = a_3(3) = \frac{1}{3}$, free.
- $1 - k_3^2 = 1 - \frac{1}{9} = \frac{8}{9}$.

$$a_2(1) = \frac{a_3(1) - k_3 a_3(2)}{8/9} = \frac{\frac{13}{24} - \frac{1}{3} \cdot \frac{5}{8}}{8/9} = \frac{\frac{13}{24} - \frac{5}{24}}{8/9} = \frac{\frac{8}{24}}{8/9} = \frac{\frac{1}{3}}{8/9} = \frac{3}{8}$$

$$a_2(2) = \frac{a_3(2) - k_3 a_3(1)}{8/9} = \frac{\frac{5}{8} - \frac{1}{3} \cdot \frac{13}{24}}{8/9} = \frac{\frac{45}{72} - \frac{13}{72}}{8/9} = \frac{\frac{32}{72}}{8/9} = \frac{\frac{4}{9}}{8/9} = \frac{1}{2}$$

Stage 2, then stage 1

- $k_2 = a_2(2) = \frac{1}{2}$.
- Use the shortcut $k_1 = \frac{a_2(1)}{1 + k_2} = \frac{3/8}{3/2} = \frac{1}{4}$.

$$k = \left\{ \frac{1}{4}, \frac{1}{2}, \frac{1}{3} \right\}$$

- **Stable:** every $|k_m| < 1$.
- Check by stepping back up: $a_2(1) = k_1 + k_2 k_1 = \frac{1}{4} + \frac{1}{8} = \frac{3}{8} \checkmark$, and $a_3(1) = a_2(1) + k_3 a_2(2) = \frac{3}{8} + \frac{1}{3} \cdot \frac{1}{2} = \frac{13}{24} \checkmark$.

74 Ch is the identical question with the coefficients written as $\frac{52}{96}$ and $\frac{25}{40}$, which reduce to $\frac{13}{24}$ and $\frac{5}{8}$. Same answer, $k = \{\frac{1}{4}, \frac{1}{2}, \frac{1}{3}\}$. So is 79 Bh, backwards: see §4.

The other FIR papers

Papers	$H(z)$	$k = \{k_1, k_2, k_3, \dots\}$	Stable?
81 Ba, 78 Bh, 74 Ch	$1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$	$\{\frac{1}{4}, \frac{1}{2}, \frac{1}{3}\}$	yes
73 Shr	$2 + 1.8z^{-1} - 1.6z^{-2} + z^{-3}$ (divide by 2 first)	$\{-2.6, -1.6667, 0.5\}$	no
76 Bh	$1 + 2z^{-1} + 0.62z^{-2} + 0.8z^{-3}$	$\{-2.4258, -2.7222, 0.8\}$	no
72 Ka, 69 Ch	$1 + 2z^{-1} - 3z^{-2} + 4z^{-3}$	$\{-0.5385, 0.7333, 4\}$	no
70 Ch	$1 + 3.1z^{-1} + 5.5z^{-2} + 4.2z^{-3} + 2.3z^{-4}$	$\{0.3381, 1.1664, 0.6830, 2.3\}$	no

Five papers ask for a lattice that does not exist 72 Ch ($1 + 2z^{-1} + z^{-2}$), 73 Bh ($1 + 2z^{-1} + 2z^{-2} + z^{-3}$), 74 Ash ($1 + 0.7z^{-1} + 1.2z^{-2} - z^{-3}$), 75 Ch and 73 Ch (both containing $1 + \frac{2}{3}z^{-1} + \frac{5}{8}z^{-2} + \frac{2}{3}z^{-3} + z^{-4}$). In each the last coefficient has magnitude 1, so $k_M = \pm 1$ and the step-down must divide by $1 - k_M^2 = 0$. **Four of the five are symmetric polynomials**, that is linear-phase filters, and symmetry forces $a_M(M) = a_M(0) = 1$. **A linear-phase FIR filter has no lattice realization**, and that is the answer: show $k_M = 1$, show the division by zero, and note that the zeros lie on the unit circle. For 72 Ch it is visible without any recursion, since $1 + 2z^{-1} + z^{-2} = (1 + z^{-1})^2$ is a double zero at $z = -1$.

4 The reverse: lattice coefficients to direct form

PIN 4/45 79 Bh 6, 71 Shr 5, 80 Ch, 74 Ma 7

QUESTION AS PRINTED 2079 Bhadra · [6]

Given a 3-stage lattice filter with coefficients $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{2}$, $K_3 = \frac{1}{3}$, obtain the system function and the FIR filter coefficients.

- Step up, one stage at a time, with $a_m(i) = a_{m-1}(i) + k_m a_{m-1}(m-i)$ and $a_m(m) = k_m$.
- $A_1(z) = 1 + \frac{1}{4}z^{-1}$.
- $a_2(1) = a_1(1) + k_2 a_1(1) = \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{3}{8}$, and $a_2(2) = k_2 = \frac{1}{2}$, so $A_2(z) = 1 + \frac{3}{8}z^{-1} + \frac{1}{2}z^{-2}$.
- $a_3(1) = a_2(1) + k_3 a_2(2) = \frac{3}{8} + \frac{1}{3} \cdot \frac{1}{2} = \frac{13}{24}$; $a_3(2) = a_2(2) + k_3 a_2(1) = \frac{1}{2} + \frac{1}{3} \cdot \frac{3}{8} = \frac{5}{8}$; $a_3(3) = k_3 = \frac{1}{3}$.

$$H(z) = A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$$

- So $h[n] = \{1, \frac{13}{24}, \frac{5}{8}, \frac{1}{3}\}$.
- **This is §3 run backwards**, on the same numbers. 71 Shr asks it identically.

80 Ch and 74 Ma change one coefficient They give $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{4}$, $K_3 = \frac{1}{3}$. The same recursion gives $a_2(1) = \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{4} = \frac{5}{16}$, $a_2(2) = \frac{1}{4}$, then $a_3(1) = \frac{5}{16} + \frac{1}{3} \cdot \frac{1}{4} = \frac{19}{48}$, $a_3(2) = \frac{1}{4} + \frac{1}{3} \cdot \frac{5}{16} = \frac{17}{48}$, $a_3(3) = \frac{1}{3}$. So $h[n] = \{1, \frac{19}{48}, \frac{17}{48}, \frac{1}{3}\}$. **The two thirds in the answer look alike but are not:** $\frac{19}{48}$ and $\frac{17}{48}$ differ, and writing both as $\frac{18}{48} = \frac{3}{8}$ is the easy mistake.

5 All-pole and lattice-ladder

TOP 21/45 80 Bh 10, 81 Ch, 72 Ash 4+2+1, 79 Ch 4+2+1, 69 Bh 8, 71 Bh 6+2, 70 Ma 6, 71 Ch 4+2+1, 80 Ba, 82 Bh 6, 82 Ba 10, 79 Ba 6+1, 81 Bh 6+4, 76 Ch 6+4, 76 Ash 6+3, 74 Bh 5+1, 75 Bh 7+1, 77 Ch 7+2, 73 Ma 6+1, 78 Ch 6+2, 70 Asa 6, 66 Ma 6, 70 Bh 8

QUESTION AS PRINTED 2080 Bhadra · [10]

Determine the lattice coefficients and draw the lattice structure for the system $H(z) = \frac{1}{1 - 0.2z^{-1} + 0.4z^{-2} + 0.6z^{-3}}$.

The denominator is the polynomial: everything else is identical to §3

- $A_3(z) = 1 - 0.2z^{-1} + 0.4z^{-2} + 0.6z^{-3}$, so $k_3 = 0.6$ and $1 - k_3^2 = 0.64$.

$$a_2(1) = \frac{-0.2 - 0.6(0.4)}{0.64} = \frac{-0.44}{0.64} = -\frac{11}{16}, \quad a_2(2) = \frac{0.4 - 0.6(-0.2)}{0.64} = \frac{0.52}{0.64} = \frac{13}{16}$$

- $k_2 = a_2(2) = \frac{13}{16} = 0.8125$.
- $k_1 = \frac{a_2(1)}{1 + k_2} = \frac{-11/16}{29/16} = -\frac{11}{29} = -0.3793$.

$$k = \left\{ -\frac{11}{29}, \frac{13}{16}, \frac{3}{5} \right\}$$

- All $|k_m| < 1$, so the system is **stable**. **An all-pole $H(z)$ is drawn as the lattice run in reverse:** the same stages, with the forward path fed from the output end.

Adding the ladder, when the numerator is not 1

- Run the step-down on $A(z)$ exactly as above and **keep every intermediate** $A_i(z)$.
- Then work down from $C_M = b_M$ with $C_m = b_m - \sum_{i=m+1}^M C_i a_i(i-m)$.
- **The k 's never depend on the numerator**, so papers sharing a denominator share their lattice: 76 Ch with 76 Ash, and 74 Bh with 75 Bh.

Papers	$H(z)$ denominator	k	C (ladder)
80 Bh	$1 - 0.2z^{-1} + 0.4z^{-2} + 0.6z^{-3}$	$\{-0.3793, 0.8125, 0.6\}$	all-pole
81 Ch, 72 Ash	$1 - 0.525z^{-1} + 0.6125z^{-2} + 0.3z^{-3}$	$\{-0.4219, 0.8462, 0.3\}$	all-pole, stable
79 Ch, 69 Bh	$1 - 0.9z^{-1} + 0.64z^{-2} - 0.576z^{-3}$	$\{-0.6728, 0.1820, -0.576\}$	all-pole, stable
71 Bh	$1 - 0.3z^{-1} + 0.5z^{-2} + 0.25z^{-3}$	$\{-0.2810, 0.6133, 0.25\}$	all-pole
70 Ma	$1 + 2z^{-1} - 3z^{-2} + 4z^{-3}$	$\{-0.5385, 0.7333, 4\}$	unstable , same as 72 Ka
71 Ch	$1 - 0.01z^{-1} - 0.23z^{-2} + 0.5z^{-3}$	$\{0.2, -0.3, 0.5\}$	all-pole, stable
80 Ba, 82 Bh	$1 - 0.3z^{-1} + 0.25z^{-2}$, $B = 2 - 0.7z^{-1} + 0.5z^{-2}$	$\{-0.24, 0.25\}$	$\{1.743, -0.55, 0.5\}$
79 Ba	$1 - 0.3z^{-1} + 0.5z^{-2}$, $B = 1 - 0.4z^{-1} + 0.25z^{-2}$	$\{-0.2, 0.5\}$	$\{0.81, -0.325, 0.25\}$
76 Ch	$1 - 0.5z^{-1} - 0.7z^{-2} + 0.3z^{-3}$, $B = 0.5 - 2z^{-1} + 3z^{-2}$	$\{-0.8056, -0.6044, 0.3\}$	$\{1.4722, -1.0440, 3\}$
76 Ash	same denominator , $B = 0.7 - 1.5z^{-1} + 0.5z^{-2}$	$\{-0.8056, -0.6044, 0.3\}$	$\{-0.0778, -1.3407, 0.5\}$
74 Bh	$1 + 0.27z^{-1} + 0.06z^{-2} - 0.75z^{-3}$, $B = 0.62 + 0.42z^{-1} - 0.25z^{-2}$	$\{0.45, 0.6, -0.75\}$	$\{0.5, 0.6, -0.25\}$
75 Bh	same denominator , $B = 0.6 - 0.45z^{-1} - 0.25z^{-2}$	$\{0.45, 0.6, -0.75\}$	$\{0.8715, -0.27, -0.25\}$
77 Ch	$1 + 0.2z^{-1} + 0.7z^{-2} - 0.75z^{-3}$, $B = 0.5 - 0.4z^{-1} - 0.2z^{-2}$	$\{0.5631, \mathbf{1.9429}, -0.75\}$	unstable , $ k_2 > 1$
73 Ma	$1 - 0.45z^{-1} + 0.3z^{-2} + 0.5z^{-3}$, $B = 0.5 + 0.65z^{-1} + 0.2z^{-2}$	$\{-0.4706, 0.7, 0.5\}$	$\{0.7412, 0.81, 0.2\}$
78 Ch	$1 + 0.2z^{-1} - 0.15z^{-2}$, $B = 1 - z^{-1} + 0.5z^{-2}$	$\{0.2353, -0.15\}$	$\{1.3338, -1.1, 0.5\}$
70 Asa	$1 - 0.3z^{-1} + 0.4z^{-2}$, $B = 0.5 + 2z^{-1} + 0.6z^{-2}$	$\{-0.2143, 0.4\}$	$\{0.7271, 2.18, 0.6\}$
66 Ma	$1 + \frac{1}{5}z^{-1} + \frac{2}{5}z^{-2} + \frac{3}{5}z^{-3}$, $B = \frac{1}{2} + \frac{1}{3}z^{-1} + \frac{1}{4}z^{-2} + \frac{1}{5}z^{-3}$	$\{-0.0435, 0.4375, 0.6\}$	$\{0.2997, 0.2665, 0.21, 0.2\}$
82 Ba	$1 - 0.001z^{-1} + 0.6546z^{-2} - 0.0775z^{-3}$	$\{0.0302, 0.6585, -0.0775\}$	$\{-0.0493, 0.3059, 0.5124, 0.2759\}$
81 Bh	$(1 + 0.5z^{-1})(1 + 0.3z^{-1})(1 + 0.4z^{-1})$ $= 1 + 1.2z^{-1} + 0.47z^{-2} + 0.06z^{-3}$	multiply out first	$B = 1 + z^{-1} + z^{-2}$

70 Bh: the all-pass, and the tidiest result in the chapter $H(z) = \frac{1 + 0.4z^{-1} - 1.2z^{-2} + 2z^{-3}}{2 - 1.2z^{-1} + 0.4z^{-2} + z^{-3}}$. Look before calculating: **the numerator is the denominator written backwards**. That is the definition of an **all-pass**, $H(z) = z^{-N}A(z^{-1})/A(z)$. The backward path of a lattice already computes $z^{-N}A(z^{-1})$, so the ladder collapses completely: every $C_m = 0$ except $C_3 = 1$. **An all-pass is a plain lattice with the output taken from the last backward node**. Normalising by $a_0 = 2$ gives $A(z) = 1 - 0.6z^{-1} + 0.2z^{-2} + 0.5z^{-3}$ and $k = \{-0.56, \frac{2}{3}, \frac{1}{2}\}$, all inside the unit circle, so it is stable.

6 Quantization, number formats and limit cycles

TOP 10/45 71 Ch **3**, 70 Asa **1+3**, 72 Ma **5**, 81 Ch, 80 Ch, 70 Bh **3+4**, 66 Ma **4**, 71 Shr **1+1+2+1**, 69 Bh **6**, 69 Bh **3+4**, 75 Ash **3**, 67 Mng

QUESTION AS PRINTED 2066 Magh · [4]

For the first order filter $y(n) = Q\{a y(n-1)\} + x(n)$, the product term $a y(n-1)$ has been quantized by rounding it to 3 bits, with $y(-1) = 0$, $x(n) = 0.875\delta(n)$ and $a = -0.5$. Show whether the filter goes into a limit cycle. What is the period?

Just run it. Three bits after the point means the step is $q = 2^{-3} = \frac{1}{8}$

n	$a y(n-1)$	$Q\{\cdot\}$ to $\frac{1}{8}$	$y(n)$	
0	$-0.5 \times 0 = 0$	0	0.875	the impulse arrives
1	-0.4375	-0.5	-0.5	rounds away from zero
2	0.25	0.25	0.25	exact, no rounding needed
3	-0.125	-0.125	-0.125	exact; now at the dead-band edge
4	0.0625	0.125	0.125	half a step, rounded up
5	-0.0625	-0.125	-0.125	and back again
6	0.0625	0.125	0.125	

– **Yes, it enters a limit cycle**, amplitude $\frac{1}{8}$, and it never leaves.

– **Period 2**, because $a < 0$ flips the sign each step.

– Confirm with the dead band: $\frac{q}{2(1-|a|)} = \frac{1/8}{2(1-0.5)} = \frac{1}{8}$, **exactly the amplitude observed**. Quote the formula and the table together.

- The mechanism is row 4 alone: $-0.5 \times (-0.125) = 0.0625$ is *half* a step, and rounding lifts it back to a whole one instead of letting it fall to zero.

75 Ash: $\pm \frac{5}{8}$ in the three fixed-point codes

- $\frac{5}{8} = 0.101_2$, so with a sign bit and 3 fraction bits:
 $+\frac{5}{8} = 0.101$ in sign magnitude, 1's complement and 2's complement alike.
 $-\frac{5}{8}$: sign magnitude 1.101 (flip the sign bit only), 1's complement 1.010 (invert the fraction bits), 2's complement 1.011 (invert and add 1 in the last place).
- **Check: 1's complement plus 1 ulp is 2's complement.** $1.010 + 0.001 = 1.011 \checkmark$.

69 Bh asks for $-192/220$ in 8-bit form, which cannot be done $192/220 = 0.87273$, and 220 is not a power of two, so **the number has no exact binary fraction** in 8 bits or any other number of bits. Rounded to 8 bits it is $-\frac{223}{256} = -0.871094$, giving sign magnitude 1.11011111, 1's complement 1.00100000 and 2's complement 1.00100001. **Almost certainly 220 is a misprint for 256**, the natural 8-bit scale: $-192/256 = -0.75$ exactly, sign magnitude 1.11000000, 1's complement 1.00111111, 2's complement 1.01000000. Give the clean version and note the other in a line.

67 Mng: show the two truncation error ranges by example Take $b_u = 6$ bits truncated to $b = 3$, so $q = \frac{1}{8}$ and $q_u = \frac{1}{64}$, and $q - q_u = \frac{7}{64}$. Sweeping every 6-bit fraction confirms both bounds exactly. **Sign magnitude:** truncating shortens the *magnitude*, so a positive number falls and a negative number rises, and e runs from $-\frac{7}{64}$ to $+\frac{7}{64}$, **symmetric**. **2's complement:** truncating always drops the value towards $-\infty$ whatever the sign, so e runs from $-\frac{7}{64}$ to 0 and **never exceeds zero**. One example each way is enough: $+0.828125 \rightarrow 0.75$ (error $-\frac{7}{64}$ in both codes) and $-0.828125 \rightarrow -0.75$ in sign magnitude (error $+\frac{7}{64}$) but $\rightarrow -0.875$ in 2's complement (error $-\frac{3}{64}$). **That asymmetry is why 2's complement truncation adds a DC bias** and rounding is preferred.

Ch 5 • FIR Filter Design

Numerical Problems & Solutions

47 of the 70 questions are one design calculation. The other 23 are the Remez algorithm, which asks for a flowchart and never for a number: that is \$5.5 of the theory, not here. Every design paper is below.

The recipe, used in §1, §2 and §3

1. **Convert the spec** to ω in radians per sample if it is given in Hz ($\omega = 2\pi f/f_s$) or rad/s ($\omega = \Omega/f_s$).
2. **Window** from the stopband attenuation A_s , taking the cheapest that clears it. If the paper gives δ_s , first $A_s = -20 \log_{10} \delta_s$.
3. **Length** $M = c\pi/\Delta\omega$ with c from the same table, then round up to the next **odd** N .
4. **Cut-off** $\omega_c = \frac{\omega_p + \omega_s}{2}$, the middle of the transition band.
5. **Coefficients** $h[n] = h_d[n] w[n]$, with $h_d[n] = \frac{\sin(\omega_c(n - \alpha))}{\pi(n - \alpha)}$, $\alpha = \frac{N - 1}{2}$, and the centre tap $h_d[\alpha] = \omega_c/\pi$ done separately.

The table, again, because everything turns on it

Window	Stop band	Transition
Rectangular	21 dB	$1.8\pi/M$
Bartlett	25 dB	$6.1\pi/M$
Hann	44 dB	$6.2\pi/M$
Hamming	53 dB	$6.6\pi/M$
Blackman	74 dB	$11\pi/M$

Checks that cost nothing

1. The centre tap is always ω_c/π exactly. **If it is not, ω_c is wrong.**
2. $h[n]$ must be **symmetric** about α : compute one half and mirror it.
3. A **Hann**-windowed filter has $h[0] = h[N - 1] = 0$ exactly, because the window is zero there. So "the first three coefficients" begins with a zero.
4. N odd, always, so α is an integer.

1 Fixed-window low-pass designs

TOP 13/45 79 Ba **5+3**, 71 Ch, 69 Ch **8**, 74 Ash **2+6**, 79 Ch **8**, 75 Bh **6**, 82 Ba **2+6**, 67 Mng **8**, 70 Asa **9**, 69 Bh **8**, 68 Bh **9**, 76 Ch **6**, 81 Ch **2+6**, 73 Ma **4+5**, 82 Bh **10**

QUESTION AS PRINTED 2079 Chaitra · 2075 Bhadra · [8] [6]

Design a low pass FIR filter using a suitable window for $\omega_p = 0.24\pi$, $\omega_s = 0.54\pi$ and stopband attenuation $\alpha_s = 42.2$ dB. Find the first three coefficients of the impulse response.

Step 1: the two lookups

- $A_s = 42.2$ dB. Hann gives **44 dB**, which clears it; Bartlett's 25 dB does not. **So Hann**, and Hamming would be over-design.
- $\Delta\omega = 0.54\pi - 0.24\pi = 0.30\pi$.

$$M = \frac{6.2\pi}{\Delta\omega} = \frac{6.2\pi}{0.30\pi} = 20.67 \Rightarrow M = 21, \quad \boxed{N = 23}$$

rounding up, then up again to odd.

$$\alpha = \frac{N - 1}{2} = \mathbf{11}.$$

Step 2: the cut-off, in the middle

$$\omega_c = \frac{0.24\pi + 0.54\pi}{2} = \mathbf{0.39\pi} = 1.2252 \text{ rad/sample}$$

$$h_d[n] = \frac{\sin(0.39\pi(n - 11))}{\pi(n - 11)}, \quad h_d[11] = \frac{\omega_c}{\pi} = \mathbf{0.39}$$

Step 3: the first three coefficients

- Hann window, $w[n] = 0.5 - 0.5 \cos \frac{2\pi n}{22}$:

$$w[0] = 0, \quad w[1] = 0.020254, \quad w[2] = 0.079373$$

$$h_d[0] = 0.022865, \quad h_d[1] = -0.009836, \quad h_d[2] = -0.035350$$

$$h[0] = 0, \quad h[1] = -0.000199, \quad h[2] = -0.002806$$

- $h[0] = 0$ is not a mistake: the Hann window is exactly zero at both ends, so the first and last taps of the filter vanish. Say so, because it looks like an error otherwise.
- The rest follows by symmetry, $h[22 - n] = h[n]$, with the centre tap $h[11] = 0.39$.

Every other fixed-window design

Papers	Spec	A_s	Window	N	ω_c
79 Ba	$\omega_p = 0.2\pi, \omega_s = 0.45\pi$	51 dB	Hamming	29	0.325π
71 Ch, 69 Ch	$\omega_p = 0.3\pi, \omega_s = 0.5\pi$	40 dB	Hann	33	0.4π
74 Ash	$\omega_p = 0.2\pi, \omega_s = 0.5\pi$	41 dB	Hann	23	0.35π
79 Ch, 75 Bh	$\omega_p = 0.24\pi, \omega_s = 0.54\pi$	42.2 dB	Hann	23	0.39π
67 Mng	$\omega_p = 0.35\pi, \omega_s = 0.45\pi$	54 dB	Blackman	111	0.4π
76 Ch, 81 Ch, 73 Ma	$\delta_p = 0.101, \delta_s = 0.01,$ $\omega_p = 0.2\pi, \omega_s = 0.4\pi$	40 dB	Hann	33	0.3π
82 Bh	2 kHz, 5 kHz at $f_s = 20$ kHz $(\omega_p = 0.2\pi, \omega_s = 0.5\pi)$	42 dB	Hann	23	0.35π
82 Ba	–3 dB at 30π , 50 dB at 45π rad/s, $f_s = 100$ Hz ($0.3\pi, 0.45\pi$)	50 dB	Hamming	45	0.375π
70 Asa	Hann named, $0.25\pi, 0.35\pi$, main lobe $8\pi/M$	–	Hann (given)	81	0.3π
69 Bh	Hann named, $0.25\pi, 0.3\pi$	–	Hann (given)	125	0.275π
68 Bh	Blackman named, $0.24\pi, 0.34\pi$, main lobe $12\pi/M$	–	Blackman (given)	121	0.29π

67 Mng is the marginal case, and that is the question It asks for "at least 54 dB". Hamming gives 53, one decibel short, so it is not allowed and Blackman is forced. The cost is brutal: Hamming would have needed $M = 6.6\pi/0.1\pi = 66$, so $N = 67$, while Blackman needs $M = 11\pi/0.1\pi = 110$ and $N = 111$. **One decibel of specification nearly doubles the filter.** That step, not a slope, is the whole point of the window table, and saying so is worth more than the arithmetic.

70 Asa and 68 Bh hand you a main-lobe width instead of a transition figure They say "main lobe width $8\pi/M$ " (Hann) and " $12\pi/M$ " (Blackman) and expect you to use *that* for the length: $M = 8\pi/\Delta\omega$ and $12\pi/\Delta\omega$ rather than the exact 6.2 and 11. For 70 Asa, $\Delta\omega = 0.1\pi$ gives $M = 80$, so $N = 81$; for 68 Bh, $\Delta\omega = 0.1\pi$ gives $M = 120$, so $N = 121$. **Use the constant the paper gives you**, and note in one line that the exact figures would give 63 and 111. Both conventions are in the textbooks and the marks follow the paper.

2 Short designs with the window named

PIN 3/45 80 Bh 10, 73 Bh 2+7, 76 Bh 1+2+7 · no window choice and no length choice: both are given

QUESTION AS PRINTED 2080 Bhadra · [10]

Design a symmetric linear-phase FIR low pass filter whose desired frequency response is $H_d(\omega) = e^{-j\omega\tau}$ for $|\omega| \leq \omega_c$ and 0 elsewhere. The length of the filter should be 7 and $\omega_c = 1$ rad/sample. Use the Hanning window.

- $N = 7$, so $\alpha = \tau = \frac{N-1}{2} = 3$, and the design is just "evaluate seven numbers twice and multiply".

$$h_d[n] = \frac{\sin(1 \cdot (n-3))}{\pi(n-3)}, \quad h_d[3] = \frac{1}{\pi} = 0.31831$$

$$w_{\text{Hann}}[n] = 0.5 - 0.5 \cos \frac{2\pi n}{6}$$

n	0	1	2	3	4	5	6
$h_d[n]$	0.01497	0.14472	0.26785	0.31831	0.26785	0.14472	0.01497
Hann $w[n]$	0	0.25	0.75	1	0.75	0.25	0
$h[n]$	0	0.03618	0.20089	0.31831	0.20089	0.03618	0
Blackman $w[n]$	0	0.13	0.63	1	0.63	0.13	0
$h[n]$	0	0.01881	0.16874	0.31831	0.16874	0.01881	0

- **73 Bh** is the same question with Hann, and **76 Bh** the same with **Blackman**, so the last two rows are that answer.
- Both windows kill the end taps, so a "7-tap" filter really has **5 non-zero coefficients**. **Worth one line:** it is why Hann and Blackman waste two taps on short filters, and why the rectangular window is not always wrong.
- Every row is symmetric about $n = 3$, confirming linear phase with a group delay of exactly 3 samples.

73 Bh and 76 Bh also ask "what is windowing, list the key points". Answer in four: **truncation is unavoidable** because h_d is infinite; truncating abruptly causes **Gibbs ripple**; a tapered window trades **ripple for transition width**; and the window is chosen by **attenuation**, the length by **transition width**.

3 Kaiser window designs

TOP 13/45 81 Ba **10**, 81 Bh **8**, 80 Bh **2+2+2**, 80 Ba **2+8**, 79 Bh **2+8**, 78 Bh **8**, 74 Ch **9+3**, 75 Ash **8+4**, 72 Ka **6+8**, 76 Ash **2+4+4**, 73 Ch **8**, 78 Ch **8**, 66 Ma

QUESTION AS PRINTED 2081 Baishakh · [10]

Design a linear phase FIR filter using the Kaiser window for $0.99 \leq |H(e^{j\omega})| \leq 1.01$ for $0 \leq |\omega| \leq 0.3\pi$ and $|H(e^{j\omega})| \leq 0.01$ for $0.35\pi \leq |\omega| \leq \pi$.

Step 1: read the ripples off the inequalities

- Passband 1 ± 0.01 , so $\delta_p = 0.01$. Stopband ≤ 0.01 , so $\delta_s = 0.01$.

$$\delta = \min(\delta_p, \delta_s) = 0.01, \quad A = -20 \log_{10}(0.01) = \mathbf{40 \text{ dB}}$$

Step 2: the shape parameter

- $21 < A \leq 50$, so the middle branch applies:

$$\beta = 0.5842(40 - 21)^{0.4} + 0.07886(40 - 21) = 0.5842(3.2474) + 1.4983 = \mathbf{3.3953}$$

Step 3: the length, and the window

- $\Delta\omega = 0.35\pi - 0.3\pi = 0.05\pi = 0.15708$.

$$M = \frac{A - 8}{2.285 \Delta\omega} = \frac{32}{2.285(0.15708)} = 89.15 \Rightarrow M = 90, \quad \boxed{N = 91}$$

- $\omega_c = \frac{0.3\pi + 0.35\pi}{2} = 0.325\pi$, and $\alpha = 45$.

$$w[n] = \frac{I_0\left(3.3953\sqrt{1 - \left(\frac{2n}{90} - 1\right)^2}\right)}{I_0(3.3953)}, \quad h[n] = h_d[n]w[n]$$

- **β and N came from different specifications**, which is the whole advantage over a fixed window: the attenuation did not force a longer filter.

Every other Kaiser paper

Eight of the ten share $\delta = 0.01$, so they share $A = 40 \text{ dB}$ and $\beta = 3.3953$ exactly. Only the transition width, and therefore N , changes. Learn one calculation.

Papers	Spec	δ	A	β	N
81 Ba, 81 Bh	$0.3\pi / 0.35\pi, \delta = 0.01$	0.01	40	3.3953	91
80 Bh	$0.19\pi / 0.21\pi, \delta_p = 0.05$	0.01	40	3.3953	225
80 Ba	$0.016\pi / 0.08\pi$	0.01	40	3.3953	71
79 Bh	$0.2\pi / 0.4\pi, \delta_p = 0.101$	0.01	40	3.3953	25
78 Bh, 74 Ch	band pass, $0.25\pi / 0.35\pi$ and $0.6\pi / 0.65\pi$	0.01	40	3.3953	47
75 Ash, 72 Ka	$0.19\pi / 0.21\pi$	0.01	40	3.3953	225
76 Ash	$0.16\pi / 0.18\pi$	0.01	40	3.3953	225
73 Ch	$0.09\pi / 0.14\pi, \delta_p = 0.02$	0.01	40	3.3953	91
78 Ch	$0.19\pi / 0.21\pi$, both $\delta = 0.02$	0.02	33.98	2.6523	183
66 Ma	$0.25\pi / 0.65\pi$, both $\delta = 0.035$	0.035	29.12	1.9903	9

66 Ma supplies a Bessel table, and it is mislabelled The paper prints a table headed $J_0(x)$ with values 1, 1.4826, 1.8926, 2.0508, 2.1675, 2.1718, 2.2642, 2.2796 at $x = 0, 1.3165, 1.7237, 1.8455, 1.9271, 1.93, 1.9903, 2$. **Every one of those eight is $I_0(x)$** , the **modified** Bessel function, not J_0 : the ordinary J_0 oscillates and never exceeds 1, while these climb past 2. It is the right table with the wrong name. **And the table gives the answer away:** $\delta = 0.035$ gives $A = 29.12$ dB and $\beta = \mathbf{1.9903}$, which is one of its own x entries, so $I_0(\beta) = 2.2642$ is supplied ready to use. The remaining entries are the x values needed for $n = 0 \dots 8$ of a 9-tap window.

73 Ch prints a passband that swallows its own stopband It gives $0.98 \leq |H| \leq 1.02$ for $0 \leq \omega \leq 0.9\pi$, then $|H| \leq 0.01$ for $0.14\pi \leq \omega \leq \pi$. **The passband edge is larger than the stopband edge**, so the two overlap and no filter exists. Read 0.9π as **0.09 π** : the transition 0.09π to 0.14π is then sensible and gives $\Delta\omega = 0.05\pi$, $\beta = 3.3953$ and $N = 91$. Note the misprint in a line and design the sensible filter.

78 Bh and 74 Ch are band pass, not low pass Their spec has three bands: stop up to 0.25π , pass from 0.35π to 0.6π , stop from 0.65π . **Use the narrower of the two transitions**, here both 0.1π , so $\Delta\omega = 0.1\pi$ and $N = 47$. The ideal response is the band-pass one, $h_d[n] = \frac{\sin(\omega_2(n - \alpha)) - \sin(\omega_1(n - \alpha))}{\pi(n - \alpha)}$ with $\omega_1 = 0.3\pi$ and $\omega_2 = 0.625\pi$, each edge again placed in the middle of its transition.

4 The bookwork, which is a third of the chapter

TOP 31/45 · Gibbs **TOP 8/45** and Remez **TOP 23/45** · **175 marks of Remez alone, and not one calculation**

- **Gibbs** is asked in eight papers for 3 to 6 marks. Three sentences: truncating with a rectangular window convolves the ideal response with a kernel whose sidelobes are only 13 dB down; the result overshoots the band edge by about **9 %** with ringing either side; and **lengthening the filter does not reduce the overshoot**, only narrows the ripples, so the cure is a tapered window.
- **Remez** is asked in twenty-three, for 4 to 10 marks, and almost always wants **the flowchart**. Learn the seven boxes in §5.5 of the theory: input, initial guess of $r + 2$ extremal frequencies, solve for δ , interpolate and evaluate $E(\omega)$ on a dense grid, exchange the extrema, test against the alternation theorem, output.
- The three "why is it better" marks: **it wastes no tolerance**, so it gives a **shorter filter** for the same spec; it allows **different passband and stopband ripple** through the weight $W(\omega)$; and it ignores the transition band instead of constraining it.
- **Write $\min_h \max_\omega |W(\omega)[H_d(\omega) - H(\omega)]|$ once** and the "mathematical expression" marks are collected.

Ch 6 · IIR Filter Design

Numerical Problems & Solutions

47 of the 50 questions are one of three calculations and the other 3 are the spectral transformation, which is also below. Every design paper in the 45 is here: 32 bilinear Butterworth, 4 bilinear Chebyshev, 11 impulse invariance (two of them sharing a statement with a bilinear paper), 3 spectral transformation.

The seven steps, used by every problem in §1, §2 and §3

1. Get ω_p, ω_s in rad/sample. If the paper gives Hz, use $\omega = 2\pi f/f_s$. If it gives them in π already, do nothing.
2. Get α_p, α_s in positive dB. From a gain, $\alpha = -20 \log_{10} |H|$. From a ripple δ_p , the edge gain is $1 - \delta_p$ first. A deviation already in dB is used as printed.
3. Translate to the analog domain, by the rule of the method: **bilinear** $\Omega = \frac{2}{T} \tan \frac{\omega}{2}$ (pre-warp), **impulse invariance** $\Omega = \omega/T$.
4. Order $N = \left\lceil \frac{\log [(10^{0.1\alpha_s} - 1)/(10^{0.1\alpha_p} - 1)]}{2 \log (\Omega_s/\Omega_p)} \right\rceil$, and cut-off $\Omega_c = \Omega_p/(10^{0.1\alpha_p} - 1)^{1/2N}$.
5. Poles $s_k = \Omega_c e^{j\pi(N+2k+1)/2N}$, $k = 0 \dots N-1$, then $H_a(s) = \Omega_c^N / \prod_k (s - s_k)$, written as real quadratic sections.
6. Map back: substitute $s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$, or expand in partial fractions and use $\frac{1}{s-p_k} \rightarrow \frac{1}{1-e^{p_k T} z^{-1}}$.
7. Check $|H(e^{j\omega_p})|$ and $|H(e^{j\omega_s})|$ against the specification. Two substitutions, and they catch almost every slip.

Checks that cost nothing

1. $H(z)$ from a bilinear design has all its zeros at $z = -1$, so its numerator is $K(1+z^{-1})^N$ and its coefficients are K times the binomials $1, N, \binom{N}{2}, \dots$. **Anything else is an algebra error.**
2. $|H(e^{j0})| = 1$ exactly for a Butterworth design, either method. Set $z = 1$, i.e. add up the coefficients: $\sum b_k / \sum a_k = 1$.
3. $|H(e^{j\omega_p})|$ must come out at exactly $10^{-\alpha_p/20}$, because Ω_c was chosen to make the passband exact.
4. Every pole must satisfy $|z| < 1$, and they come in conjugate pairs, so all the printed coefficients are real.

The three shortcuts worth taking in the exam

1. **Set $T = 1$ in any bilinear design and say why.** The $2/T$ from the pre-warp and the $2/T$ from the substitution cancel, so $H(z)$ does not depend on T (§6.5 of the theory). Nine papers give a sampling frequency that changes nothing.
2. **Do not expand $\prod (s - s_k)$ into a single polynomial** unless asked. Leave $H_a(s)$ as a product of quadratics; the bilinear substitution then runs one quadratic at a time and the arithmetic stays small.
3. For $N = 2$ use the tabulated normalised form $H_a(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$ with $s \rightarrow s/\Omega_c$; for $N = 3$, $\frac{1}{(s+1)(s^2+s+1)}$. **Both are just the pole formula pre-computed.**

1 Bilinear transformation, Butterworth prototype

TOP 32/45 80 Bh 12, 81 Bh 12+3, 81 Ba 2+10, 75 Ch, 70 Ch 10, 82 Ba 12+3, 78 Ch 10, 73 Ch 11+4, 73 Shr 12, 80 Ba 11, 79 Ba 11+3, 79 Ch 15, 72 Ash 12, 73 Bh 11, 71 Bh 9, 72 Ka 15, 71 Shr 15, 77 Ch 10, 73 Ma 11, 79 Bh, 71 Ch, 69 Ch 11+4, 76 Ash 12, 74 Ash 15, 66 Ma 15, 68 Bh 18, 75 Ash 12, 74 Bh 9, 76 Bh 1+10, 76 Ch 10, 70 Bh 9, 82 Bh 3+12 · 405 marks, the single biggest question in the subject

QUESTION AS PRINTED 2079 Bhadra · 2071 Chaitra · 2069 Chaitra · [11+4]

Design a digital Butterworth low pass filter using the bilinear transformation method for $W_p = 0.25\pi$ rad, $W_s = 0.55\pi$ rad, $\delta_p = 0.11$, $\delta_s = 0.21$, sampling frequency 0.5 Hz. Then convert the obtained digital low-pass filter to a high-pass filter with new pass band frequency $W'_p = 0.45\pi$ using the digital domain transformation.

Step 1: the specification in dB

- $\delta_p = 0.11$ is a **deviation**, so the passband edge gain is $1 - 0.11 = 0.89$. $\delta_s = 0.21$ is already a **gain**.

$$\alpha_p = -20 \log_{10} 0.89 = \mathbf{1.0122 \text{ dB}}, \quad \alpha_s = -20 \log_{10} 0.21 = \mathbf{13.5556 \text{ dB}}$$

- Sampling frequency 0.5 Hz means $T = 2$ s. **It will cancel**, so take $T = 1$ and note it. (Working it through with $T = 2$ gives the identical $H(z)$; only the intermediate Ω values change, by a factor of 2.)

Step 2: pre-warp the band edges

$$\Omega_p = 2 \tan \frac{0.25\pi}{2} = 2 \tan(22.5^\circ) = \mathbf{0.828427}, \quad \Omega_s = 2 \tan \frac{0.55\pi}{2} = 2 \tan(49.5^\circ) = \mathbf{2.341699}$$

- **This is the step that is skipped**, and skipping it puts the finished band edges in the wrong place.

Step 3: order and cut-off

$$10^{0.1\alpha_p} - 1 = 10^{0.10122} - 1 = 0.262467, \quad 10^{0.1\alpha_s} - 1 = 10^{1.35556} - 1 = 21.675737$$

$$N \geq \frac{\log_{10}(21.675737/0.262467)}{2 \log_{10}(2.341699/0.828427)} = \frac{\log_{10} 82.5847}{2 \log_{10} 2.82668} = \frac{1.91690}{2(0.45128)} = 2.1239$$

$$\boxed{N = 3} \quad (\text{round up, always})$$

$$\Omega_c = \frac{\Omega_p}{(10^{0.1\alpha_p} - 1)^{1/2N}} = \frac{0.828427}{(0.262467)^{1/6}} = \frac{0.828427}{0.800164} = \mathbf{1.035321}$$

Step 4: the poles, and $H_a(s)$

- $N = 3$, so three left-half-plane poles at radius Ω_c , spaced $\pi/3$, at angles $\pi \frac{3+2k+1}{6}$ for $k = 0, 1, 2$, i.e. $120^\circ, 180^\circ, 240^\circ$:

k	Angle	$s_k = \Omega_c e^{j\theta}$	Factor
0	120°	$-0.517661 + j0.896615$	$s^2 + 1.035321s + 1.071891$
2	240°	$-0.517661 - j0.896615$	
1	180°	-1.035321	$s + 1.035321$

- The conjugate pair multiplies to a real quadratic: sum of roots -1.035321 , and product $= \Omega_c^2 = 1.071891$. **For a Butterworth filter every quadratic section has constant term Ω_c^2** , which is a free check.

$$H_a(s) = \frac{\Omega_c^3}{(s + 1.035321)(s^2 + 1.035321s + 1.071891)} = \frac{1.109751}{(s + 1.035321)(s^2 + 1.035321s + 1.071891)}$$

Step 5: the bilinear substitution

- Put $s = 2 \frac{1 - z^{-1}}{1 + z^{-1}}$ into each factor, multiplying numerator and denominator of the whole expression by $(1 + z^{-1})^3$:

$$s + 1.035321 \rightarrow \frac{2(1 - z^{-1}) + 1.035321(1 + z^{-1})}{1 + z^{-1}} = \frac{3.035321 - 0.964679z^{-1}}{1 + z^{-1}}$$

$$\begin{aligned} s^2 + 1.035321s + 1.071891 &\rightarrow \frac{4(1 - z^{-1})^2 + 2.070642(1 - z^{-2}) + 1.071891(1 + z^{-1})^2}{(1 + z^{-1})^2} \\ &= \frac{7.142534 - 5.856219z^{-1} + 3.001248z^{-2}}{(1 + z^{-1})^2} \end{aligned}$$

- Multiply the two denominators:

$$21.679886 - 24.665755z^{-1} + 14.759120z^{-2} - 2.895239z^{-3}$$

- Divide through by the constant term 21.679886, and the $(1 + z^{-1})^3$ left over on top carries Ω_c^3 , giving $K = 1.109751/21.679886 = 0.051188$:

$$\begin{aligned} H(z) &= \frac{0.051188 (1 + z^{-1})^3}{1 - 1.137725z^{-1} + 0.680775z^{-2} - 0.133545z^{-3}} \\ &= \frac{0.051188 + 0.153564z^{-1} + 0.153564z^{-2} + 0.051188z^{-3}}{1 - 1.137725z^{-1} + 0.680775z^{-2} - 0.133545z^{-3}} \end{aligned}$$

- **The numerator coefficients are $0.051188 \times (1, 3, 3, 1)$** . If they are not the binomials, the substitution went wrong.

Step 6: check, before writing anything else

- At $z = 1$ every z^{-1} becomes 1, so $H(1)$ is just the sum of the numerator coefficients over the sum of the denominator's:

$$H(1) = \frac{0.051188(1+3+3+1)}{1-1.137725+0.680775-0.133545} = \frac{0.409504}{0.409505} = \mathbf{1.0000}$$

✓ **Two additions, and they catch a wrong K instantly.**

- $|H(e^{j0.25\pi})| = \mathbf{0.8900}$, which is $1 - \delta_p$ **exactly**, as the choice of Ω_c guaranteed. ✓
- $|H(e^{j0.55\pi})| = \mathbf{0.0861} \leq 0.21$. ✓ **Comfortably over-satisfied**, which is what rounding N up from 2.12 to 3 bought.

Step 7: the low pass to high pass conversion (the 4 marks)

- Prototype edge $\omega_p = 0.25\pi$, new edge $\omega'_p = 0.45\pi$:

$$\alpha = -\frac{\cos \frac{\omega_p + \omega'_p}{2}}{\cos \frac{\omega_p - \omega'_p}{2}} = -\frac{\cos(0.35\pi)}{\cos(0.10\pi)} = -\frac{0.45399}{0.95106} = \mathbf{-0.47735}$$

- Substitute $z^{-1} \rightarrow -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}} = -\frac{z^{-1} - 0.47735}{1 - 0.47735z^{-1}}$ into $H(z)$ above:

$$H_{hp}(z) = \frac{0.276238(1 - z^{-1})^3}{1 - 0.683740z^{-1} + 0.457627z^{-2} - 0.068533z^{-3}}$$

$$= \frac{0.276238 - 0.828713z^{-1} + 0.828713z^{-2} - 0.276238z^{-3}}{1 - 0.683740z^{-1} + 0.457627z^{-2} - 0.068533z^{-3}}$$

- Checks: $|H_{hp}(e^{j0})| = \mathbf{0}$ exactly, because every zero moved from $z = -1$ to $z = +1$; and $|H_{hp}(e^{j0.45\pi})| = \mathbf{0.8900}$, the same $1 - \delta_p$ the prototype had at its own edge. **Both checks are free and both are worth writing down.**

Every other bilinear Butterworth paper

Same seven steps, different numbers. **T is taken as 1 throughout**, since it cancels, so a paper's own sampling frequency changes the Ω_c column but not the $H(z)$ column. Ω_p, Ω_s are the *pre-warped* edges. Every numerator is $K(1 + z^{-1})^N$ with $K = (\sum a_k)/2^N$.

Papers	Spec, as printed	α_p	α_s	N	Ω_c	$H(z)$ denominator
80 Bh, 81 Bh	0.9 to 0.5π ; 0.2 from 0.75π	0.9151	13.9794	3	2.54674	$1 + 0.43938z^{-1}$ $+ 0.38450z^{-2}$ $+ 0.041621z^{-3}$
81 Ba	0.6 to 0.35π ; 0.1 from 0.7π	4.4370	20	2	1.06140	$1 - 0.70699z^{-1}$ $+ 0.26136z^{-2}$
75 Ch, 70 Ch, 82 Ba	0.8 to 0.2π ; 0.2 from 0.6π	1.9382	13.9794	2	0.75037	$1 - 1.02819z^{-1}$ $+ 0.36508z^{-2}$
78 Ch	0.82 to 0.22π ; 0.18 from 0.58π	1.7237	14.8945	2	0.86185	$1 - 0.90725z^{-1}$ $+ 0.32103z^{-2}$
73 Shr, 77 Ch, 76 Bh	0.89125 to 0.2π ; 0.17783 from 0.3π (= 1 dB and 15 dB)	1	15	6	0.72729	see the cascade in §1.2

Papers	Spec, as printed	α_p	α_s	N	Ω_c	$H(z)$ denominator
80 Ba	0.26 π , 0.99 dB; −14.99 dB at 0.58 π	0.99	14.99	3	1.08611	$1 - 1.06340z^{-1} + 0.63628z^{-2} - 0.12114z^{-3}$
79 Ba	0.24 π , 0.98 dB; −14.95 dB at 0.57 π	0.98	14.95	3	0.99560	$1 - 1.19710z^{-1} + 0.71861z^{-2} - 0.14406z^{-3}$
79 Ch, 72 Ash	0.24 π , 1 dB; −14.9 dB at 0.57 π	1	14.9	3	0.99186	$1 - 1.20280z^{-1} + 0.72232z^{-2} - 0.14509z^{-3}$
73 Bh	0.25 π , 0.99 dB; −14.85 dB at 0.59 π	0.99	14.85	3	1.03961	$1 - 1.13140z^{-1} + 0.67685z^{-2} - 0.13245z^{-3}$
71 Bh	0.25 π , 1 dB; −15 dB at 0.55 π	1	15	3	1.03767	$1 - 1.13430z^{-1} + 0.67862z^{-2} - 0.13295z^{-3}$
72 Ka	0.25 π , 1 dB; −15 dB at 0.45 π	1	15	4	0.98086	$1 - 1.64760z^{-1} + 1.35630z^{-2} - 0.52524z^{-3} + 0.083384z^{-4}$
71 Shr	0.2 π , 1 dB; −15 dB at 0.4 π	1	15	3	0.81397	$1 - 1.48260z^{-1} + 0.92964z^{-2} - 0.20333z^{-3}$
73 Ma	0.2 π , 0.98 dB; 20 dB at 0.5 π	0.98	20	3	0.81704	$1 - 1.47760z^{-1} + 0.92551z^{-2} - 0.20214z^{-3}$
79 Bh, 71 Ch, 69 Ch	worked in full above	1.0122	13.5556	3	1.03532	$1 - 1.13773z^{-1} + 0.68078z^{-2} - 0.13355z^{-3}$

Papers	Spec, as printed	α_p	α_s	N	Ω_c	$H(z)$ denominator
76 Ash	0.22 π , 0.54 π ; $\delta_p=0.11$, $\delta_s=0.22$	1.0122	13.1515	2	1.00598	$1 - 0.76056z^{-1} + 0.27575z^{-2}$
74 Ash	0.27 π , 0.58 π ; $\delta_p=0.11$, $\delta_s=0.21$	1.0122	13.5556	3	1.12856	$1 - 1.00260z^{-1} + 0.60224z^{-2} - 0.11157z^{-3}$
68 Bh	0.25 π , 0.45 π ; $\delta_p=0.17$, $\delta_s=0.27$	1.6184	11.3727	3	0.94579	$1 - 1.27310z^{-1} + 0.77005z^{-2} - 0.15836z^{-3}$
75 Ash, 74 Bh	0.7 dB to 0.15 π ; 14 dB from 0.6 π	0.7	14	2	0.74249	$1 - 1.03700z^{-1} + 0.36853z^{-2}$

Every numerator is $K(1 + z^{-1})^N$ with K fixed by $|H(e^{j0})| = 1$, i.e. $K = (\sum a_k)/2^N$. The four papers whose calculation genuinely differs are split out below.

1.1 Three papers whose specification is given in Hz

HOT 3/45 76 Ch **10**, 82 Bh **3+12**, 70 Bh **9** · only step 1 changes

QUESTION AS PRINTED 2076 Chaitra · [10]

Design a digital Butterworth low pass filter using the bilinear transformation method for passband peak-to-peak ripple ≤ 1 dB and passband edge frequency 1.2 kHz; stopband attenuation ≥ 40 dB and stopband edge frequency 2.5 kHz; sample rate 8 kHz.

$$\omega_p = \frac{2\pi(1200)}{8000} = \mathbf{0.3\pi}, \quad \omega_s = \frac{2\pi(2500)}{8000} = \mathbf{0.625\pi}$$

- Pre-warp: $\Omega_p = 2 \tan(0.15\pi) = 1.01905$, $\Omega_s = 2 \tan(0.3125\pi) = 2.99321$.
- $\alpha_p = 1$, $\alpha_s = 40$. $N = 4.9010 \Rightarrow \boxed{N = 5}$, $\Omega_c = 1.16648$.
- Sections of $H_a(s)$: $(s + 1.16648)$, $(s^2 + 0.72093s + 1.36068)$, $(s^2 + 1.88741s + 1.36068)$, constant term $\Omega_c^2 = 1.36068$ in both quadratics.

$$H(z) = \frac{0.010975(1 + z^{-1})^5}{(1 - 0.26323z^{-1})(1 - 0.77598z^{-1} + 0.57608z^{-2})(1 - 0.57782z^{-1} + 0.17359z^{-2})}$$

- Check: $|H(e^{j0.3\pi})| = 0.89125$ (exactly −1 dB) and $|H(e^{j0.625\pi})| = 0.00899 \leq 0.01$. ✓

82 Bh 3+12

350 Hz and 1000 Hz, -3 dB and -10 dB, $f_s = 5000$ Hz, plus "explain the frequency warping effect in detail" (§6.4 of the theory).

- $\omega_p = 2\pi(350)/5000 = 0.14\pi$, $\omega_s = 2\pi(1000)/5000 = 0.4\pi$; $\Omega_p = 0.44705$, $\Omega_s = 1.45309$; $\alpha_p = 3$, $\alpha_s = 10$.
- $N = 0.9340 \Rightarrow \boxed{N = 1}$, $\Omega_c = 0.44812$. **The smallest filter in the chapter:** one real pole, no quadratic section at all.

$$H_a(s) = \frac{0.44812}{s + 0.44812} \Rightarrow H(z) = \frac{0.183045(1 + z^{-1})}{1 - 0.633910z^{-1}}$$

- Check $|H(e^{j0.14\pi})| = 0.70795 \approx 10^{-3/20} = 0.70795$. ✓ With $\alpha_p = 3$ dB the passband edge is the 3 dB point, so $\Omega_c \approx \Omega_p$ and the order formula barely clears 1.

70 Bh 9

defective as printed

QUESTION AS PRINTED 2070 Bhadra · [9]

Design a digital low pass Butterworth filter by applying bilinear transformation technique for: pass band edge = 120 Hz, pass band attenuation = 1 dB, stop band edge = 170 Hz, stop band attenuation = 16 dB. Assume sampling frequency of 256 Hz.

- With $f_s = 256$ Hz the **Nyquist frequency is 128 Hz**. The stopband edge of 170 Hz is **above it**, so it does not exist as a digital frequency: $\omega_s = 2\pi(170)/256 = 1.3281\pi > \pi$.
- Pre-warping it gives $\tan(0.664\pi) < 0$, a **negative** Ω_s , and the order formula takes the logarithm of a negative number. **No filter meets this specification, because the specification is not a specification.**
- Aliased, 170 Hz lands at $256 - 170 = 86$ Hz, which is *below* the 120 Hz passband edge. The stopband would sit inside the passband.
- **Say that, then solve the workable reading:** treat the two edges as *analog* frequencies, skip the pre-warp, and map with $T = 1/256$.

$$\Omega_p = 2\pi(120) = 753.98, \quad \Omega_s = 2\pi(170) = 1068.14, \quad N = 7.19 \Rightarrow \boxed{N = 8}, \quad \Omega_c = 820.42$$

- An 8th-order filter for a 9-mark question is itself a sign the paper is wrong. **The marks are in naming the defect**, not in grinding out eight poles.

1.2 The textbook filter: 73 Shr, 77 Ch and 76 Bh

HOT 3/45 73 Shr 12, 77 Ch 10, 76 Bh 1+10 · one spec stated three ways

- 73 Shr prints gains: 0.89125 to 0.2π , 0.17783 from 0.3π . 77 Ch and 76 Bh print the same thing in dB: 1 dB and 15 dB. **They are the same filter**, and it is Oppenheim's own worked example.
- $\Omega_p = 0.64984$, $\Omega_s = 1.01905$, $N = 5.3044 \Rightarrow \boxed{N = 6}$, $\Omega_c = 0.72729$.
- $H_a(s)$, three quadratics, each with constant term $\Omega_c^2 = 0.52895$:

$$H(z) = \frac{0.147996}{(s^2 + 0.37647s + 0.52895)(s^2 + 1.02854s + 0.52895)(s^2 + 1.40502s + 0.52895)} \cdot \frac{0.00057969(1 + z^{-1})^6}{(1 - 1.31432z^{-1} + 0.71490z^{-2})(1 - 1.05406z^{-1} + 0.37532z^{-2})(1 - 0.94592z^{-1} + 0.23422z^{-2})}$$

- Check: $|H(e^{j0.2\pi})| = 0.89125$ exactly, $|H(e^{j0.3\pi})| = 0.13101 \leq 0.17783$. ✓
- **A textbook difference worth knowing.** Oppenheim places the 3 dB point so the *stopband* is met exactly, giving $\Omega_c = 0.766229$ and a different (equally valid) $H(z)$. The local formula meets the *passband* exactly, as above. **State which convention you used** and either is right.
- 76 Bh's extra mark is the Butterworth/Chebyshev comparison: on this very spec Chebyshev needs $N = 4$ against Butterworth's $N = 6$.

1.3 Two papers where the arithmetic collapses

PIN 2/45 73 Ch 11+4, 66 Ma 15

73 Ch 11+4

QUESTION AS PRINTED 2073 Chaitra · [11+4]

Design a digital Butterworth low pass filter using bilinear transformation for $0.707 \leq |H(e^{jw})| \leq 1$ for $0 \leq w \leq \pi/2$; $|H(e^{jw})| \leq 0.2$ for $3\pi/4 \leq w \leq \pi$; $T = 1$ s. Realize the filter using the most convenient realization form.

- 0.707 is $1/\sqrt{2}$, so the passband edge is the **3 dB point** and $\Omega_c = \Omega_p$ with no formula at all:

$$\Omega_p = 2 \tan \frac{\pi}{4} = 2, \quad \Omega_s = 2 \tan \frac{3\pi}{8} = 4.82843, \quad \Omega_c = \Omega_p = 2$$

- $\alpha_p = 3.0103$ dB, $\alpha_s = 13.9794$ dB, $N = 1.8026 \Rightarrow \boxed{N = 2}$.

$$H_a(s) = \frac{\Omega_c^2}{s^2 + \sqrt{2}\Omega_c s + \Omega_c^2} = \frac{4}{s^2 + 2\sqrt{2}s + 4}$$

- Substituting $s = 2 \frac{1-z^{-1}}{1+z^{-1}}$ the whole thing stays **exact**:

$$H(z) = \frac{4(1+z^{-1})^2}{(8+4\sqrt{2}) + (8-4\sqrt{2})z^{-2}} = \frac{\frac{1}{2+\sqrt{2}}(1+z^{-1})^2}{1 + \frac{2-\sqrt{2}}{2+\sqrt{2}}z^{-2}} = \frac{0.292893(1+z^{-1})^2}{1 + 0.171573z^{-2}}$$

- **The z^{-1} term vanishes identically.** That happens only because $\omega_p = \pi/2$ puts the poles at $\pm 135^\circ$, symmetric about the imaginary axis, so their sum is purely imaginary and cancels.
- Taking 0.707 literally instead of $1/\sqrt{2}$ gives $\Omega_c = 1.99970$ and a residual $-0.000177z^{-1}$. **That is rounding, not a term.** Write the exact form.
- **The 4 marks for realization:** with one quadratic section, direct form II is the most convenient, and it needs 2 delays, 3 multipliers (the numerator's 1, 2, 1 are shifts and adds) and 4 adders. Cross-reference chapter 4.

66 Ma 15

$\omega_p = 0.3\pi$, $\omega_s = 0.4\pi$, $\delta_p = 0.11$, $\delta_s = 0.21$, and the paper asks for all five parts by name.

- (a) $\Omega_p = 1.01905$, $\Omega_s = 1.45309$, $\alpha_p = 1.0122$, $\alpha_s = 13.5556$; $N = 6.2199 \Rightarrow \boxed{N = 7}$. **The narrowest transition in the chapter** (0.3π to 0.4π) buys the highest order.
- (b) $\Omega_c = 1.01905/(0.262467)^{1/14} = \mathbf{1.12122}$.
- (c) The paper asks for the poles of the squared magnitude response, which is the full set of $2N = 14$: $s_k = \Omega_c e^{j\pi(2k+1)/14}$ for $k = 0 \dots 13$, equally spaced by $\pi/7 = 25.71^\circ$ on the circle of radius 1.12122. **Half are discarded**; only the 7 with $\sigma < 0$ belong to $H_a(s)$.
- (d) $H_a(s) = \frac{\Omega_c^7}{(s + 1.12122) \prod_{i=1}^3 (s^2 + b_i s + 1.25713)}$ with $b_i = 0.49899, 1.39814, 2.02037$ and $\Omega_c^2 = 1.25713$ throughout.
- (e) $H(z) = \frac{0.0015237(1+z^{-1})^7}{(1-0.28155z^{-1})(1-0.87700z^{-1}+0.68091z^{-2})(1-0.68117z^{-1}+0.30557z^{-2})(1-0.59000z^{-1}+0.13083z^{-2})}$
- Check $|H(e^{j0.3\pi})| = 0.8900$ and $|H(e^{j0.4\pi})| = 0.16074 \leq 0.21$. ✓

2 Impulse invariance

TOP 11/45 78 Bh 12+3, 72 Ch 3+12, 81 Ch, 80 Ch, 69 Bh 12, 70 Asa 13, 74 Ch 12, 67 Mng 15, 69 Bh 15, 73 Shr 3, 70 Ma 3+6 · 133 marks

Steps 1, 4 and 5 are unchanged from §1. Step 3 becomes $\Omega = \omega/T$ with no pre-warp, and step 6 becomes a partial fraction expansion. **T no longer cancels**, so it must be carried.

The pair that answers "compare the two methods" 12

75 Ash and 74 Bh print exactly the same specification as 74 Ch, and ask for the other method. Nothing else in the 45 papers demonstrates the difference so cleanly, so solve them together.

QUESTION AS PRINTED 2075 Ashwin / 2074 Bhadra (bilinear) · 2074 Chaitra (impulse invariance) · [12] [9] [12]

Design a digital Butterworth low pass filter whose passband magnitude is constant to 0.7 dB below the frequency 0.15π and whose stopband attenuation is at least 14 dB for frequencies between 0.6π and π .

Step	75 Ash / 74 Bh: bilinear	74 Ch: impulse invariance
Spec in dB	$\alpha_p = 0.7, \alpha_s = 14$, so $10^{0.07} - 1 = 0.174898$ and $10^{1.4} - 1 = 24.118864$	
Analog edges	pre-warp: $\Omega_p = 2 \tan(0.075\pi) = 0.480158$ $\Omega_s = 2 \tan(0.30\pi) = 2.752764$	linear: $\Omega_p = 0.15\pi = 0.471239$ $\Omega_s = 0.6\pi = 1.884956$
Ω_s/Ω_p	5.733043	4.000000
N exact	1.4106	1.7769
N	2	2
Ω_c	0.742485	0.728694
Poles	$-0.52502 \pm j0.52502$ 0.551284	$-0.51526 \pm j0.51526$ 0.530995
$H_a(s)$	$\frac{s^2 + 1.050033s + 0.551284}{\text{not needed}}$	$\frac{s^2 + 1.030529s + 0.530995}{A_{1,2} = \pm j0.51526}$
Residues	from the substitution	$e^{p_k T} = 0.51979 \mp j0.29435$
z poles	$0.082883(1 + z^{-1})^2$	$0.303336 z^{-1}$
$H(z)$	$\frac{1 - 1.036997z^{-1} + 0.368530z^{-2}}{0.922571 \text{ (spec } \geq 0.922571) \checkmark}$	$\frac{1 - 1.039570z^{-1} + 0.356818z^{-2}}{0.898611 \text{ (spec } \geq 0.922571) \text{ fails}}$
$ H $ at ω_p	0.922571 (spec ≥ 0.922571) $\checkmark$	0.898611 (spec ≥ 0.922571) fails
$ H $ at ω_s	0.072559 (spec ≤ 0.199526) $\checkmark$	0.191753 (spec ≤ 0.199526) barely
$ H $ at DC	1.000000 exactly	0.956147

- **Read the last three rows: that is the whole comparison, in numbers.** The bilinear filter meets the passband exactly and clears the stopband by a factor of three. The impulse-invariant filter of the same order misses the passband edge, scrapes the stopband, and does not even have unity DC gain.
- Every one of those shortfalls is **aliasing**: the analog tail beyond π/T folds back and adds to the response. It is worst where the response is smallest, which is why the stopband suffers most.
- The cure is to **over-design**: raise N or tighten α_s before designing. Nothing can remove the aliasing itself.
- **Note $b_0 = 0$** : the numerator starts at z^{-1} . That is general, not specific: when $H_a(s)$ is strictly proper by two or more degrees, $h[0] = h_a(0) = 0$, so the impulse-invariant filter always has a one-sample delay built in. **It is not a dropped term.**

81 Ch, 80 Ch, 69 Bh 12

$\omega_p = 0.25\pi$ at 0.5 dB; -15 dB at $\omega_s = 0.55\pi$; $f_s = 1$ Hz so $T = 1$.

- $\Omega_p = 0.25\pi = 0.78540$, $\Omega_s = 0.55\pi = 1.72788$ (**no pre-warp**).
- $\alpha_p = 0.5$, $\alpha_s = 15$; $N = 3.5039 \Rightarrow \boxed{N = 4}$, $\Omega_c = 1.02161$.
- $H_a(s) = \frac{\Omega_c^4}{(s^2 + 0.78191s + 1.04369)(s^2 + 1.88770s + 1.04369)}$, constant term $\Omega_c^2 = 1.04369$ in both.

$$H(z) = \frac{0.088750z^{-1} + 0.174840z^{-2} + 0.023521z^{-3}}{1 - 1.513201z^{-1} + 1.180022z^{-2} - 0.449386z^{-3} + 0.069280z^{-4}}$$

- Checks: $|H(e^{j0.25\pi})| = 0.94338$ against a required $10^{-0.5/20} = 0.94406$, so it **misses the passband by 0.0007**; $|H(e^{j0.55\pi})| = 0.11982 \leq 0.17783 \checkmark$; and $|H(e^{j0})| = 1.00139$, **slightly above 1**.
- Both anomalies are aliasing, and both are worth one line in the answer. **A DC gain above 1 is the signature of impulse invariance** and never happens with the bilinear method.

70 Asa 13

$\omega_p = 0.2\pi$ at 0.5 dB; -15 dB at 0.35π ; $T = 1$.

- $\Omega_p = 0.62832$, $\Omega_s = 1.09956$; $N = 4.9367 \Rightarrow \boxed{N = 5}$, $\Omega_c = 0.77542$.
- N odd, so there is a real pole at $-\Omega_c$ and two quadratic sections, $(s + 0.77542)(s^2 + 0.47924s + 0.60128)(s^2 + 1.25466s + 0.60128)$.

$$H(z) = \frac{0.0069166z^{-1} + 0.044472z^{-2} + 0.027021z^{-3} + 0.0015381z^{-4}}{1 - 2.584442z^{-1} + 2.999693z^{-2} - 1.857044z^{-3} + 0.603063z^{-4} - 0.081324z^{-5}}$$

- $|H(e^{j0.2\pi})| = 0.94403$ (need 0.94406), $|H(e^{j0.35\pi})| = 0.17189$ (need ≤ 0.17783), $|H(e^{j0})| = 1.00002$. **The same three aliasing signatures**, smaller because Ω_s/Ω_p is smaller here.

78 Bh, 72 Ch 12+3

200 Hz and 500 Hz, 5 dB and 12 dB, $f_s = 5000$ Hz. **The only impulse-invariance paper where $T \neq 1$** , so the one where the convention matters. 78 Bh's rider is pre-warping, 72 Ch's is the bilinear/impulse-invariance comparison.

- $T = 1/5000$. $\Omega_p = 2\pi(200) = 1256.64$, $\Omega_s = 2\pi(500) = 3141.59$ (rad/s, **not** divided by anything: $\Omega = \omega/T = 2\pi f$).
- $10^{0.5} - 1 = 2.1623$, $10^{1.2} - 1 = 14.849$, **ratio 6.8672**; $N = 1.0514 \Rightarrow \boxed{N = 2}$, $\Omega_c = 1036.29$.
- Poles $-732.769 \pm j732.769$; residues $A_{1,2} = \mp j732.769$; $e^{p_k T} = 0.854421 \mp j0.126123$.

$$H(z) = \frac{184.838 z^{-1}}{1 - 1.708842z^{-1} + 0.745942z^{-2}} \quad (\text{Proakis, } h[n] = h_a(nT))$$

$$H(z) = \frac{0.0369676 z^{-1}}{1 - 1.708842z^{-1} + 0.745942z^{-2}} \quad (\text{Oppenheim, } h[n] = T h_a(nT))$$

- **The first has a DC gain of 4982.** That is $\approx 1/T = 5000$, and it is exactly why the second convention exists. The poles are identical; only the numerator scales.
- With the T scaling: $|H(e^{j0.08\pi})| = 0.5633$ against a required $10^{-5/20} = 0.5623$ ✓ (by a hair), and $|H(e^{j0.2\pi})| = 0.1114$ against $10^{-12/20} = 0.2512$ ✓.
- With $T = 1$ the two conventions coincide, which is why the other eight impulse-invariance papers never raise the question. **Here you must say which you used.**

67 Mng 15

QUESTION AS PRINTED 2067 Mangsir · [15]

Find the system function for a digital filter using the impulse invariance technique from the analog Butterworth filter transfer function $H(s) = \frac{1}{(s+1.3)(s-1.3e^{j2\pi/3})(s-1.3e^{-j2\pi/3})}$ with $T = 1$ second, and draw the block diagram of the system function $H(z)$ realized in terms of second order sections.

- **Recognise it before computing.** $1.3e^{\pm j2\pi/3} = -0.65 \pm j1.12583$, so the three poles sit at $120^\circ, 180^\circ, 240^\circ$ on a circle of radius 1.3. **This is a third-order Butterworth filter with $\Omega_c = 1.3$** , already designed. Steps 1 to 4 are done for you.
- Expanded: $H_a(s) = \frac{1}{s^3 + 2.6s^2 + 3.38s + 2.197}$, and $2.197 = 1.3^3$, which confirms the reading.
- Residues $A_k = 1/\prod_{j \neq k} (p_k - p_j)$:

Pole p_k	Residue A_k	z pole $e^{p_k T}, T = 1$
-1.30000	+0.591716	0.272532
-0.65000 + $j1.12583$	-0.295858 + $j0.170814$	0.224701 - $j0.471212$
-0.65000 - $j1.12583$	-0.295858 - $j0.170814$	0.224701 + $j0.471212$

- The residues sum to zero, which is the general reason $b_0 = 0$ below.

$$\begin{aligned} H(z) &= \frac{0.591716}{1 - 0.272532z^{-1}} + \frac{-0.591716 + 0.293938z^{-1}}{1 - 0.449403z^{-1} + 0.272532z^{-2}} \\ &= \frac{0.189281z^{-1} + 0.081154z^{-2}}{1 - 0.721935z^{-1} + 0.395008z^{-2} - 0.074274z^{-3}} \end{aligned}$$

- **The block diagram the paper asks for is the parallel form**, and it falls out of the partial fraction expansion with no extra work: one first-order section and one second-order section, both fed by $x[n]$, their outputs summed.
- **Draw it as two direct-form-II boxes in parallel:** the real section has one delay and one feedback multiplier 0.272532; the quadratic section has two delays, feedback multipliers 0.449403 and -0.272532, and feedforward multipliers -0.591716 and 0.293938.
- Why parallel and not cascade: impulse invariance *produces* a sum of terms. Forcing it into a cascade means factorising the numerator for no reason. Cross-reference chapter 4 for the structures themselves.

3 Bilinear transformation, Chebyshev prototype

HOT 4/45 72 Ma **11**, 74 Ma **9**, 70 Ma **9**, 75 Bh **10** · **39 marks, all four on the EX 753 papers**

Steps 1, 2, 6 and 7 are identical to §1. **Only steps 3 to 5 change:** a different order formula, $\Omega_c = \Omega_p$ with no formula, and poles on an ellipse instead of a circle.

QUESTION AS PRINTED 2072 Magh · [11]

Design a low pass discrete time Chebyshev filter by applying bilinear transformation for $0.707 \leq |H(e^{jw})| \leq 1$ for $|w| \leq 0.2\pi$ and $|H(e^{jw})| \leq 0.1$ for $0.5\pi \leq |w| \leq \pi$.

Step 1 to 3: spec, pre-warp, ripple factor

- $\alpha_p = -20 \log_{10} 0.707 = 3.0116$ dB, $\alpha_s = -20 \log_{10} 0.1 = 20$ dB.
- $\Omega_p = 2 \tan(0.1\pi) = \mathbf{0.64984}$, $\Omega_s = 2 \tan(0.25\pi) = \mathbf{2}$ exactly.

$$\varepsilon = \sqrt{10^{0.1\alpha_p} - 1} = \sqrt{10^{0.30116} - 1} = \mathbf{1.00030} \approx 1$$

- $\varepsilon = 1$ **because 0.707 is the 3 dB point**, the same coincidence 73 Ch exploits in §1.3.
- $\Omega_c = \Omega_p = 0.64984$. A Chebyshev filter ripples right up to the passband edge, so there is no separate cut-off to compute.

Step 4: order, by the $\cosh^{-1}$ formula

$$N \geq \frac{\cosh^{-1} \sqrt{(10^2 - 1)/(10^{0.30116} - 1)}}{\cosh^{-1}(2/0.64984)} = \frac{\cosh^{-1}(9.94687)}{\cosh^{-1}(3.07769)} = 1.6694$$

$$\boxed{N = 2}$$

- Use $\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$ if the calculator lacks the key: $\ln(19.8433) = 2.98787$ and $\ln(5.9884) = 1.78982$.
- **Compare: a Butterworth filter on this same spec needs $N = 3$.** That one saved order is the whole argument for Chebyshev.

Step 5: the pole ellipse

$$\alpha = \frac{1}{\varepsilon} + \sqrt{1 + \frac{1}{\varepsilon^2}} = 1.0003 + 1.4145 = \mathbf{2.41370}$$

$$a = \frac{1}{2}(\alpha^{1/2} - \alpha^{-1/2}) = \mathbf{0.45497}, \quad b = \frac{1}{2}(\alpha^{1/2} + \alpha^{-1/2}) = \mathbf{1.09864}$$

- Semi-axes: $a\Omega_c = 0.29566$ (real direction), $b\Omega_c = 0.71394$ (imaginary direction). **Taller than wide, always.**
- $N = 2$, so $\theta_k = \frac{(2k+1)\pi}{4} = 45^\circ, 135^\circ$:

$$s_k = -a\Omega_c \sin \theta_k + j b\Omega_c \cos \theta_k = \mathbf{-0.20906 \pm j0.50483}$$

$$H_a(s) = \frac{K}{s^2 + 0.418125s + 0.298560}, \quad K = \frac{|s_k|^2}{\sqrt{1 + \varepsilon^2}} = \frac{0.298560}{1.41443} = \mathbf{0.211082}$$

- **The $\sqrt{1 + \varepsilon^2}$ divisor is because N is even.** Omit it and the passband sits a factor of 1.414 too high everywhere.

Step 6 and 7: map back and check

$$\boxed{H(z) = \frac{0.041108 (1 + z^{-1})^2}{1 - 1.441705z^{-1} + 0.674282z^{-2}}}$$

- $|H(e^{j0.2\pi})| = \mathbf{0.70700}$, exactly the specified 0.707. ✓
- $|H(e^{j0.5\pi})| = \mathbf{0.05563} \leq 0.1$. ✓
- $|H(e^{j0})| = \mathbf{0.70700}$, **not 1**. The passband oscillates between 0.707 and 1 and, with N even, it **starts at the bottom of the ripple**. This is correct and looks wrong: say so in the answer.

The other three Chebyshev papers

Paper	Spec	ϵ	N	Poles	$H(z)$
74 Ma	0.2π , 1 dB; -15 dB at 0.3π ; $T=1$	0.50885	4	$-0.0907 \pm j0.6390$ $-0.2189 \pm j0.2647$	$K = 0.0018356$, numerator $K(1 + z^{-1})^4$, denominator $1 - 3.054340z^{-1}$ $+ 3.828999z^{-2}$ $- 2.292452z^{-3}$ $+ 0.550745z^{-4}$
70 Ma	0.25π , 0.55π ; $\alpha_{\max}=1.01$, $\alpha_{\min}=13.55$; $f_s=0.5$ Hz	0.51169	2	$-0.45286 \pm j0.74042$	$K = 0.102154$, numerator $K(1 + z^{-1})^2$, denominator $1 - 0.989132z^{-1}$ $+ 0.448133z^{-2}$

- 74 Ma is the §1.2 spec again ($0.2\pi/0.3\pi$, 1 dB/15 dB) done with Chebyshev: $N = 4$ where Butterworth needed $N = 6$. **That is the comparison 76 Bh asks for, with real numbers.**
- 70 Ma prints $\alpha_{\max}$ for the passband and $\alpha_{\min}$ for the stopband, which is the reverse of the usual naming. It means $\alpha_p = 1.01$, $\alpha_s = 13.55$: the $\delta_p = 0.11$, $\delta_s = 0.21$ spec yet again, in a third notation.
- Both have even N , so both have DC gain $1/\sqrt{1 + \epsilon^2}$, not 1.

75 Bh 10

an analog design first

QUESTION AS PRINTED 2075 Bhadra · [10]

Design an *analog* filter using Chebyshev approximation with maximum passband attenuation 5 dB at 20 rad/sec and stopband attenuation 20 dB at 50 rad/sec; then, taking a sampling frequency of 0.5 Hz, convert the obtained analog filter to digital using the bilinear transformation method.

- **The edges are already analog**, in rad/s. **Do not pre-warp**: the question specifies the analog filter directly, so pre-warping would design the wrong one. This is the only paper in the chapter where the pre-warp is skipped legitimately.
- $\Omega_p = 20$, $\Omega_s = 50$, $\alpha_p = 5$, $\alpha_s = 20$, $\epsilon = \sqrt{10^{0.5} - 1} = \mathbf{1.47047}$.

$$N \geq \frac{\cosh^{-1}(6.76647)}{\cosh^{-1}(2.5)} = 1.6592 \Rightarrow \boxed{N = 2}$$

- $\alpha = 1.88938$, $a = 0.32352$, $b = 1.05103$; semi-axes 6.47037 and 21.02060; poles $-\mathbf{4.57524} \pm \mathbf{j14.86381}$.

$$H_a(s) = \frac{136.011}{s^2 + 9.15049s + 241.866}$$

- Now $T = 2$ s, and **here T does NOT cancel**, because the analog filter was fixed before the mapping instead of being derived through a pre-warp. Substitute $s = \frac{2}{2} \frac{1-z^{-1}}{1+z^{-1}} = \frac{1-z^{-1}}{1+z^{-1}}$:

$$H(z) = \frac{0.539692(1 + z^{-1})^2}{1 + 1.911510z^{-1} + 0.927382z^{-2}}$$

- **Both denominator coefficients are positive**, which in §1 never happened. The poles are at $-0.956 \pm j0.118$, near $z = -1$: sampling a 20 rad/s filter at 0.5 Hz is sampling far below Nyquist, and the filter that results is a high-frequency one. **Say that the sampling rate is inconsistent with the analog design** rather than presenting the answer as a good low-pass filter.

4 Spectral transformation

HOT 6/45 70 Ch 5, 80 Ba 4, 70 Asa 2, 79 Bh, 71 Ch, 69 Ch 4 · the 79 Bh family's 4 marks are worked at the end of §1

QUESTION AS PRINTED 2070 Chaitra · [5]

A digital LPF with cut off frequency $w_c = 0.2575\pi$ is given as $H(z) = \frac{0.1 + 0.4z^{-1}}{1 - 0.6z^{-1} + 0.1z^{-2}}$. Design a digital high pass filter with $w'_c = 0.3567\pi$.

- **Sanity check the data first**, because it is free: $|H(e^{j0.2575\pi})| = 0.70789 \approx 1/\sqrt{2}$, so 0.2575π really is this filter's 3 dB point. **The paper's numbers are consistent.**

$$\alpha = -\frac{\cos \frac{\omega_c + \omega'_c}{2}}{\cos \frac{\omega_c - \omega'_c}{2}} = -\frac{\cos(0.3071\pi)}{\cos(0.0496\pi)} = -\frac{0.569595}{0.987880} = -\mathbf{0.57658}$$

- $|\alpha| < 1 \checkmark$, so the map is stable.
- Substitute $z^{-1} \rightarrow -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}} = -\frac{z^{-1} - 0.57658}{1 - 0.57658z^{-1}}$, multiplying top and bottom by $(1 - 0.57658z^{-1})^2$ to clear the fractions:

$$H_{hp}(z) = \frac{0.48106 - 0.94325z^{-1} + 0.38393z^{-2}}{1 - 0.68240z^{-1} + 0.12585z^{-2}}$$

- Check: $|H_{hp}(e^{j0.3567\pi})| = \mathbf{0.70789}$, **the same value the low pass had at its own cut-off**. That is exactly what the transformation promises, and it is the one line that proves the answer.
- **Note the numerator's signs alternate** where the prototype's did not. A high pass built this way pushes the zeros towards $z = +1$.

80 Ba, 70 Asa 4 2

Both are bookwork, answered in §6.6 of the theory: why the transformation is required (impulse invariance cannot make a high pass; doing it in the z domain avoids aliasing entirely), the all-pass form the map must take, and the LP $\rightarrow$ HP row of the table with its α .

Ch 7 · Discrete Fourier Transform and FFT

Numerical Problems & Solutions

82 of the 99 questions are two calculations: an N -point DFT by DIT or DIF (43 questions, 340 marks) and a circular convolution (39 questions, 253 marks). The other 17 are bookwork, answered in the theory chapter. Every computational paper in the 45 is below.

The DIT recipe

1. Pad $x[n]$ with zeros up to N , a power of 2. Eleven papers print six or seven values for an 8-point transform; **that is the zero padding, not a misprint.**
2. Reorder the input into bit-reversed order. For $N = 8$ that is $x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]$.
3. Stage 1: $N/2$ butterflies on *adjacent* pairs, twiddle $W_N^0 = 1$, so it is pure add and subtract.
4. Stage 2: butterflies spanning 2, twiddles W_N^0 and W_N^2 .
5. Stage 3: butterflies spanning 4, twiddles $W_N^0 \dots W_N^3$.
6. Read the output in natural order: it is $X[0]$ to $X[N-1]$.

The DIF recipe, which is the same thing backwards

1. Pad to N as before, but **leave the input in natural order.**
2. Stage 1: butterflies spanning $N/2$, twiddle applied to the *difference*, twiddles $W_N^0 \dots W_N^3$.
3. Stage 2: span $N/4$, twiddles W_N^0 and W_N^2 .
4. Stage 3: span 1, twiddle W_N^0 , pure add and subtract.
5. **Unscramble the output:** what comes out in position i is $X[\text{bitrev}(i)]$.

The twiddle factors, which is all the arithmetic there is

W_8^k	Exact	Decimal
W_8^0	1	1
W_8^1	$\frac{1}{\sqrt{2}}(1-j)$	0.7071 – $j0.7071$
W_8^2	$-j$	$-j$
W_8^3	$-\frac{1}{\sqrt{2}}(1+j)$	$-0.7071 -$ $j0.7071$

And W_4^k is 1, $-j$, -1 , $+j$. Beyond $k = 3$ use $W_8^{k+4} = -W_8^k$; **there are only four numbers to know.**

Checks that cost nothing

1. $X[0] = \sum x[n]$. One addition, and it catches a wrong first stage.
2. $X[N-k] = X^*[k]$ for real $x[n]$. Compute $X[1]$ to $X[N/2]$ and **mirror the rest**, which halves the work and is not a shortcut the examiner can object to.
3. $X[N/2] = \sum (-1)^n x[n]$, the alternating sum.
4. Parseval: $\sum |x[n]|^2 = \frac{1}{N} \sum |X[k]|^2$.
5. For a convolution, $\sum y[n] = (\sum x_1)(\sum x_2)$.

1 The 8-point DFT by DIT and by DIF

TOP 43/45 80 Bh, 81 Ba, 82 Ba, 80 Ba, 70 Ch, 78 Bh, 79 Bh, 81 Bh, 79 Ba, 76 Ch, 76 Ash, 82 Bh, 75 Ch, 75 Ash, 74 Ch, 74 Ash, 70 Bh, 81 Ch, 76 Bh, 70 Ma, 73 Ch, 73 Shr, 72 Ch, 72 Ka, 71 Ch, 70 Asa, 71 Shr, 71 Bh, 69 Ch, 80 Ch, 73 Bh, 72 Ma, 78 Ch, 77 Ch, 75 Bh, 74 Bh, 73 Ma, 74 Ma, 66 Ma, 68 Bh, 69 Bh, 72 Ash · **340 marks, the largest single question in the subject**

QUESTION AS PRINTED 2080 Baishakh · 2070 Chaitra · [2+6]

Find the 8-point DFT of the sequence $x[n] = \{1, 1, 0, 0, 1, 1, 2\}$ using the decimation in frequency FFT algorithm. 2080 Baishakh also asks for the same sequence by the decimation in time algorithm.

This is the one sequence a paper asks BOTH ways, so it is worked both ways here. The answers are identical, which is the point.

Step 0: pad, and say so

- Seven values are printed for an 8-point transform, so $x[7] = 0$: $x[n] = \{1, 1, 0, 0, 1, 1, 2, 0\}$.
- Check target: $X[0] = \sum x[n] = 1 + 1 + 0 + 0 + 1 + 1 + 2 + 0 = 6$.

Route A: decimation in time

- Bit-reversed input order 0, 4, 2, 6, 1, 5, 3, 7 gives $\{x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]\} = \{1, 1, 0, 2, 1, 1, 0, 0\}$.

	0	1	2	3	4	5	6	7
input (bit-rev)	1	1	0	2	1	1	0	0
after stage 1	2	0	2	-2	2	0	0	0
after stage 2	4	j^2	0	$-j^2$	2	0	2	0
after stage 3	6	j^2	$-j^2$	$-j^2$	2	j^2	j^2	$-j^2$

- **Stage 1** is pure add/subtract ($W^0 = 1$) on adjacent pairs: $1 \pm 1 = 2, 0$; $0 \pm 2 = 2, -2$; $1 \pm 1 = 2, 0$; $0 \pm 0 = 0, 0$.
- **Stage 2** spans 2, with twiddles $W_8^0 = 1$ and $W_8^2 = -j$. The $-j$ acting on -2 is what produces the $\pm j^2$.
- **Stage 3** spans 4, twiddles $W_8^0, W_8^1, W_8^2, W_8^3$. The last row is already $X[k]$ in **natural order**.

Route B: decimation in frequency, same sequence

	0	1	2	3	4	5	6	7
input (natural)	1	1	0	0	1	1	2	0
after stage 1	2	2	2	0	0	0	j^2	0
after stage 2	4	2	0	$-j^2$	j^2	0	$-j^2$	0
after stage 3	6	2	$-j^2$	j^2	j^2	j^2	$-j^2$	$-j^2$
unscrambled	6	j^2	$-j^2$	$-j^2$	2	j^2	j^2	$-j^2$

- **Stage 1** pairs $x[n]$ with $x[n+4]$: sums 2, 2, 2, 0 on top; the differences 0, 0, -2, 0 are then multiplied by $W_8^0, W_8^1, W_8^2, W_8^3$, and $(-2)(W_8^2) = (-2)(-j) = j^2$ lands in position 6.
- **The last row is in bit-reversed order.** Position i holds $X[\text{bitrev}(i)]$, so reading positions 0, 4, 2, 6, 1, 5, 3, 7 gives the natural order shown in the final row.

$$X(k) = \{6, j^2, -j^2, -j^2, 2, j^2, j^2, -j^2\}$$

The checks, and a pleasant surprise

- $X[0] = 6$. ✓ $X[4] = \sum (-1)^n x[n] = 1 - 1 + 0 - 0 + 1 - 1 + 2 - 0 = 2$. ✓
- $X[7] = X^*[1]: -j^2 = (j^2)^*$. ✓ And $X[6] = X^*[2], X[5] = X^*[3]$. ✓
- $|X(k)| = \{6, 2, 2, 2, 2, 2, 2, 2\}$: every non-zero frequency has the same magnitude 2, and only the phase differs. Worth writing down, because it looks like an error and is not.
- Parseval: $\sum |x[n]|^2 = 1 + 1 + 0 + 0 + 1 + 1 + 4 + 0 = 8$, and $\frac{1}{8}(36 + 7 \times 4) = \frac{64}{8} = 8$. ✓

Every other FFT paper

Same recipe, different sequences. **DIT and DIF give the same $X(k)$** , so the algorithm column records only what the paper asked for. Sequences shorter than N are zero padded, which is shown as printed.

Papers	N	Alg	$x[n]$ as printed	$X(k)$
80 Bh	8	DIF	{2, 1, 2, 1, 1, 2, 1, 2}	{12, $1+j0.4142$, 0, $1+j2.414$, 0, $1-j2.414$, 0, $1-j0.4142$ }
78 Bh	8	DIT	{1, 1, 1, 1, 0, 0, 0, 0}	{4, $1-j2.414$, 0, $1-j0.4142$, 0, $1+j0.4142$, 0, $1+j2.414$ }
81 Bh	8	DIT	{1, -2, -4, 3, 0, 1}	{-1, $-3.243+j4$, $5+j4$, $5.243-j4$, -5, $5.243+j4$, $5-j4$, $-3.243-j4$ }
79 Ba	8	DIF	{1, 2, 4, 3, 5, -1, 3}	{17, $-4-j5.243$, $-1+j2$, $-4-j3.243$, 9, $-4+j3.243$, $-1-j2$, $-4+j5.243$ }
76 Ch	8	DIF	{1, 2, 3, 3, 5, 0, 4, 6}	{24, $-0.4645+j1.707$, $-1+j7$, $-7.536-j0.2929$, 2, $-7.536+j0.2929$, $-1-j7$, $-0.4645-j1.707$ }
76 Ash	8	DIF	{1, 2, 3, 3, 5, 1, 4, 2}	{21, $-4-j0.4142$, $-1+j2$, $-4-j2.414$, 5, $-4+j2.414$, $-1-j2$, $-4+j0.4142$ }
82 Bh	8	DIT	{1, 0.5, -1, -0.5, 2, -1.5}	{0.5, $0.7678-j0.06066$, $4+j0.5$, $-2.768-j2.061$, 3.5, $-2.768+j2.061$, $4-j0.5$, $0.7678+j0.06066$ }
75 Ash	8	DIT	{1, 2, 3, 4, 5, 6, 7, 8}	{36, $-4+j9.657$, $-4+j4$, $-4+j1.657$, -4, $-4-j1.657$, $-4-j4$, $-4-j9.657$ }
74 Ch	8	DIF	{1.5, -1, 1.8, 0.6, 3, 1.7}	{7.6, $-3.833-j0.3151$, $2.7-j0.1$, $0.8335+j3.285$, 5, $0.8335-j3.285$, $2.7+j0.1$, $-3.833+j0.3151$ }

Papers	N	Alg	$x[n]$ as printed	$X(k)$
74 Ash, 70 Bh	8	DIF	$\{0.5, 0.5, 0.5, 0.5, 0, 0, 0, 0\}$	$\{2, 0.5-j1.207, 0, 0.5-j0.2071, 0, 0.5+j0.2071, 0, 0.5+j1.207\}$
81 Ch	8	DIF	$\{-0.5, -0.5, 0.5, 0.5, 0, 0, 0, 0\}$	$\{0, -1.207-j0.5, -1+j1, 0.2071+j0.5, 0, 0.2071-j0.5, -1-j1, -1.207+j0.5\}$
73 Ch	8	DIF	$\{1, -1, 3, 2, 1, 1, 3, -2\}$	$\{8, -4.243-j1.414, -4, 4.243-j1.414, 8, 4.243+j1.414, -4, -4.243+j1.414\}$
73 Shr	8	DIT	$\{1, -1, 2, 2, 1, 1, 2, 2\}$	$\{10, -1.414+j1.414, -2+j4, 1.414+j1.414, 2, 1.414-j1.414, -2-j4, -1.414-j1.414\}$
72 Ch	8	DIT	$\{1, 1, 2, 4, 3, 1, 2, 1\}$	$\{15, -4.121-j2.121, j3, 0.1213-j2.121, 1, 0.1213+j2.121, -j3, -4.121+j2.121\}$
72 Ka	8	DIT	$\{1, 1, 0, 1, 0, 1, 2\}$	$\{6, 0.2929+j1.293, -1-j1, 1.707-j2.707, 0, 1.707+j2.707, -1+j1, 0.2929-j1.293\}$
71 Shr, 71 Bh	8	DIT	$\{1, 1, 2, 0, 1, 2, 0, 1\}$	$\{8, -j0.5858, -j2, j3.414, 0, -j3.414, j2, j0.5858\}$
69 Ch	8	DIF	$\{1, 1, 2, 2, 1, 1, 2, 1\}$	$\{11, -0.7071-j0.7071, -2+j1, 0.7071-j0.7071, 1, 0.7071+j0.7071, -2-j1, -0.7071+j0.7071\}$
80 Ch, 73 Bh	8	DIT	$\{1, 0, 2, 0, -1, 1, 1\}$	$\{4, 1.293-j0.2929, -3-j1, 2.707+j1.707, 2, 2.707-j1.707, -3+j1, 1.293+j0.2929\}$
72 Ma	8	DIF	$\{1, 0, 0, 0, -1, 1, 1\}$	$\{2, 1.293+j1.707, -1-j1, 2.707-j0.2929, 0, 2.707+j0.2929, -1+j1, 1.293-j1.707\}$
74 Bh	8	DIT	$\{1, 1, 0, 0, 0\}$	$\{2, 1.707-j0.7071, 1-j1, 0.2929-j0.7071, 0, 0.2929+j0.7071, 1+j1, 1.707+j0.7071\}$
73 Ma	8	DIT	$\{1, -1, 2\}$	$\{2, 0.2929-j1.293, -1+j1, 1.707+j2.707, 4, 1.707-j2.707, -1-j1, 0.2929+j1.293\}$
74 Ma	8	DIT	$\{1, 1, 0, -1, 2\}$	$\{3, 0.4142, 3-j2, -2.414, 3, -2.414, 3+j2, 0.4142\}$

1.1 The four-point papers

HOT 8/45 81 Ba **2+6**, 82 Ba **1+2+6**, 75 Ch **2+8**, 76 Bh **2+5**, 70 Ma **7**, 66 Ma **6**, 68 Bh **8**, 69 Bh **5** · **two stages, not three, and the twiddles are only 1 and $-j$**

Paper	Alg	$x[n]$	$X(k)$	Note
81 Ba	DIT	$\{2, 2, 4\}$	$\{8, -2-j2, 4, -2+j2\}$	padded to 4
82 Ba	DIF	$\{1, 4, -1\}$	$\{4, 2-j4, -4, 2+j4\}$	padded to 4
75 Ch	DIT	$\{1, -2, 3, 2\}$	$\{4, -2+j4, 4, -2-j4\}$	asks $X(3)$ and $X(5)$
76 Bh	DIT	$\{0.5, 0.5, 0.5, 0.5\}$	$\{2, 0, 0, 0\}$	constant $\rightarrow$ impulse
70 Ma	DIT	$\{2, 0, 1, 2\}$	$\{5, 1+j2, 1, 1-j2\}$	
66 Ma	DIT	$\{1, 3, 4, 5\}$	$\{13, -3+j2, -3, -3-j2\}$	"show the computation"
68 Bh	DIF	$\{1, -2, 2, 1\}$	$\{2, -1+j3, 4, -1-j3\}$	
69 Bh	DIT	$\{3.6, 5.5, 3.3, 6.3\}$	$\{18.7, 0.3+j0.8, -4.9, 0.3-j0.8\}$	

- **66 Ma worked, since it says "show the computation".** Bit-reversed input $\{x[0], x[2], x[1], x[3]\} = \{1, 4, 3, 5\}$. Stage 1: $1 \pm 4 = 5, -3$ and $3 \pm 5 = 8, -2$, giving $\{5, -3, 8, -2\}$. Stage 2 with $W_4^0 = 1, W_4^1 = -j$: $5 \pm 8 = 13, -3$ and $-3 \pm (-j)(-2) = -3 \pm j2$, giving $\{13, -3+j2, -3, -3-j2\}$. ✓ $X[0] = 1 + 3 + 4 + 5 = 13$.
- **75 Ch asks for $X(5)$ of a 4-point DFT.** The DFT is periodic in k with period N , so $X(5) = X(5-4) = X(1) = -2+j4$, and $X(3) = -2-j4 = X^*(1)$. **That periodicity is the whole question**; computing a 5th value would be a mistake, not extra credit.
- **76 Bh's constant sequence transforms to an impulse:** a flat signal has energy at DC and nowhere else. Free check on a DIT you can do in your head.

1.2 Papers whose sequence is a formula, an impulse, or missing

HOT 6/45 78 Ch **7**, 77 Ch **8**, 75 Bh **2+8**, 71 Ch, 70 Asa **2+6**, 79 Bh **2+6**

- **Sample the formula first, then run the ordinary recipe.** All three are 8-point DIF.

Paper	Formula	$x[n], n = 0 \dots 7$	$X(k)$
78 Ch	$0.5 \sin(n\pi/6)$	$\{0, 0.25, 0.433, 0.5, 0.433, 0.25, 0, -0.25\}$	$\{1.616, -0.9633-j0.9633, -j0.25, 0.09732-j0.09732, 0.116, 0.09732+j0.09732, j0.25, -0.9633+j0.9633\}$
77 Ch	$\sin(3\pi n/8)$	$\{0, 0.9239, 0.7071, -0.3827, -1, -0.3827, 0.7071, 0.9239\}$	$\{1.497, 2.848, -2.414, -0.8478, -0.6682, -0.8478, -2.414, 2.848\}$
75 Bh	$0.2n$	$\{0, 0.2, 0.4, 0.6, 0.8, 1, 1.2, 1.4\}$	$\{5.6, -0.8+j1.931, -0.8+j0.8, -0.8+j0.3314, -0.8, -0.8-j0.3314, -0.8-j0.8, -0.8-j1.931\}$

- **77 Ch's spectrum is entirely real**, because its sampled sequence happens to be symmetric about $n = 4$. A real spectrum is a legitimate answer; do not hunt for a missing j .
- **75 Bh asks only for $X(4)$ and $X(7)$.** $X(4) = -0.8$ is the alternating sum $0 - 0.2 + 0.4 - 0.6 + 0.8 - 1 + 1.2 - 1.4$; $X(7) = X^*(1) = -0.8 - j1.931$. **Two butterflies, not twenty-four.**

- **71 Ch and 70 Asa** give $x(n) = \{1, 0, 0, 0, 0, 0, 0, 0\}$, a unit impulse, and ask for $X(7)$. Every $X(k) = 1$, so $X(7) = 1$. The flow graph is still worth drawing, because that is where the marks are: the answer itself is one line.
- **79 Bh prints no sequence at all.** The paper says only "find the 8-point DFT of the following sequence using radix-2 DIT-FFT", and nothing follows. Verified on both available scans. **Answer the method:** draw the general 8-point DIT flow graph, state the bit-reversed input order and the three stages, and say the numeric answer needs the data that is missing.

1.3 72 Ash: the IDFT through a forward FFT

PIN 1/45 72 Ash 2+6

QUESTION AS PRINTED 2072 Ashwin · [2+6]

Use the FFT algorithm to compute the IDFT of the sequence $X(k) = \{6, -2 + 2j, -2, -2 - 2j\}$.

- Use $x[n] = \frac{1}{N} (\text{DFT}\{X^*[k]\})^*$ from §7.3.
- **Conjugate:** $X^*(k) = \{6, -2 - 2j, -2, -2 + 2j\}$.
- **Forward 4-point DFT** of that, by DIT: bit-reversed input $\{6, -2, -2 - 2j, -2 + 2j\}$; stage 1 gives $\{4, 8, -4, -4j\}$; stage 2 with $W_4^1 = -j$ gives $\{0, 4, 8, 12\}$.
- **Conjugate and divide by $N = 4$:**

$$x[n] = \frac{1}{4} \{0, 4, 8, 12\}^* = \{0, 1, 2, 3\}$$

- Check forwards: $\sum x[n] = 6 = X(0)$. ✓ And $x[n]$ came out real, as it must, because $X(k)$ was conjugate symmetric ($X(3) = X^*(1)$).

2 Circular convolution

TOP 39/45 15 papers direct, 12 as $x_3 = \text{IDFT}(X_1 X_2)$, 4 as a linear convolution · 253 marks, and all 31 problems are one calculation

QUESTION AS PRINTED 2080 Bhadra · 2080 Baishakh · [6] [2+5]

Compute the circular convolution of $x_1 = \{1, 2, 3, 1\}$ and $x_2 = \{4, 3, 2, 2\}$.

Route A: the matrix, which is what to do in the exam

- $N = 4$. Build the circulant of x_2 : row n , column m holds $x_2[((n - m))_4]$, so each row is the previous one shifted right by one.

$$\begin{bmatrix} 4 & 2 & 2 & 3 \\ 3 & 4 & 2 & 2 \\ 2 & 3 & 4 & 2 \\ 2 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4+4+6+3 \\ 3+8+6+2 \\ 2+6+12+2 \\ 2+4+9+4 \end{bmatrix} = \begin{bmatrix} 17 \\ 19 \\ 22 \\ 19 \end{bmatrix}$$

$$y[n] = \{17, 19, 22, 19\}$$

- **Check:** $\sum y = 77$, and $(\sum x_1)(\sum x_2) = 7 \times 11 = 77$. ✓ **Two additions and a multiplication.**

Route B: through the DFT, if the paper insists

- $X_1(k) = \{7, -2 - j1, 1, -2 + j1\}$ and $X_2(k) = \{11, 2 - j1, 1, 2 + j1\}$.
- Multiply term by term: $X_3(k) = \{77, -5, 1, -5\}$.
- IDFT with W_4^{-nk} , i.e. $1, j, -1, -j$ as nk runs $0, 1, 2, 3$:

$$y[0] = \frac{1}{4}(77 - 5 + 1 - 5) = \frac{68}{4} = 17$$

$$y[1] = \frac{1}{4}(77 + (-5)(j) + (1)(-1) + (-5)(-j)) = \frac{76}{4} = 19$$

$$y[2] = \frac{1}{4}(77 + (-5)(-1) + (1)(1) + (-5)(-1)) = \frac{88}{4} = 22$$

$$y[3] = \frac{1}{4}(77 + (-5)(-j) + (1)(-1) + (-5)(j)) = \frac{76}{4} = 19$$

The imaginary parts cancel in pairs at every n , as they must for a real answer. Same $\{17, 19, 22, 19\}$.

- **This route costs three transforms to reach a result the matrix gives in one table.** Use it only when the question names the DFT.

Every other direct circular convolution

Papers	N	x_1	x_2	$y[n]$
81 Ba	4	{1, 2, 4, 5}	{2, 1, 6, 8}	{47, 67, 56, 34}
72 Ch, 82 Bh	5	{1, 2, 3, 1}	{1, 2, 1, -1, 1}	{1, 6, 9, 8, 4}
79 Ba	5	{1, -1, -2, 3, -1}	{1, 2, 3}	{8, -2, -1, -4, -1}
82 Ba	5	{1, 1, -1, -1, 2}	{1, -1, -2}	{1, -4, -4, -2, 5}
78 Bh, 74 Ma	4	{2, 1, 2, 1}	{1, 2, 3, 4}	{14, 16, 14, 16}
78 Ch	4	{2, 1, 2, 1}	{1, -2, -1, 3}	{1, 2, 1, 2}
75 Ash	4	{1, 2, -1, 1}	{1, 3, 5, 7}	{13, 3, 17, 15}
75 Ch	4	{0, 0, 1, 1}	{1, 1, 1, 1}	{2, 2, 2, 2}
76 Bh	4	{1, 0.5, 1, 0.5}	{0, 1, 2, 3}	{4, 5, 4, 5}
71 Bh	5	{1, 2, 0, 3, 4}	{2, -1, 2, -1, 2}	{8, 8, 2, 17, 5}
67 Mng	4	{1, 2, 1}	{1, 2, 0, 1}	{3, 5, 5, 3}
66 Ma	4	{1, 2}	{1, 3, 4, 5}	{11, 5, 10, 13}
69 Bh	4	{1, 0, 1}	{1, 0, 2, 1}	{3, 1, 3, 1}
71 Ch, 70 Asa, 79 Ch	4	{1, 2}	{1, 1, 1, 1}	{3, 3, 3, 3}

75 Ch's answer is constant because x_2 is constant: convolving anything with a flat sequence circularly gives $\sum x_1$ in every position. Same reason 71 Ch's answer is {3, 3, 3, 3} with $\sum x_1 = 3$. Both are free marks once you see it.

2.1 "Find $x_3[n]$ if $X_3(k) = X_1(k)X_2(k)$ "

TOP 12/45 76 Ch, 69 Ch, 72 Ma, 79 Bh, 81 Bh, 81 Ch, 75 Bh, 73 Ch, 73 Shr, 80 Ch, 73 Bh, 73 Ma, 74 Ch, 72 Ka, 71 Shr · **this IS the circular convolution; take no transform at all**

By the convolution property of §7.2, multiplying the DFTs is circular convolution in time. Write "by the circular convolution property, $x_3[n] = x_1[n] \otimes x_2[n]$ ", then build the matrix. **One line of theory replaces three DFTs and an IDFT.**

Papers	N	x_1	x_2	$x_3[n]$
76 Ch, 69 Ch	4	{1, 2, -2}	{1, 2, 3, -1}	{-7, 6, 5, 1}
72 Ma	4	{1, 2, 3, -1}	{2, 1, -3}	{-8, 8, 5, -5}
79 Bh	4	{1, 0, 0, 1}	{2, 0, 2}	{2, 2, 2, 2}
72 Ka	4	{1, 2, 4}	{-1, 2, 3, 1}	{13, 4, 3, 15}
71 Shr	4	{1, 2, 3, 4}	{1, 3, 5, 7}	{42, 46, 42, 30}
81 Bh, 81 Ch, 75 Bh	5	{1, 3, 9, 27}	{1, 2, 4, 8, 16}	{229, 365, 451, 65, 130}
73 Ch	5	{1, -2, 5, 1, 2}	{1, 2, -3, -2}	{-8, -8, -6, 15, -7}
73 Shr	5	{1, -2, 2, 1, 4}	{2, 1, -3, -1}	{1, -16, -5, 9, 5}
80 Ch, 73 Bh	5	{1, 2, 3, -1, 5}	{2, 1, -3}	{10, -10, 5, -5, 0}
73 Ma	5	{3, 2, 1, -1, -2}	{1, 2, 3}	{-4, 2, 14, 7, -1}
74 Ch	5	{1, 2, 0, 1, -2}	{1, 0, 1, 1, 2}	{6, 1, 1, 0, 2}

- **71 Shr is the same question in disguise again:** "input $x[n]$, impulse response $h[n]$, find $y[n]$ if $G[k] = X[k]H[k]$ ". It is the 4-point circular convolution, {42, 46, 42, 30}, and **not** the linear one, because no padding was asked for.
- **79 Bh's answer is constant** for the same reason as §2: x_1 padded is {1, 0, 0, 1} and x_2 is {2, 0, 2, 0}; every output picks up $\sum x_1$ · the same pairing.

The defect: 81 Ch asks for a 4-point DFT of a 5-point sequence **1+6 1+5**

81 Ch, 75 Bh and 81 Bh set identical sequences and only one states a workable N

- All three give $x_1[n] = (3)^n$ for $0 \leq n \leq 3$ (four values, {1, 3, 9, 27}) and $x_2[n] = (2)^n$ for $0 \leq n \leq 4$ (**five values**, {1, 2, 4, 8, 16}).
- **81 Bh** says "5-point DFT". Fine, and the answer is the table row above.
- **81 Ch** says "**4-point** DFT". A 4-point DFT of a five-entry sequence **does not exist**; there is nowhere for $x_2[4] = 16$ to go. As printed the question is unanswerable.
- **75 Bh** states **no N at all**. With none given, N must be at least $\max(L, M) = 5$, so it is 81 Bh's question.
- **What to write:** take $N = 5$, say in one line why (a 5-entry sequence needs $N \geq 5$; the paper's own "what is zero padding" part is the justification), and give $x_3 = \{229, 365, 451, 65, 130\}$. **Naming the defect is worth more than forcing an answer.**
- Note for the PYQ document: its heading groups 75 Bh with 81 Ch under "4-pt". That is an inference from 81 Ch's wording, and it is the wrong one.

2.2 Linear convolution through circular convolution

HOT 4/45 74 Ch 1+5 , 74 Bh 2+6 , 72 Ash 5 , 77 Ch 6+2 · the only difference is the choice of N

QUESTION AS PRINTED 2074 Chaitra · [1+5]

- What is zero padding? Find the linear convolution of $x = \{1, 1, 1, 1\}$ and $h = \{2, 3\}$ using circular convolution.
- $L = 4, M = 2$, so the linear result has $L + M - 1 = 5$ samples. **Zero pad both to length 5:** $x = \{1, 1, 1, 1, 0\}, h = \{2, 3, 0, 0, 0\}$.
 - The 5×5 circulant of h against x gives

$y[n] = \{2, 5, 5, 5, 3\}$

- Check: $\sum y = 20 = (\sum x)(\sum h) = 4 \times 5. \checkmark$
- **Why the padding matters, in one demonstration.** Convolving the same two sequences circularly at $N = 4$ instead gives $\{5, 5, 5, 5\}$. The 5th sample, 3, had nowhere to go, so it **wrapped onto** $y[0]$: $2 + 3 = 5$. That is the time-domain aliasing the whole question is about, and it is the answer to the "what is zero padding" part.

Paper	x	h	N	$y[n]$
74 Ch	$\{1, 1, 1, 1\}$	$\{2, 3\}$	5	$\{2, 5, 5, 5, 3\}$
74 Bh	$\{-1, 1\}$	$\{2, 3, 1, -2\}$	5	$\{-2, -1, 2, 3, -2\}$
72 Ash	$\{1, 1, 1\}$	$\{2, 2, 2\}$	5	$\{2, 4, 6, 4, 2\}$
77 Ch	$\{1, 1, 1\}$	$\{1, 0, -3\}$	5	$\{1, 1, -2, -3, -3\}$

- **77 Ch's $x[n]$ is misprinted** as " $u[n] - u[n] - [n-3]$ ". It means $u[n] - u[n-3] = \{1, 1, 1\}$, and $h[n] = \delta[n] - 3\delta[n-2] = \{1, 0, -3\}$.
- **72 Ash's answer is a triangle**, $\{2, 4, 6, 4, 2\}$, because convolving a rectangle with a rectangle always gives a trapezoid, and with two equal lengths it degenerates to a triangle. A shape check worth two seconds.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. → reads “leads to / therefore”, + reads “and”, = reads “is”. ✓ means the property holds, × that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months**: 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II). Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 **Bh** is EX 753 Regular, 74 **Ma** is EX 753 Back.
- **69 Bh and 69 Bh** are two different papers, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.